

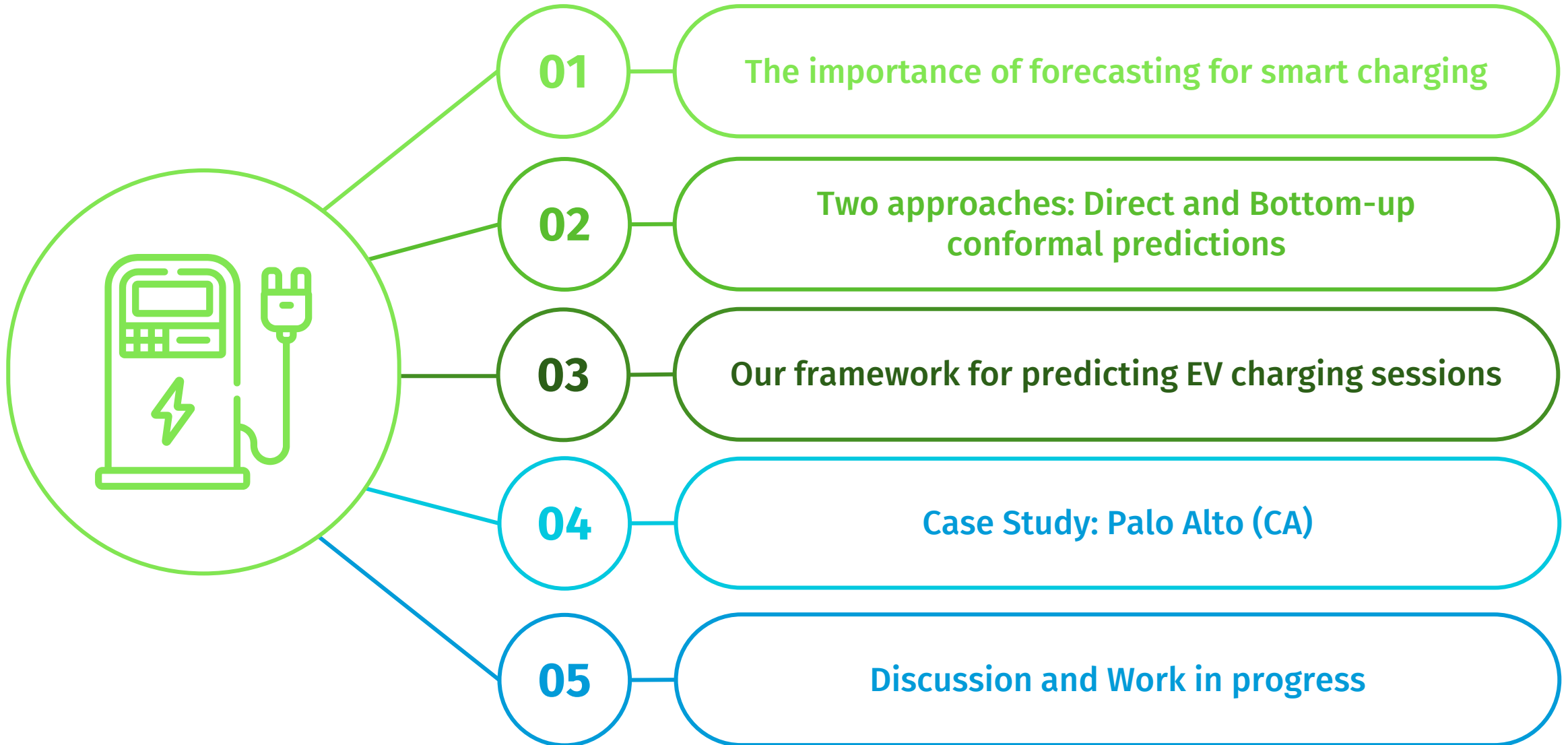
OSIRIS TECH

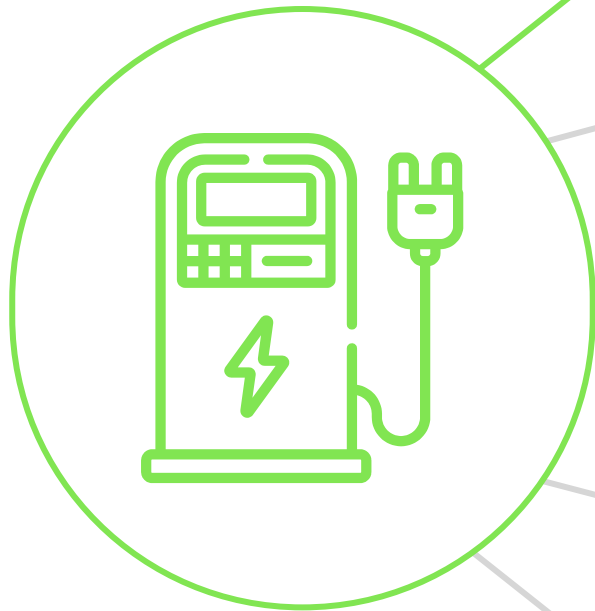
17/11/2024

Modèles de prévision pour le smart charging

R39

Yvenn Amara-Ouali, Bachir Hamrouche, Guillaume Principato





01

The importance of forecasting for smart charging

02

Two approaches: Direct and Bottom-up
conformal predictions

03

Our framework for predicting EV charging sessions

04

Case Study: Palo Alto (CA)

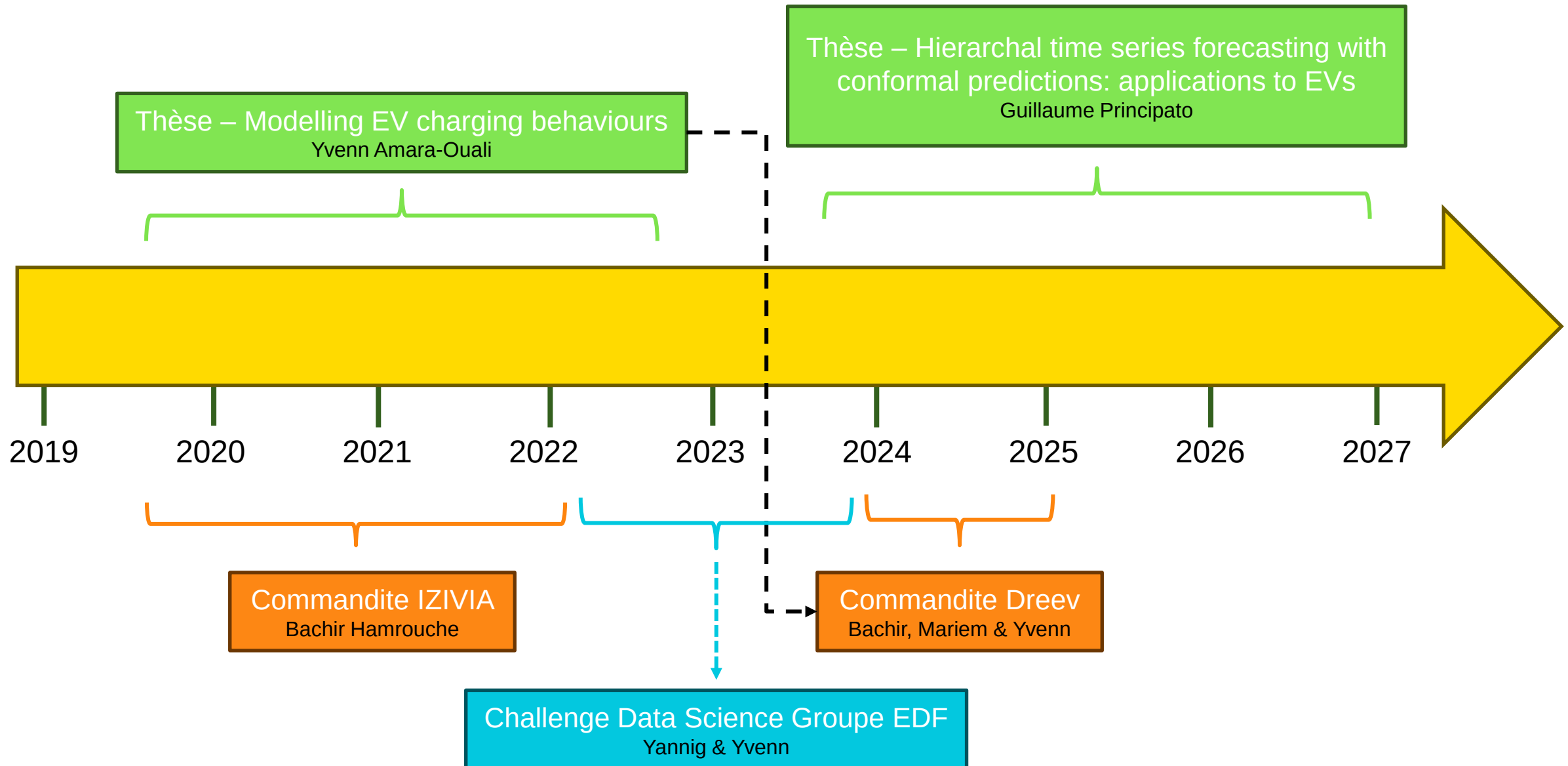
05

Discussion and Work in progress

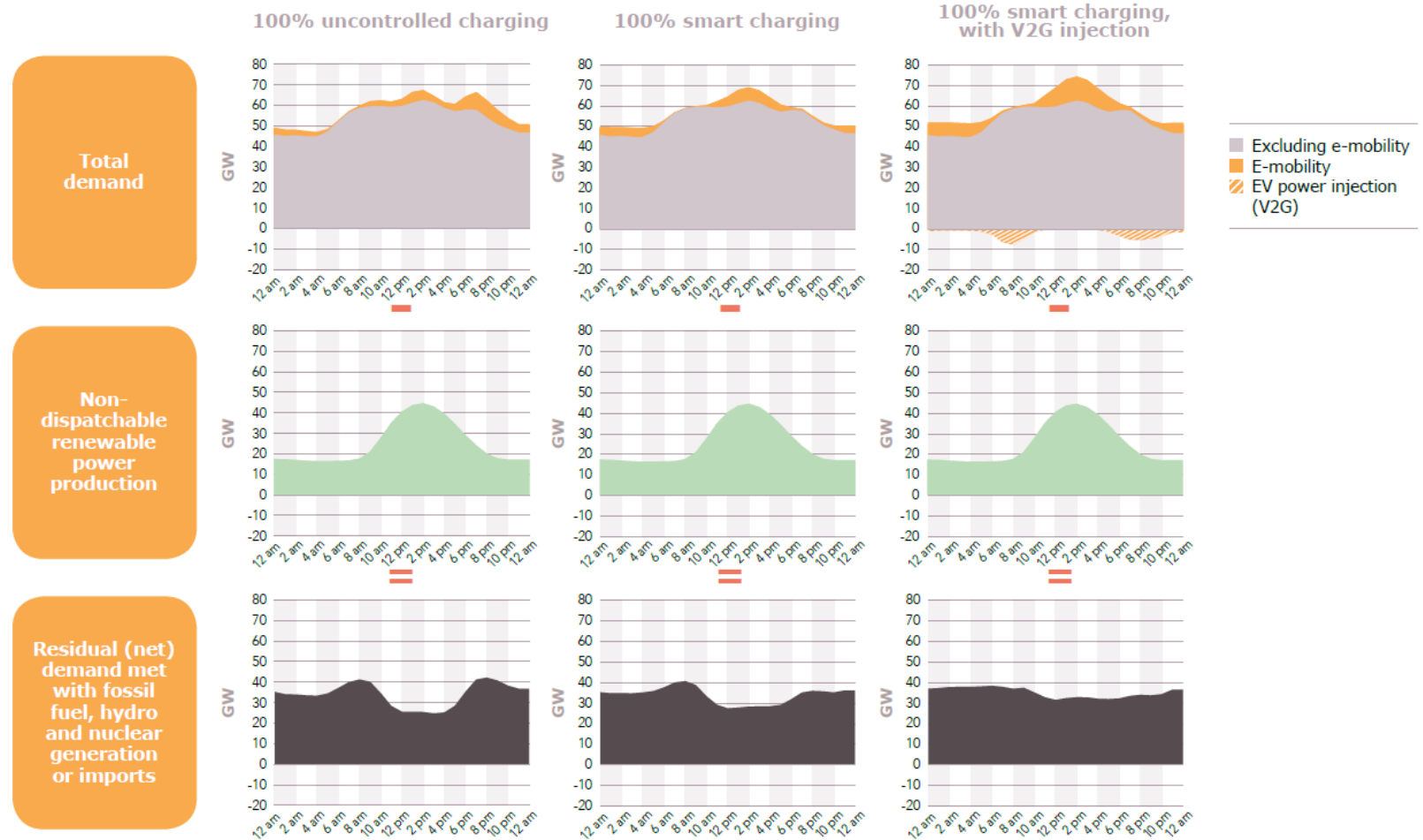
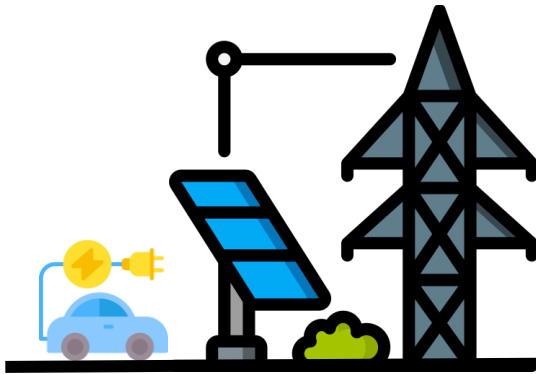
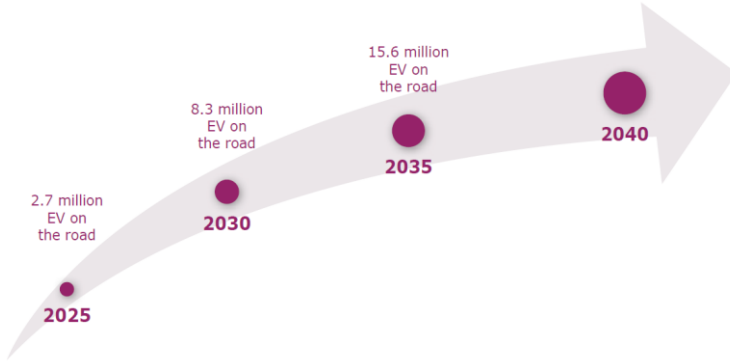
A long-term research path

Projet Energy Data Lab
Manel Boumghar

Projet Smart Charging
Jean-Baptiste Bréal



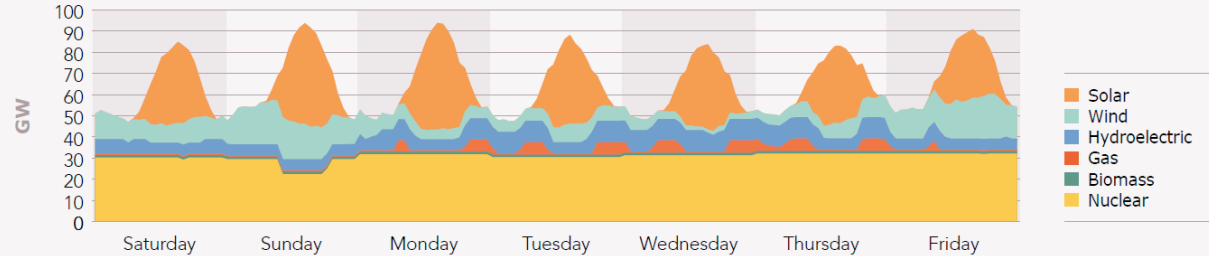
The importance of understanding EV charging behaviours



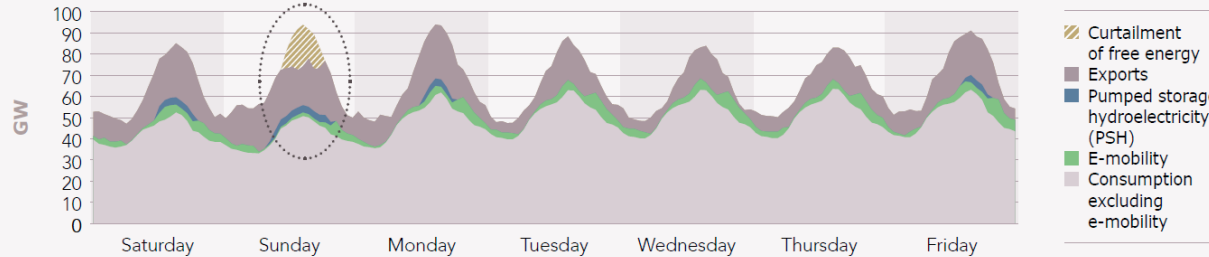
The importance of understanding EV charging behaviours

100% UNCONTROLLED CHARGING

Production in France during one week in July

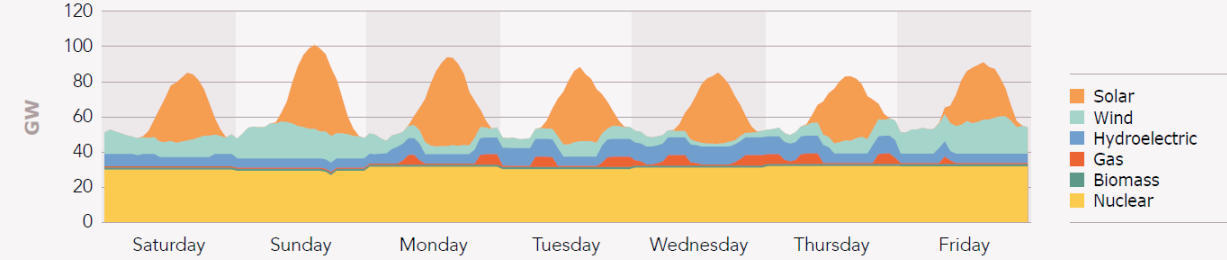


Demand in France during one week in July

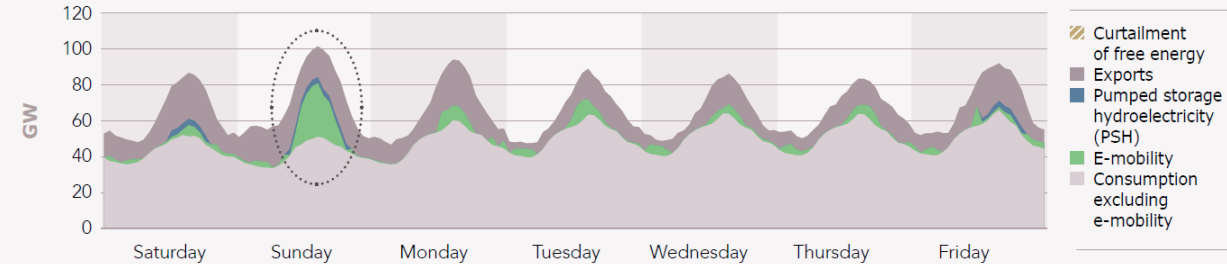


100% SMART CHARGING

Production in France during one week in July



Demand in France during one week in July



Problematic

Producing **accurate forecasts** of EV charging demand
to enable **resilience** for the grid

EV transactions at charging points

Sessions

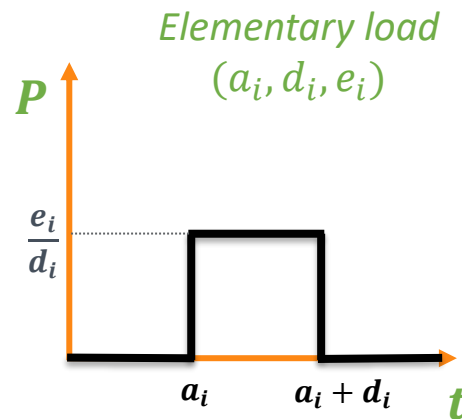
- The data often comes as a history of charging sessions
- Charging session = charging event for one vehicle

$$\text{Session } i = (a_i, d_i, p_i, e_i)$$

- a_i : arrival time
- d_i : charge duration
- p_i : parking duration
- e_i : energy demand

Elementary load profiles

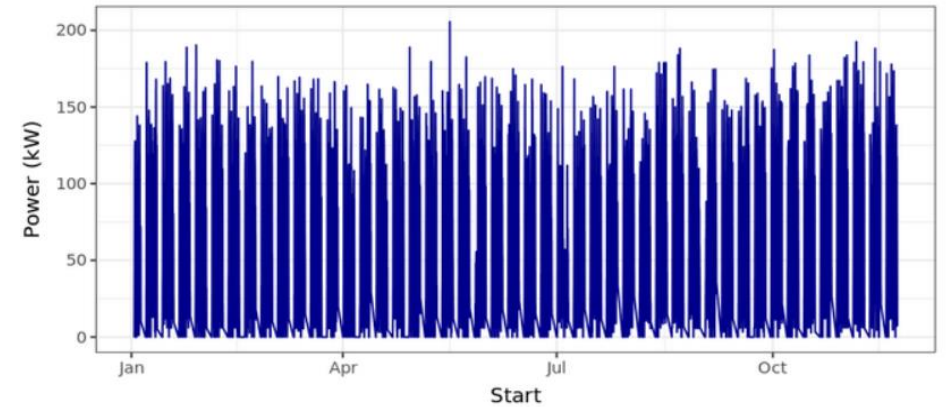
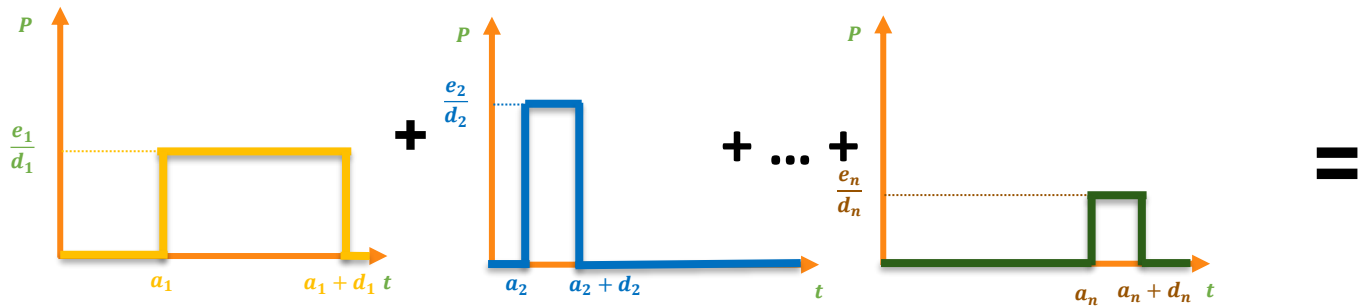
- For each session, an elementary load curve can be reconstructed from the session variables



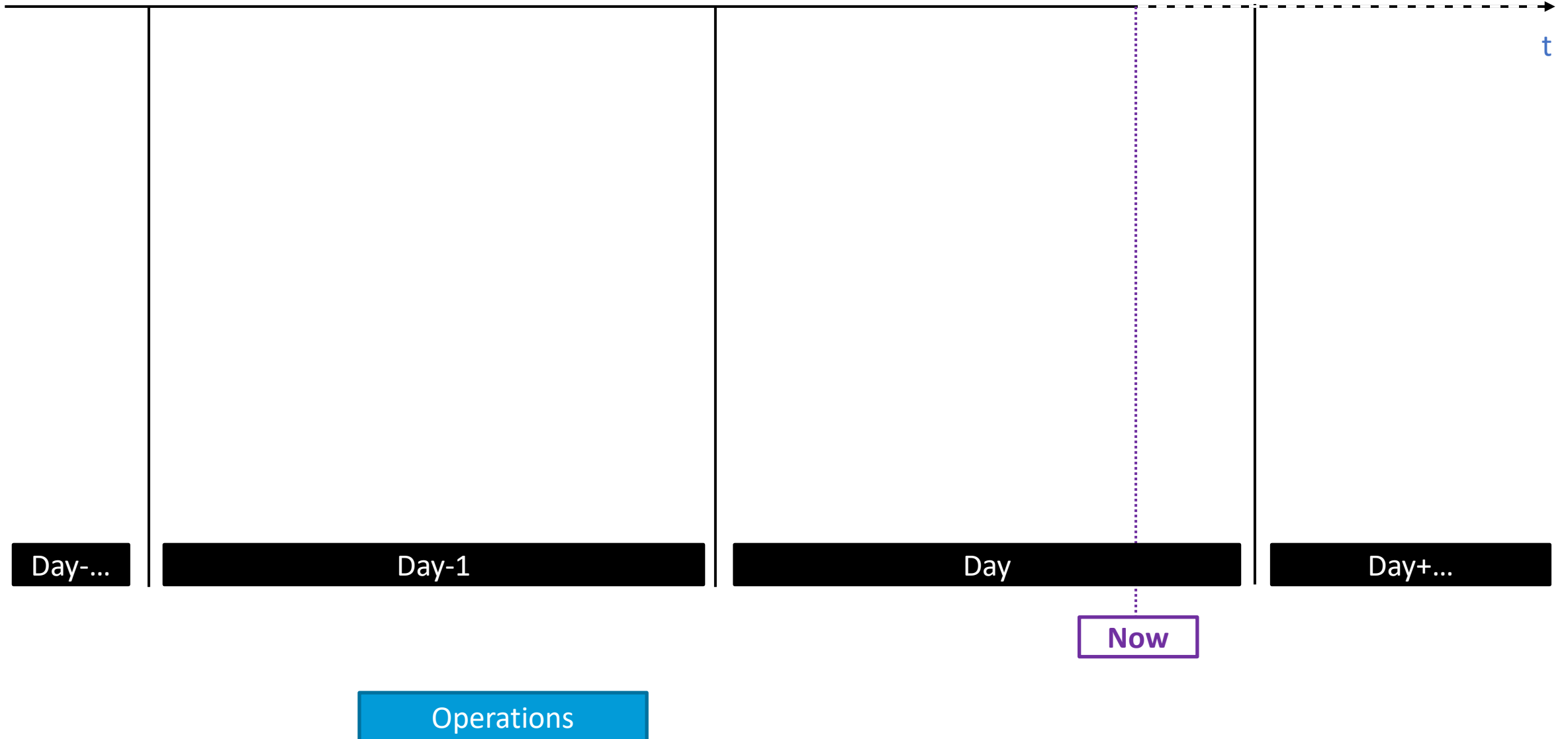
Profile reconstruction

Global load

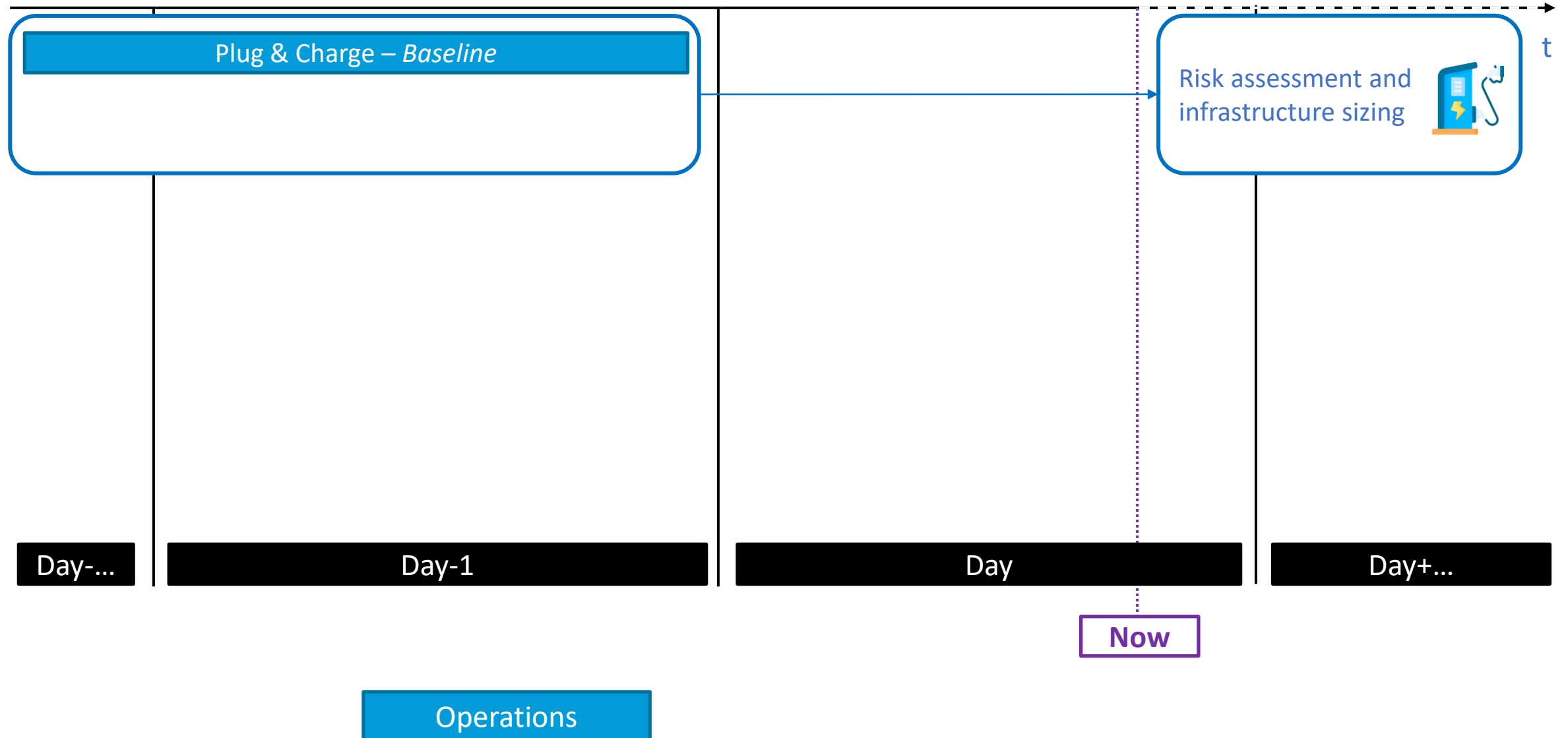
- Adding the elementary profiles leads to global profiles of interest at the scope of a station or dataset



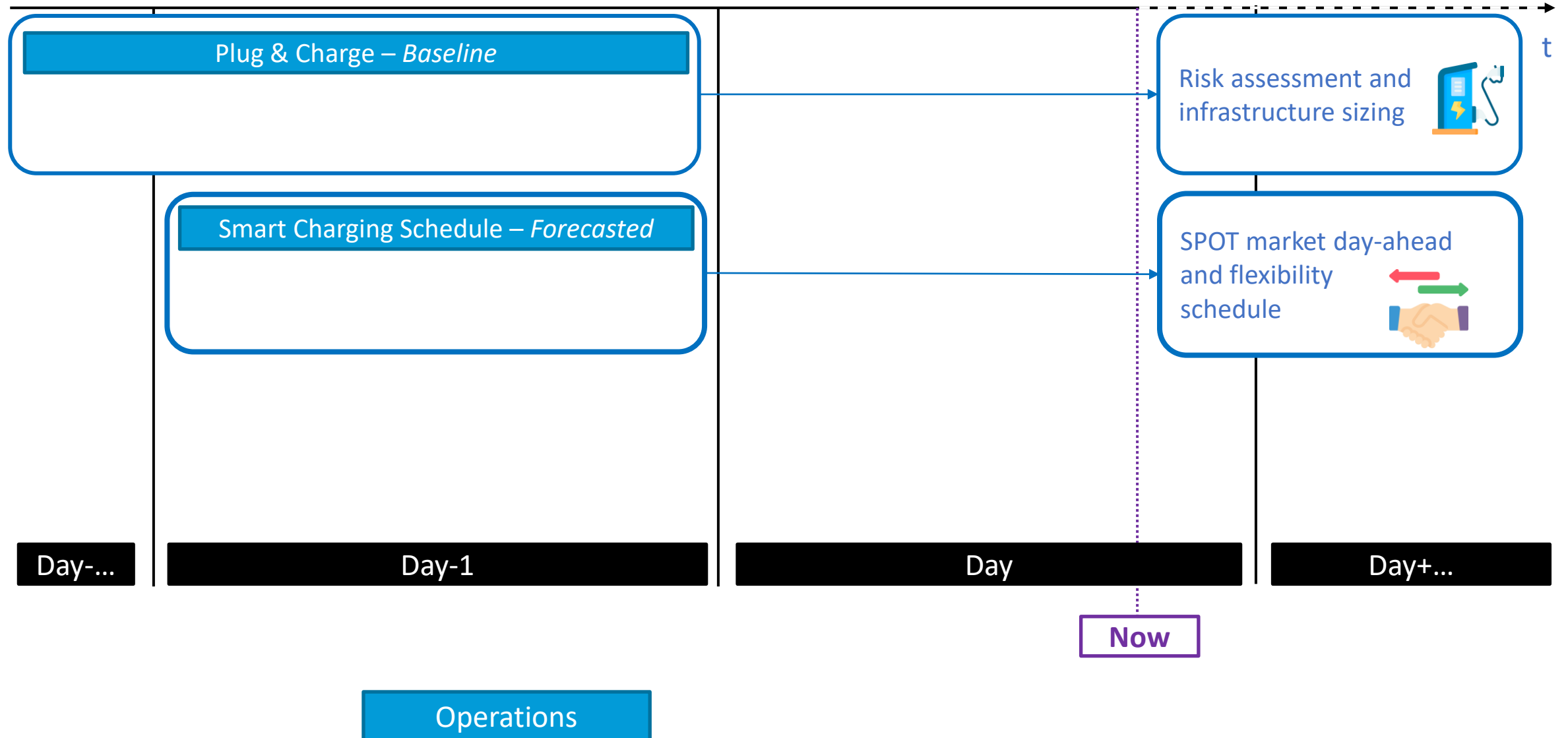
Operations



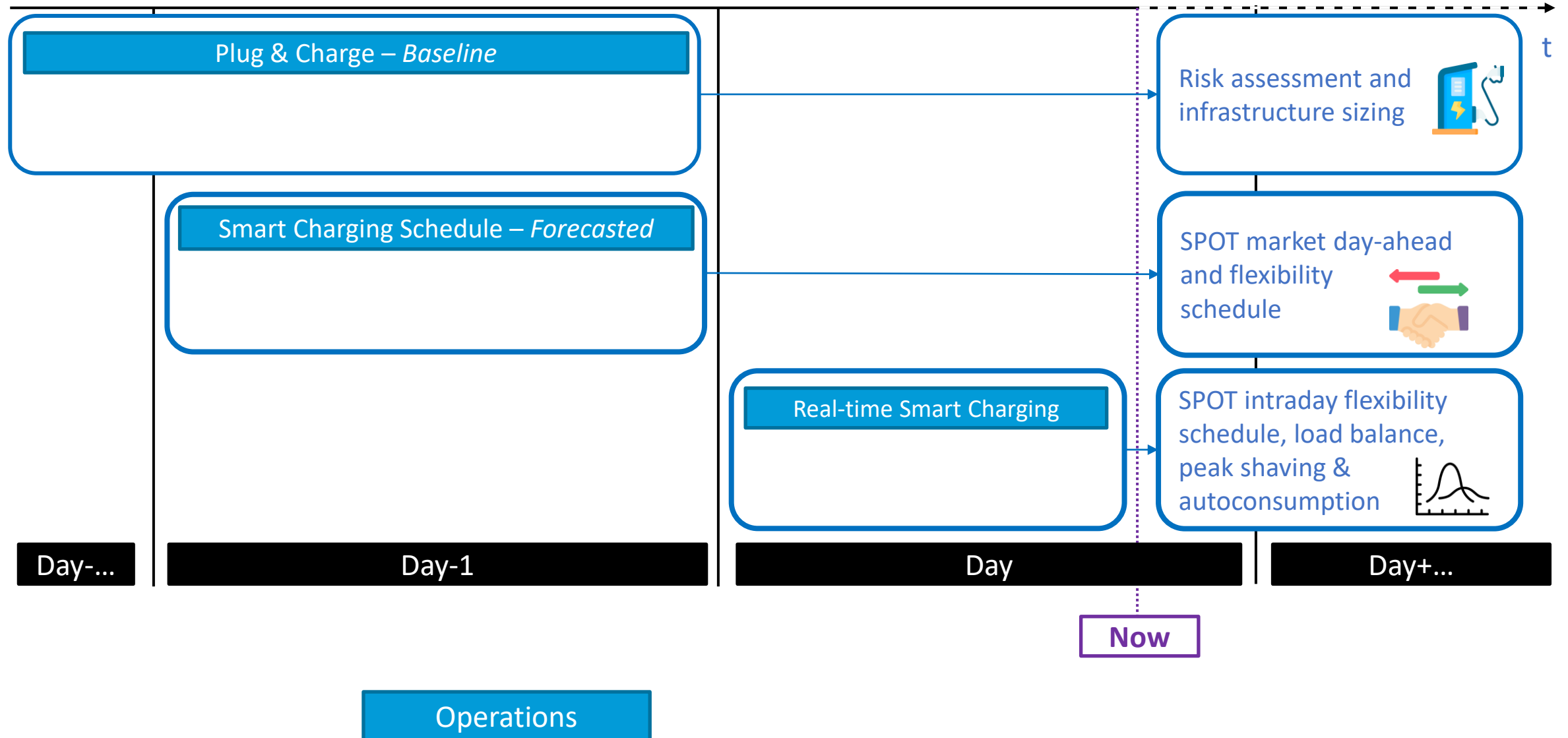
Operations



Operations



Operations



Building up on previous work

Main Results

- Automatic framework for assessing the performances of a wide variety of models across all datasets
- Direct approaches perform better than bottom-up approaches overall
- Bottom-up approaches outperform direct approaches on residential data

Limitations

- Exogenous information such as traffic and weather not considered

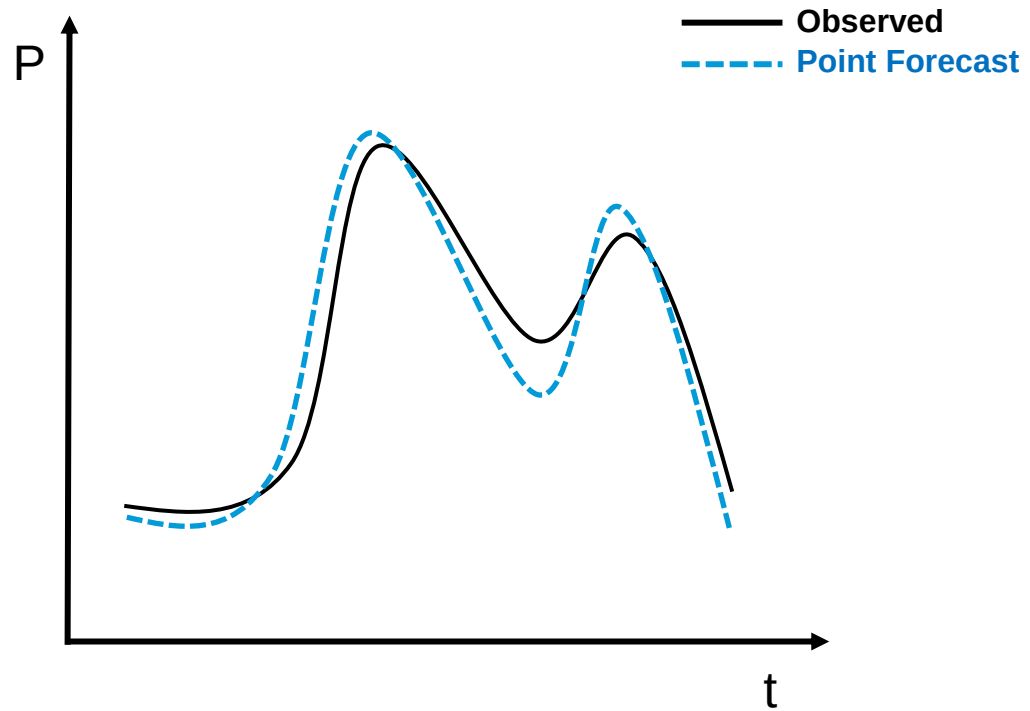
Perspectives

- Using the hierarchical structure of the data to improve bottom-up forecasts
- **Probabilistic forecasts of the EV charging demand**
- Improving the modelling performance on different datasets altogether (e.g., transfer learning)

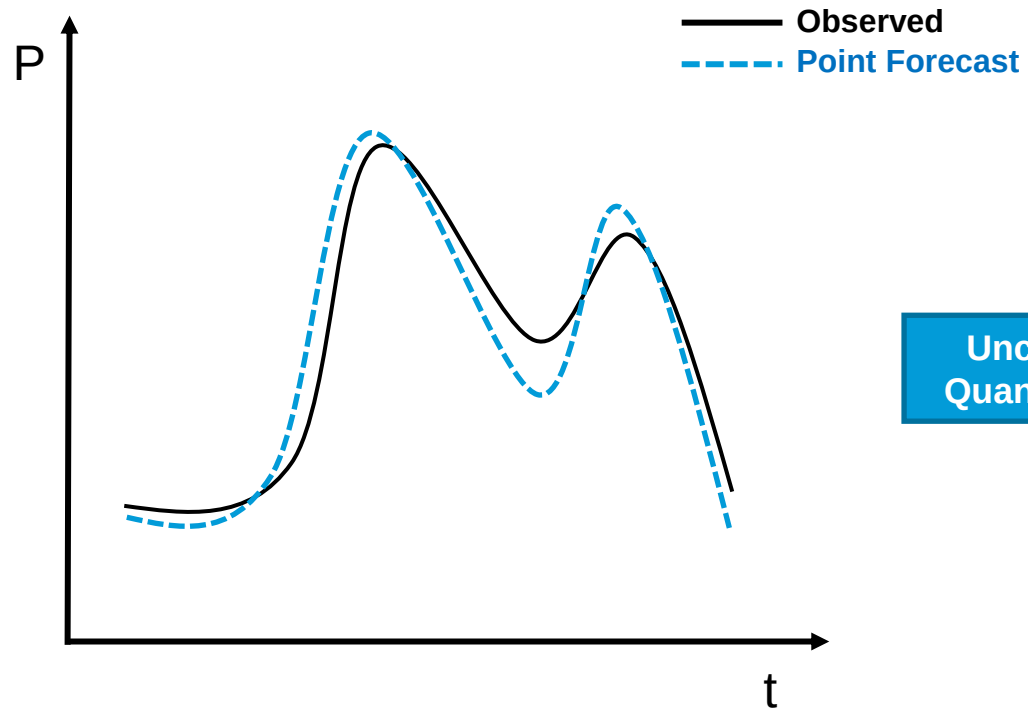
Publication

Amara-Ouali, Y., Goude, Y., Hamrouche, B., & Bishara, M. (2022). A benchmark of electric vehicle load and occupancy models for day-ahead forecasting on open charging session data. In *Proceedings of the Thirteenth ACM International Conference on Future Energy Systems* (pp. 193-207).

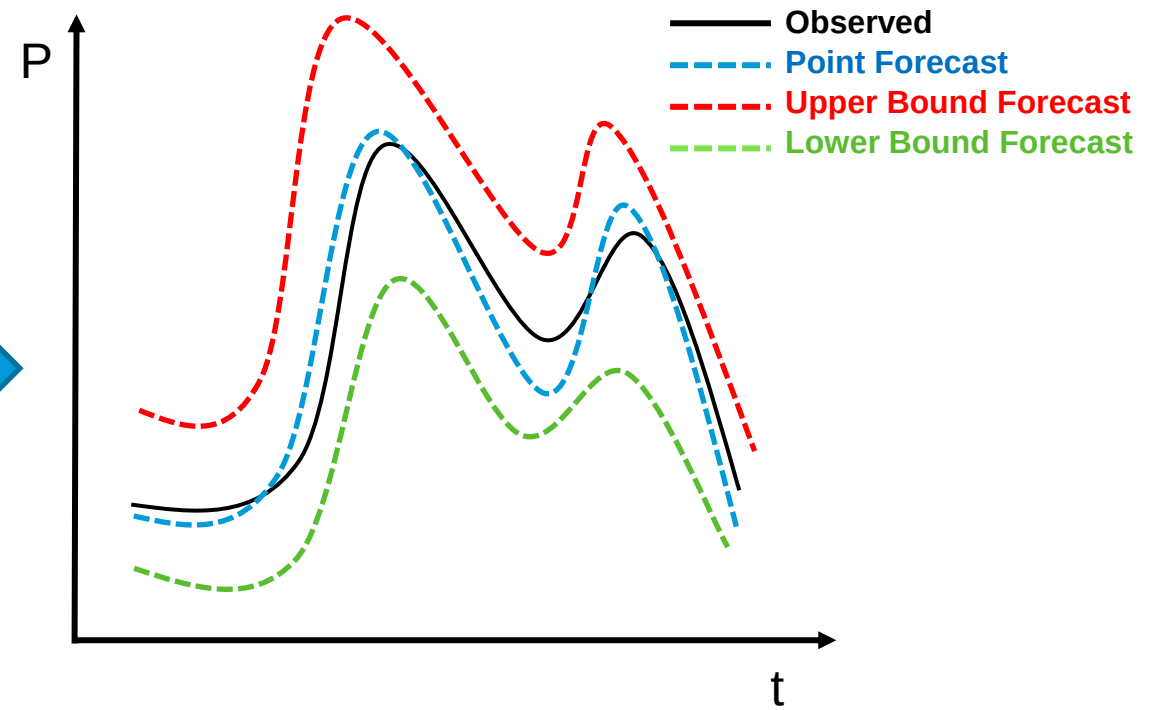
Predicting intervals



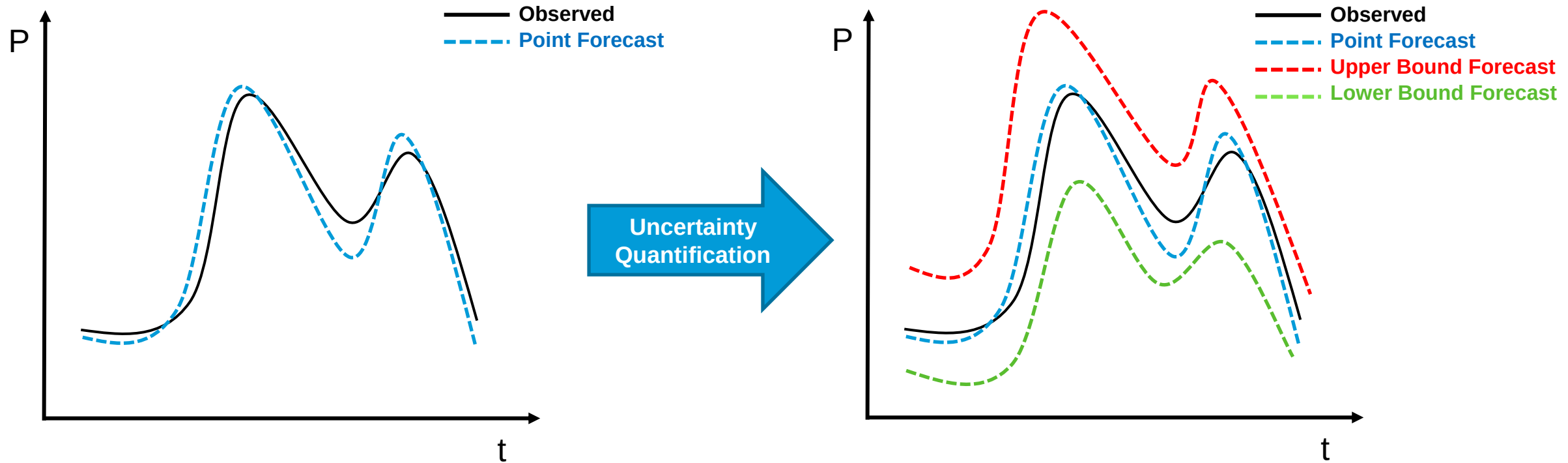
Predicting intervals



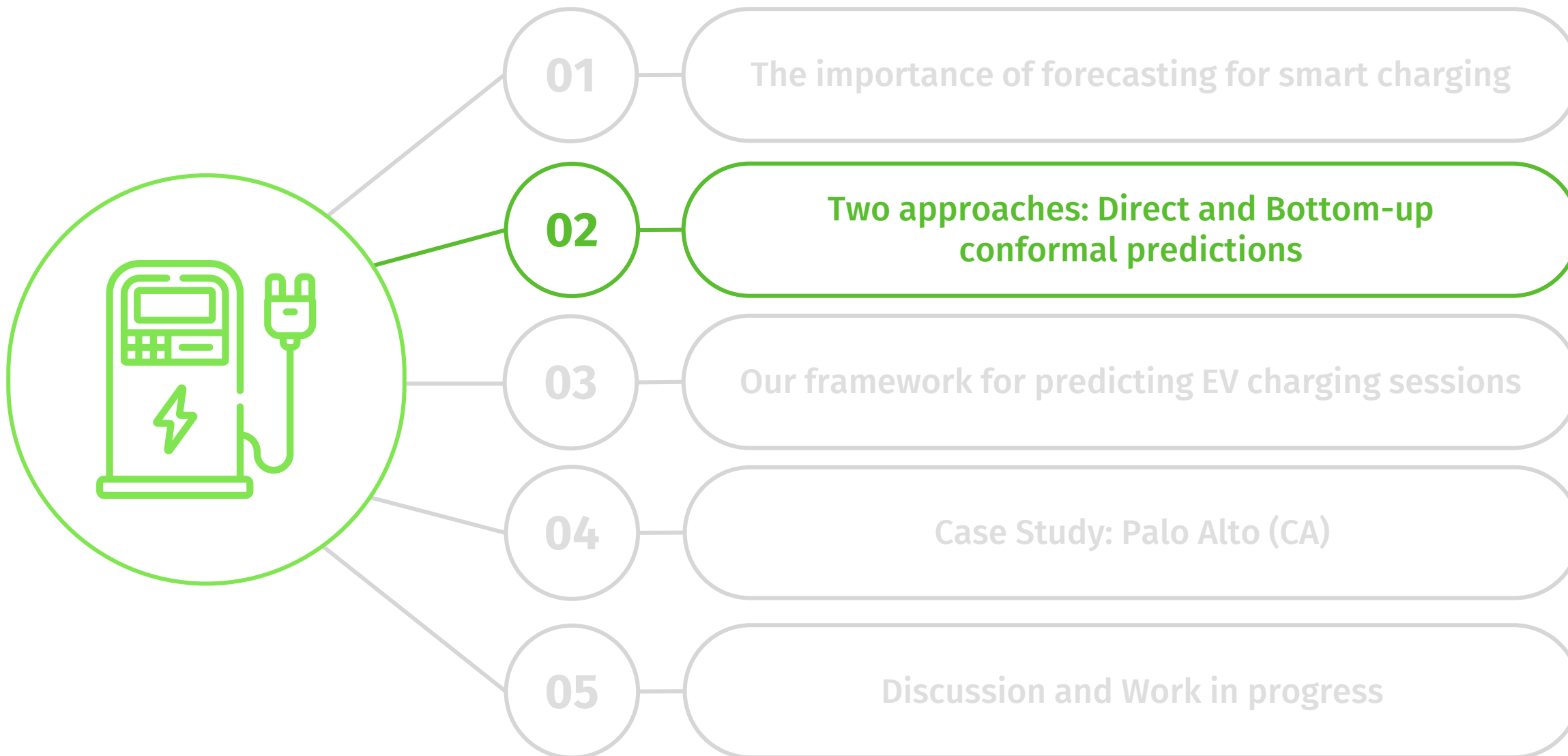
Uncertainty
Quantification



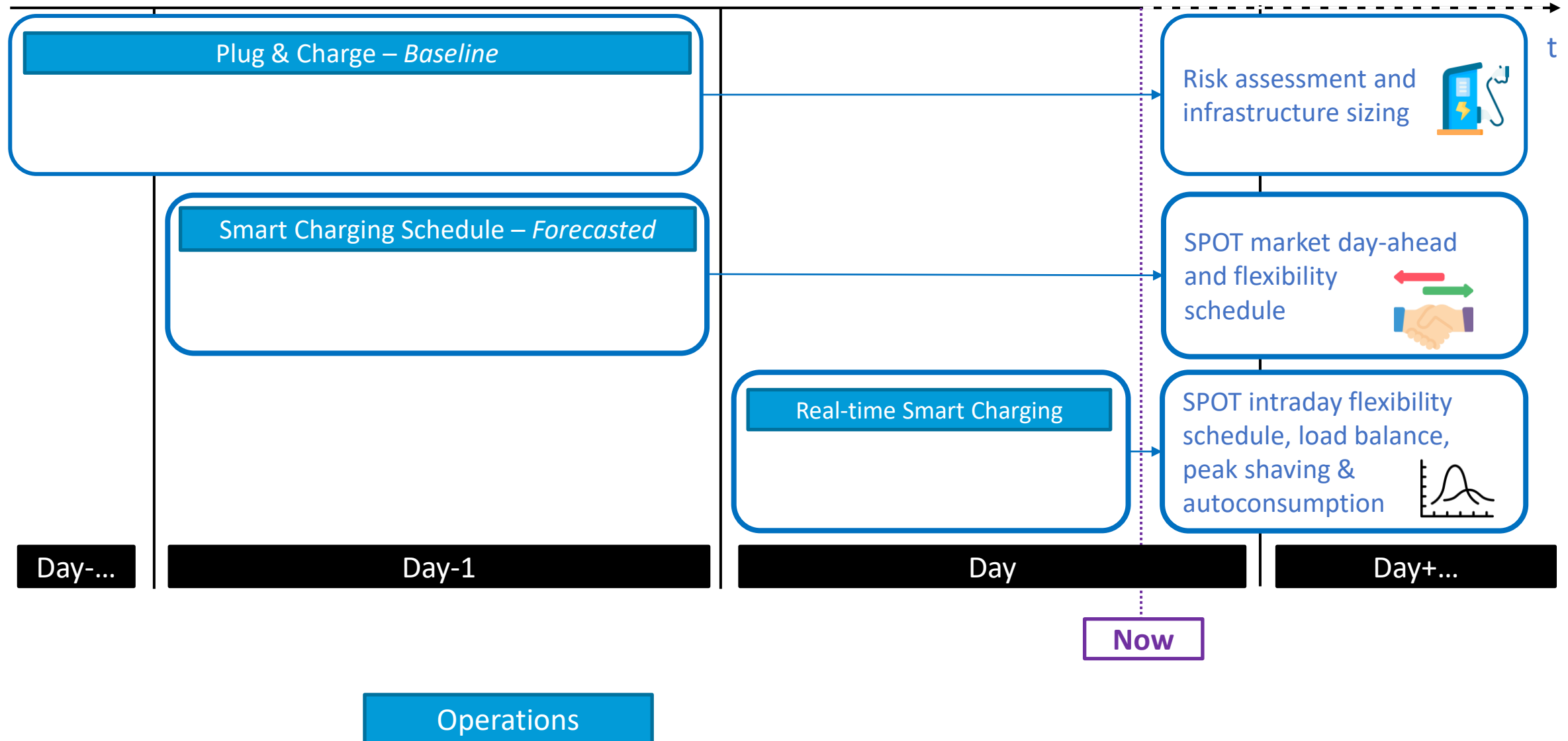
Predicting intervals



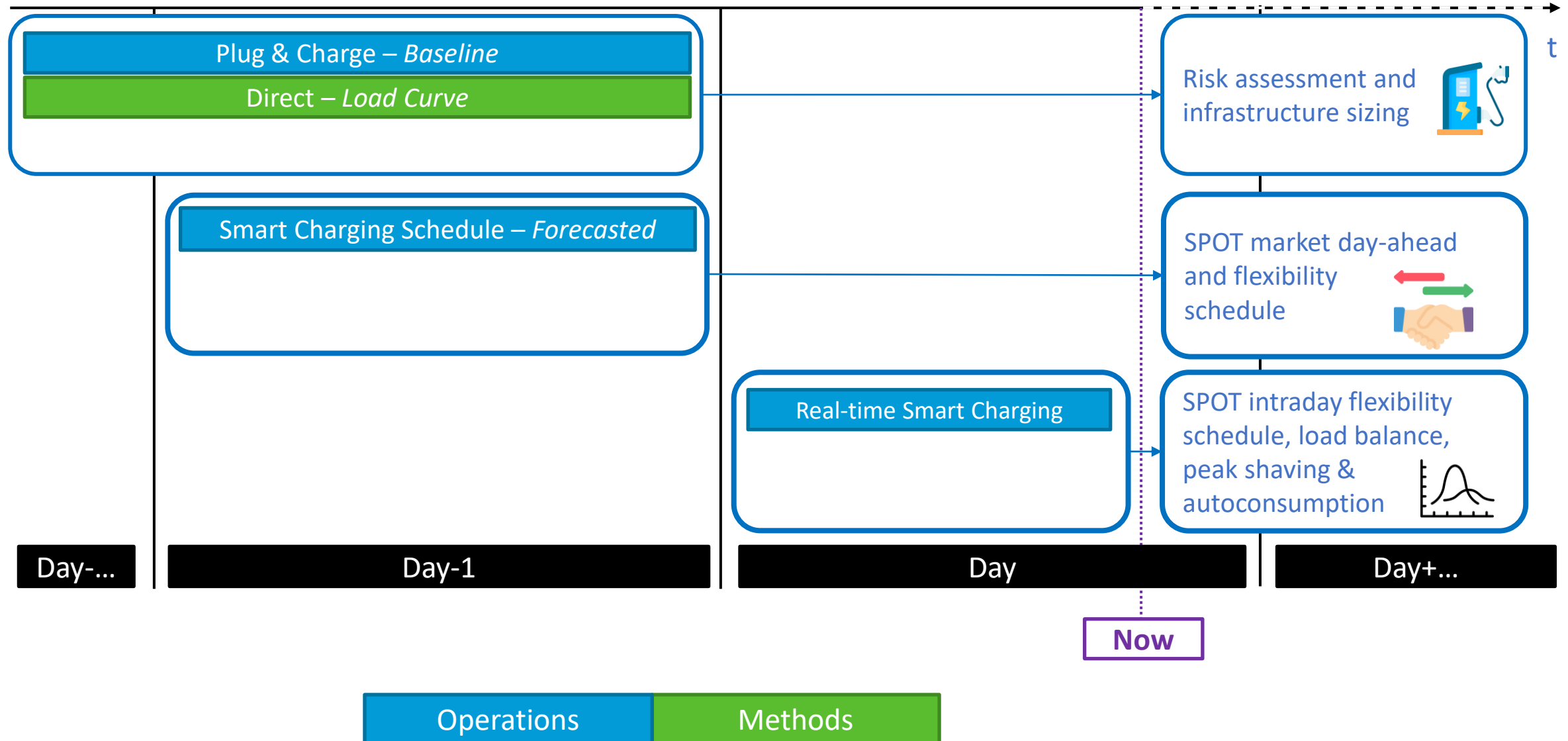
Accurately anticipating incoming EV charging demand is not enough, we need some sense of the uncertainty around the predictions to accommodate for different scenarios.



Operations and methods across the timeline



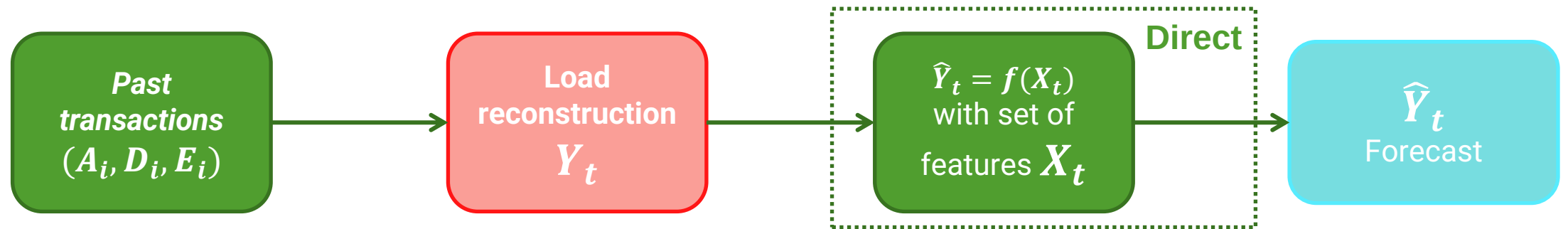
Operations and methods across the timeline



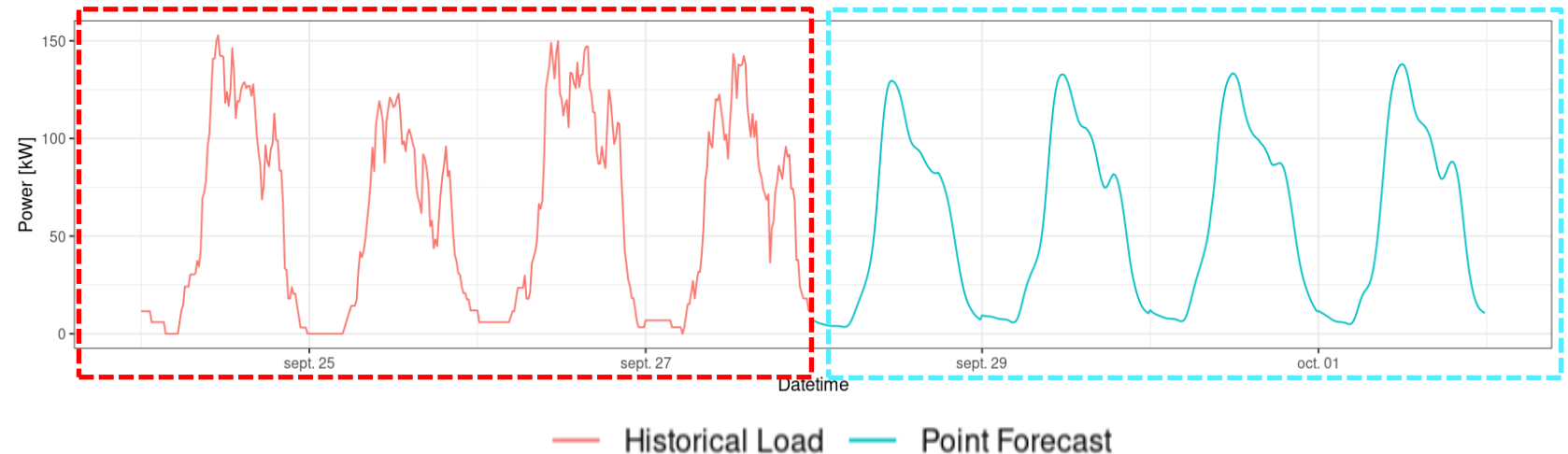
Direct approaches

Interests

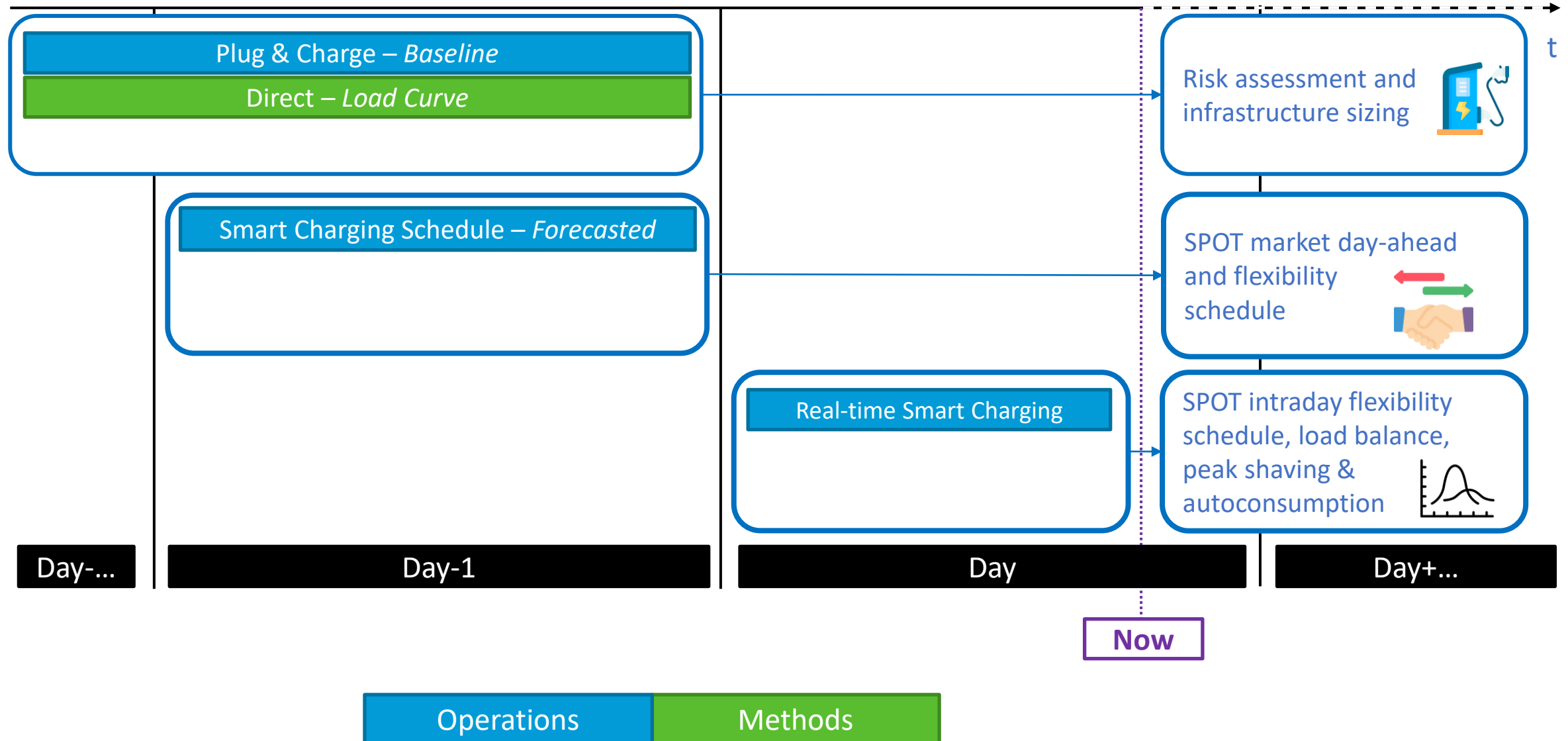
- Plug & Charge load forecasting
- Baseline to which other approaches can be compared



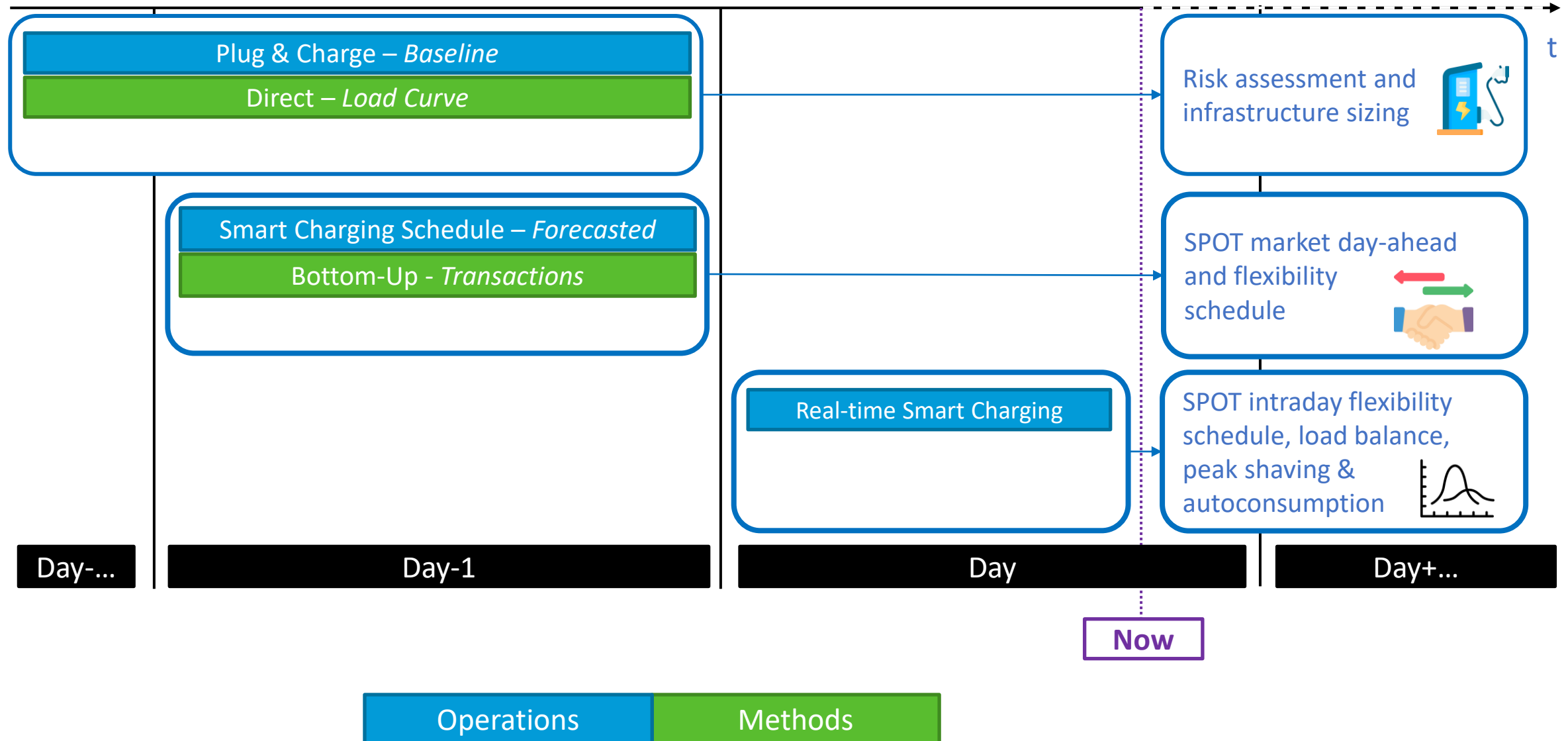
i	A_i	D_i	E_i
1	9h	5h	7kW h
2	12h	2h	3kW h
...	...	...	...
N	17h	1h	1kW h



Operations and methods across the timeline



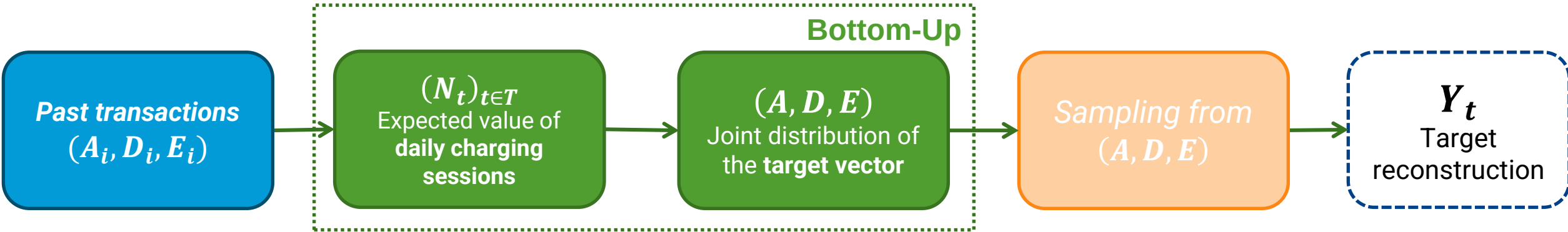
Operations and methods across the timeline



Bottom-Up approaches

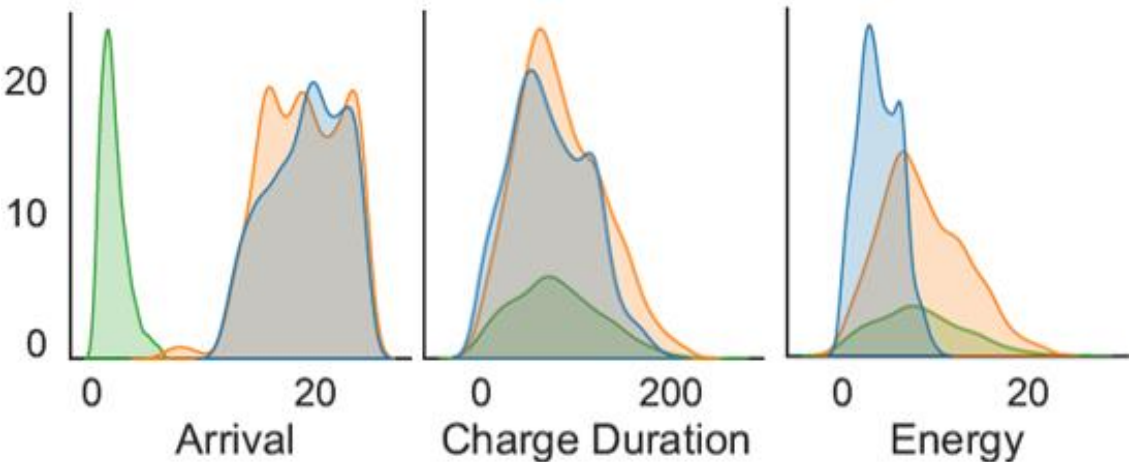
Interests

- **Finer granularity of forecasting (at the transaction level)**
- **Allows smart charging applications**
 - More leverage to optimise at the individual curve level
 - Optimisation on intra-day markets
 - Self-consumption and V2H applications



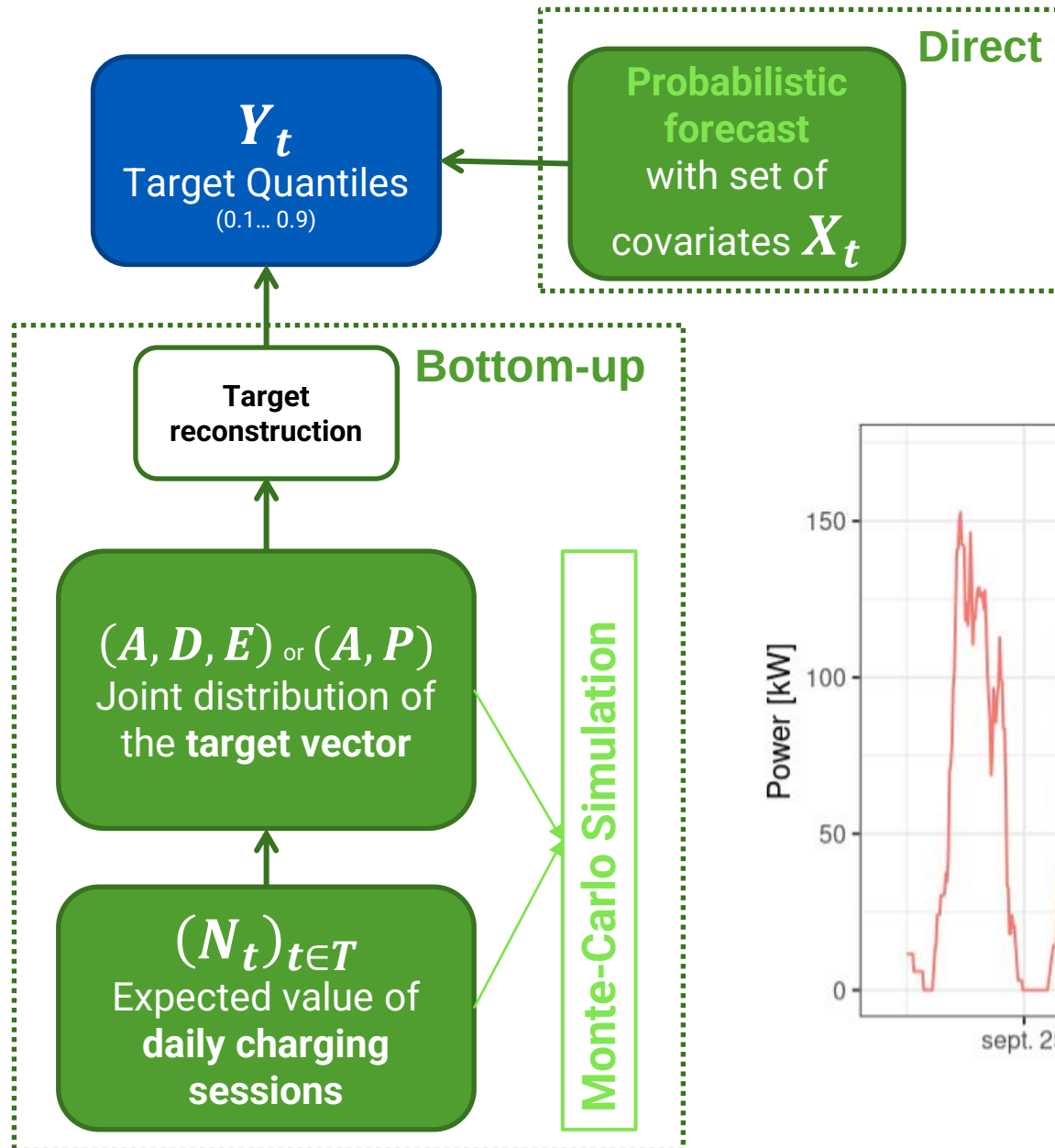
Main clusters

i	A_i	D_i	E_i
1	9h	5h	7kWh
2	12h	2h	3kWh
...	...	...	...
N	17h	1h	1kWh



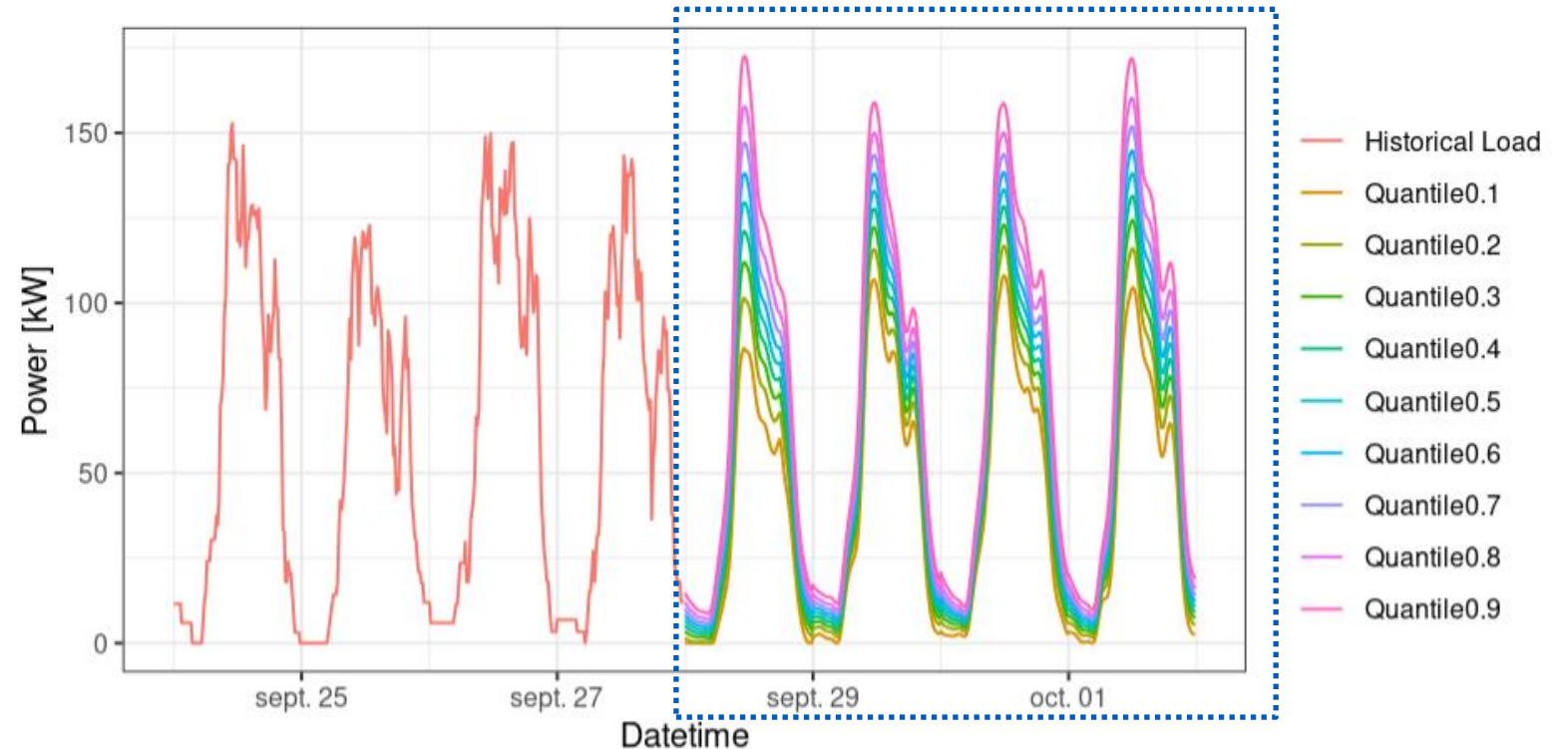
i	A_i	D_i	E_i
1	8h	4h	9kWh
2	11h	1h	1kWh
...	...	...	...
$\hat{N}$	13h	2h	1kWh

From point forecasts to probabilistic forecasts

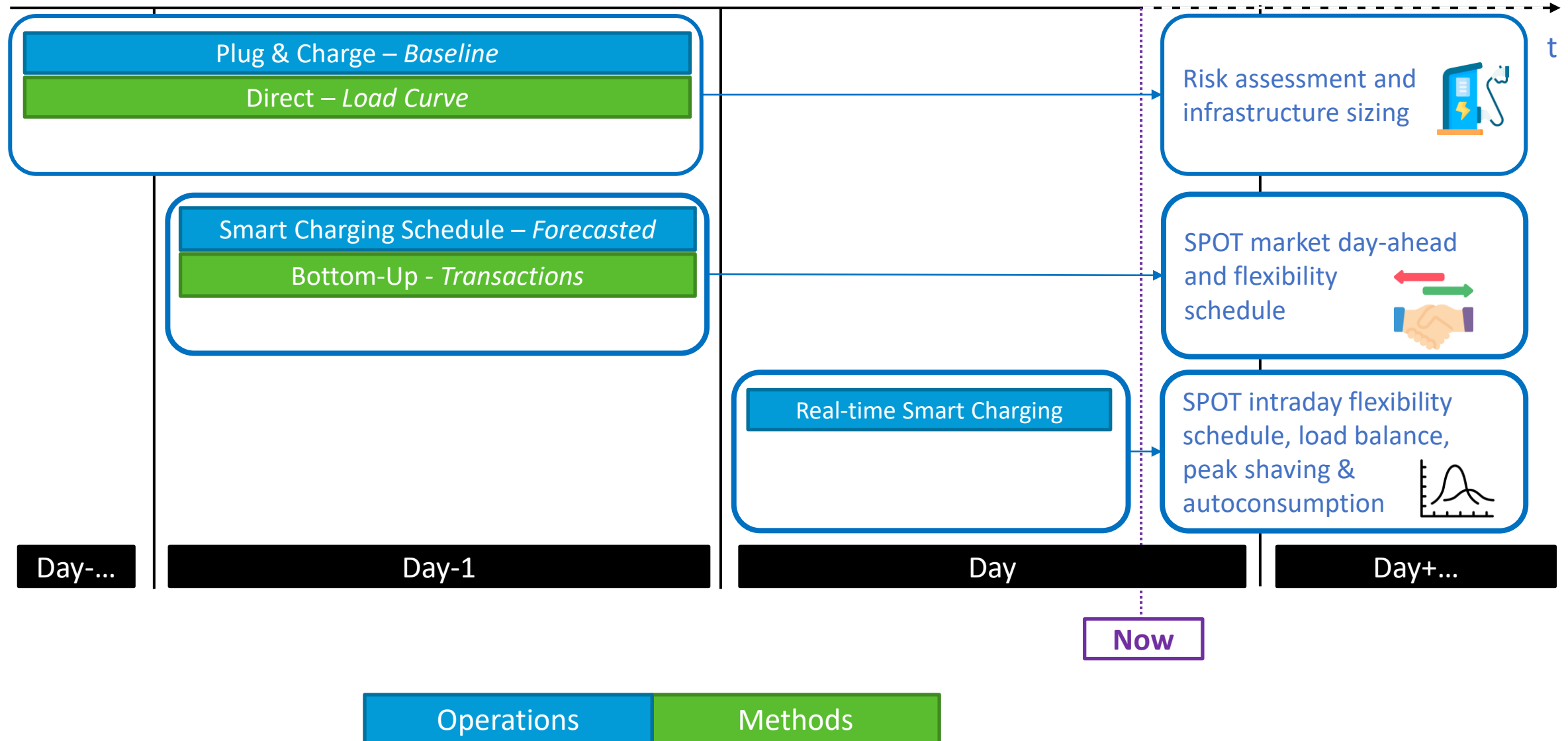


Interests

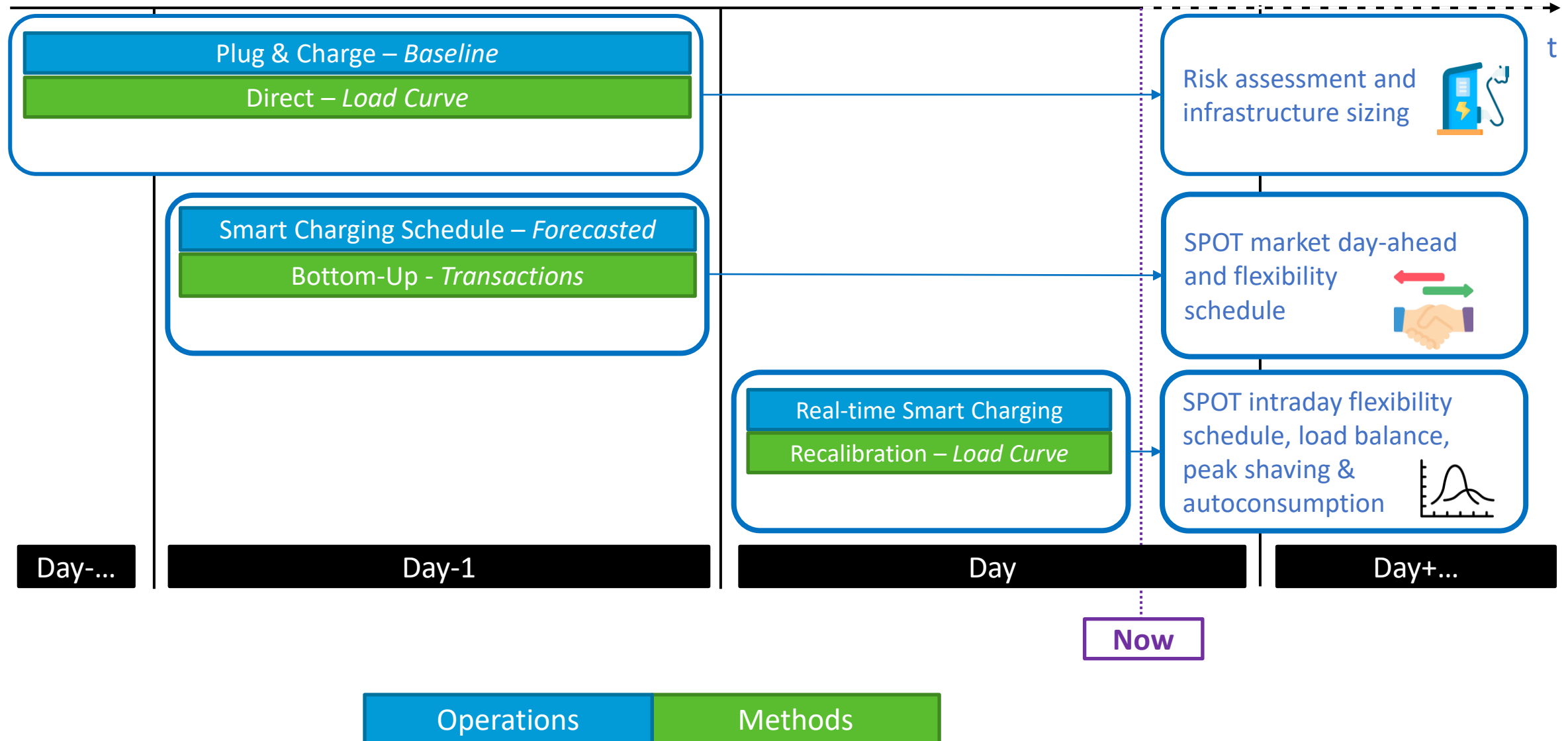
- Need to quantify the uncertainty around these forecasts to assess the risk around the smart charging operations
- The optimisation is different if there is an important uncertainty around the expected load (higher energy, shorter durations etc.)



Operations and methods across the timeline



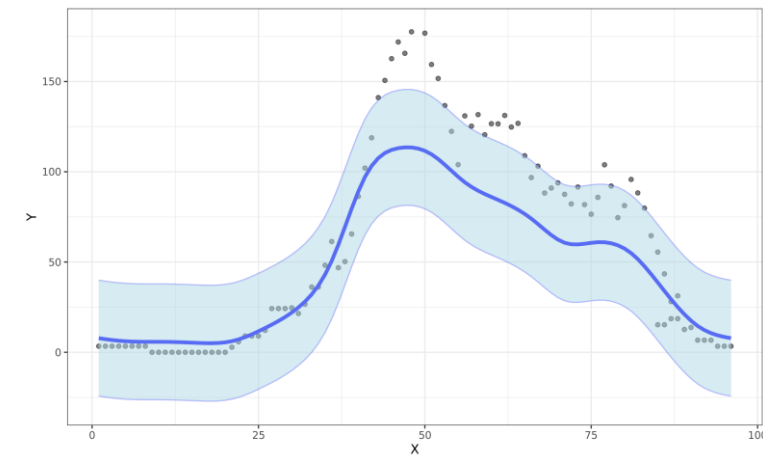
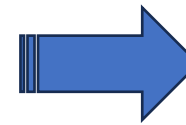
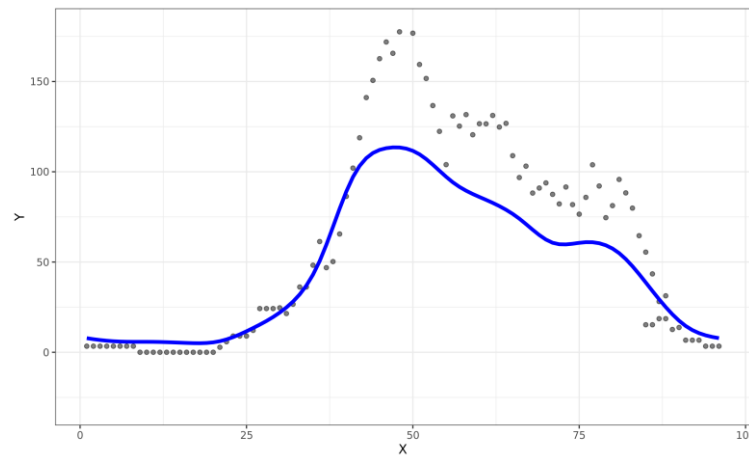
Operations and methods across the timeline



Theoretical Focus on Conformal Prediction

- Introduction to Split Conformal Prediction (SCP)
- Extension to Time Series with Adaptive Conformal Inference (ACI)

- Turns a point forecast into a prediction set



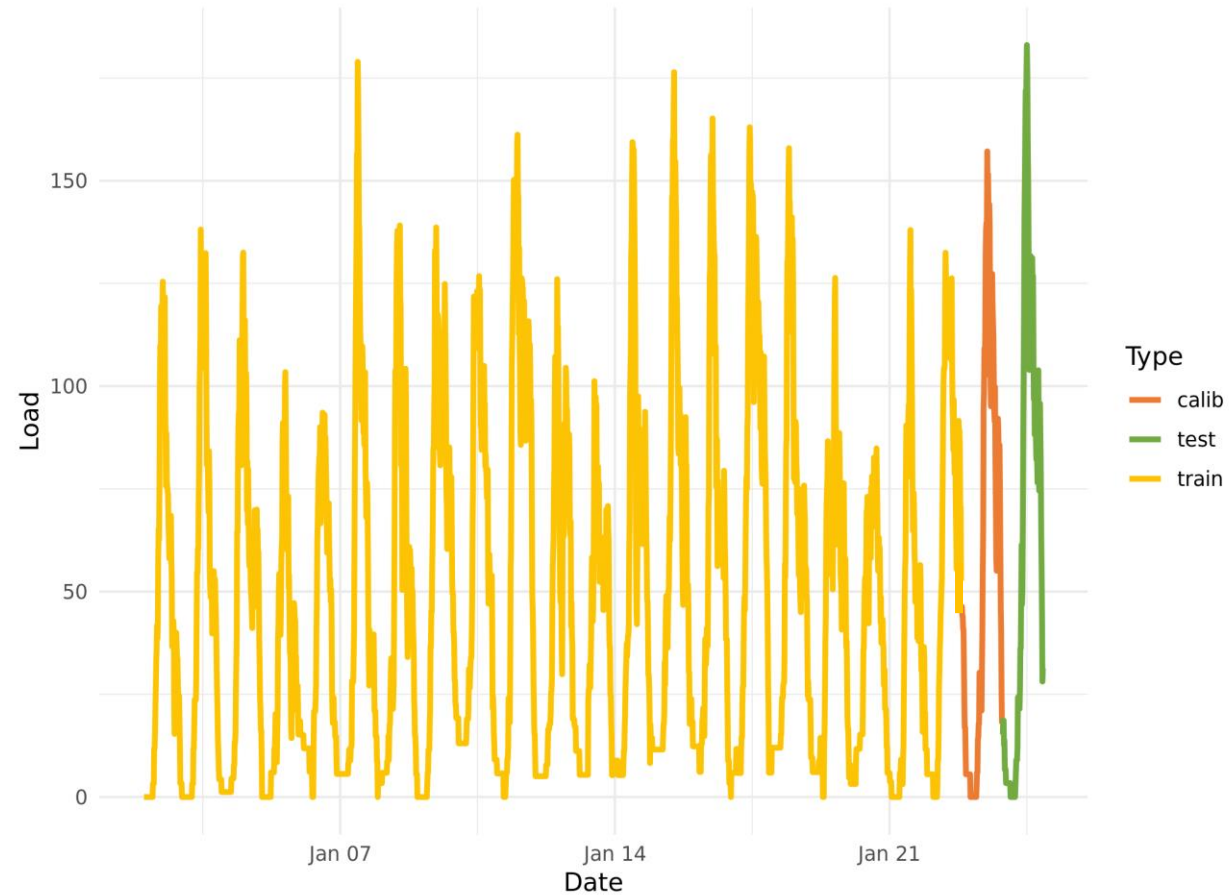
- Corrects a miscalibrated prediction set

Split Conformal Prediction (SCP)

Training

Calibration

Test

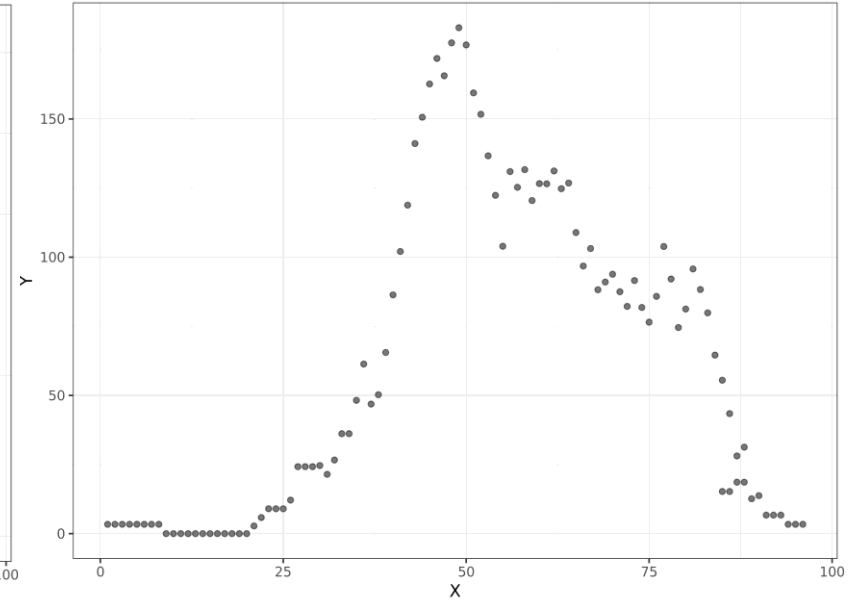
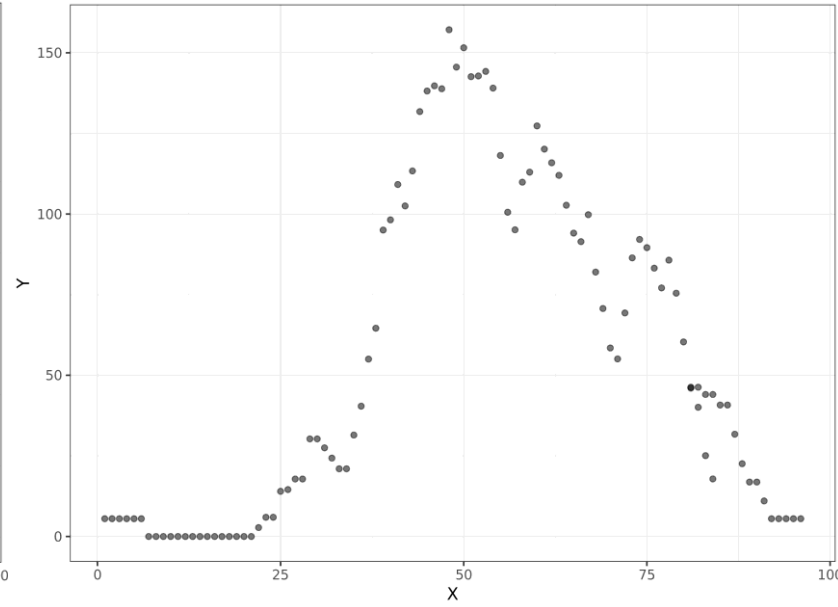
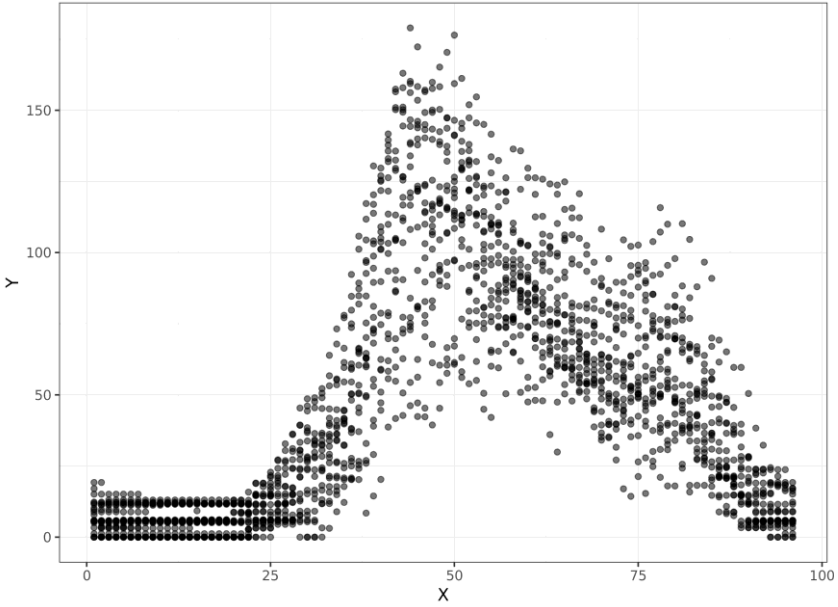


Split Conformal Prediction (SCP)

Training

Calibration

Test

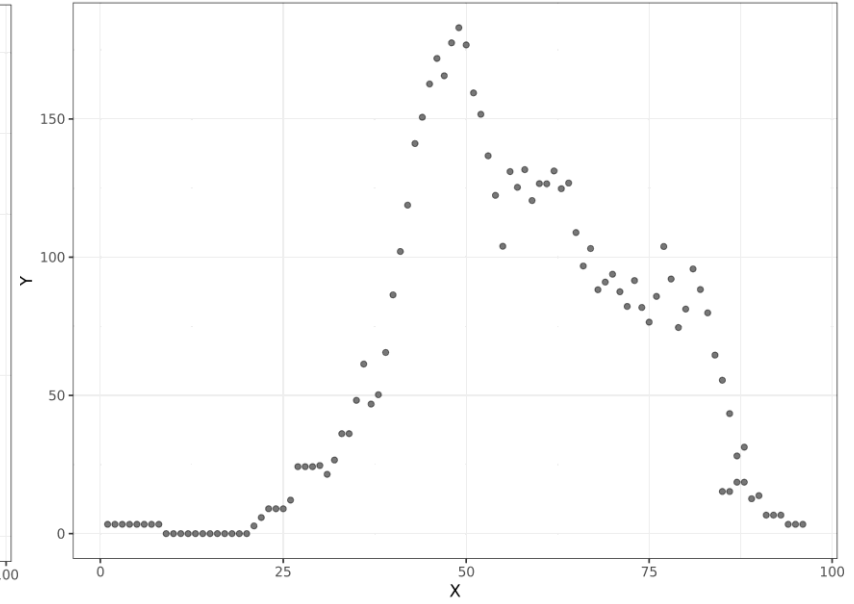
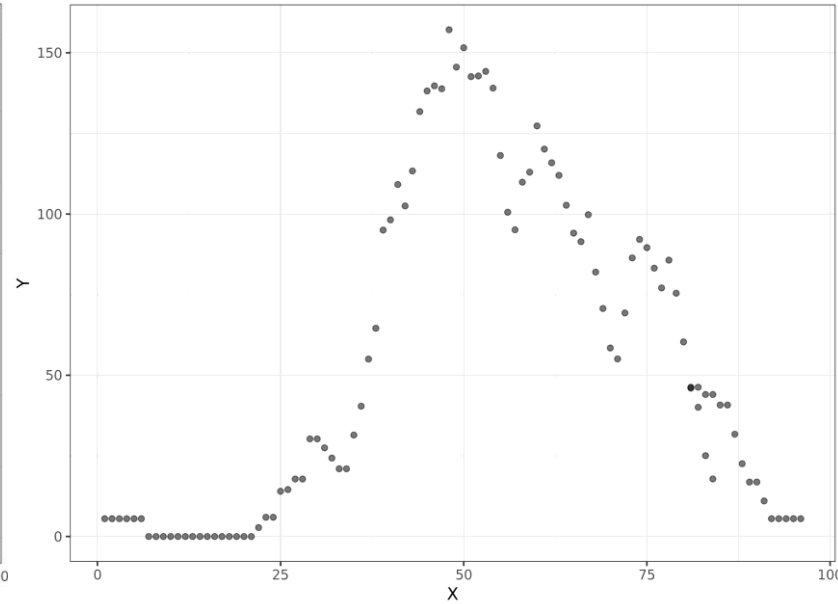
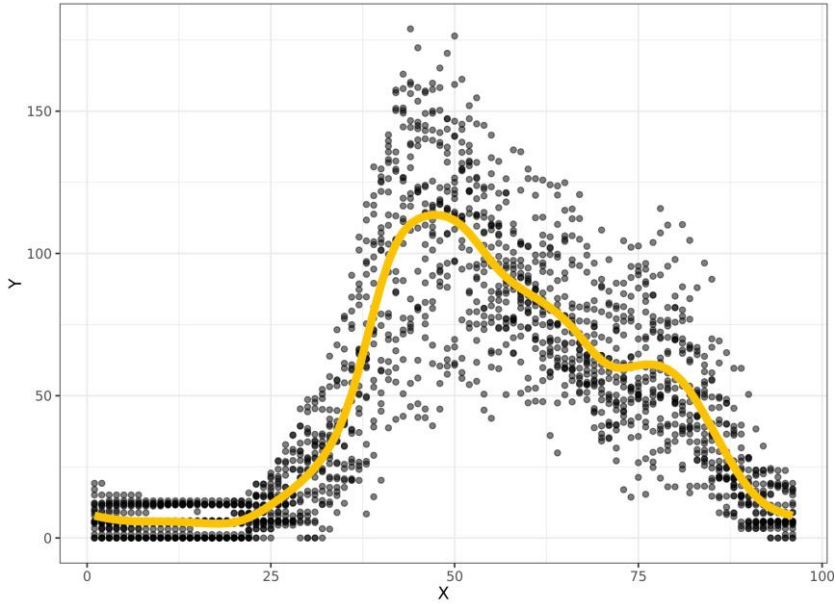


Split Conformal Prediction (SCP)

Training

Calibration

Test



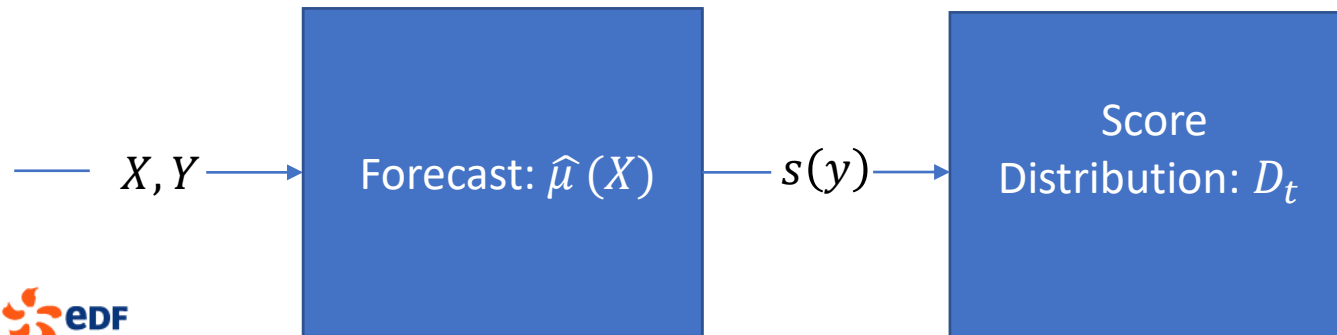
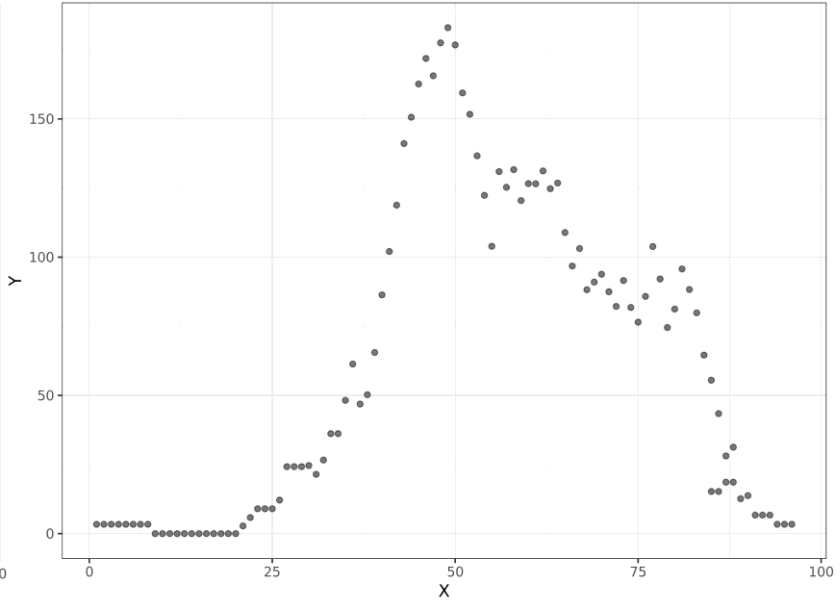
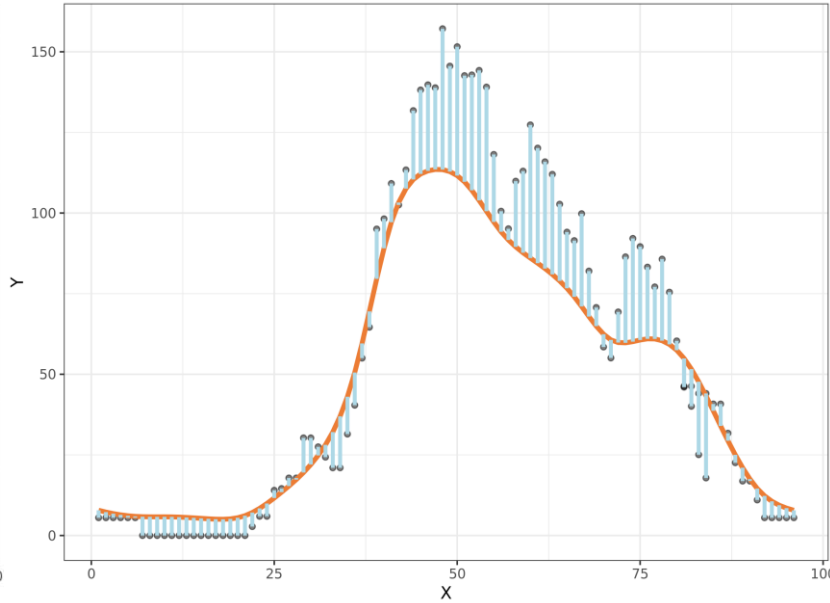
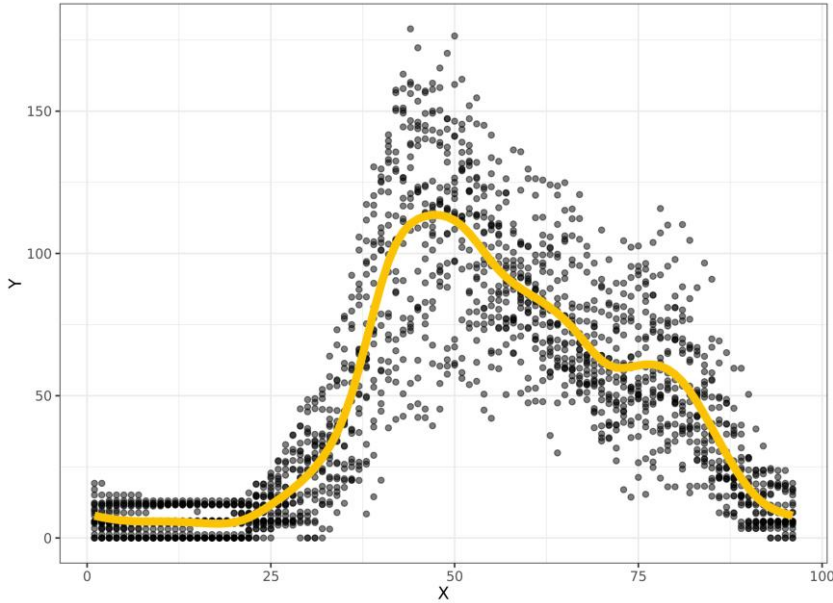
— X, Y —→ Forecast: $\hat{\mu}(X)$

Split Conformal Prediction (SCP)

Training

Calibration

Test

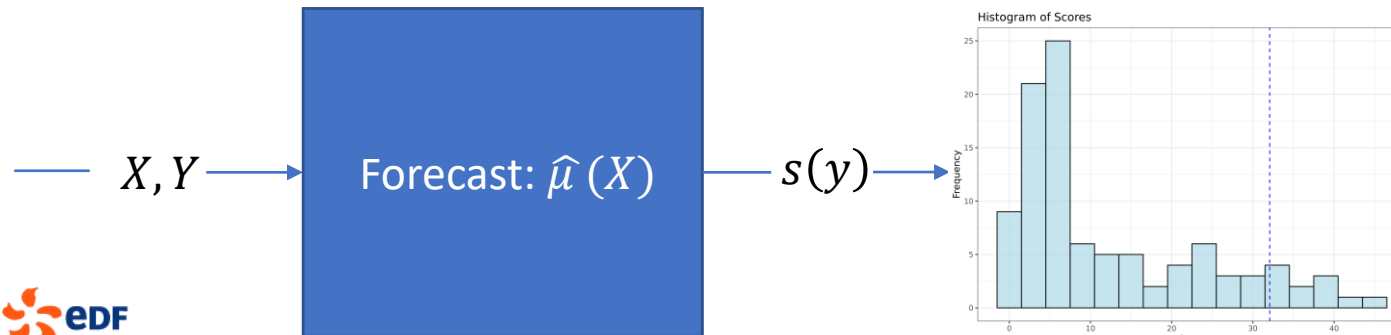
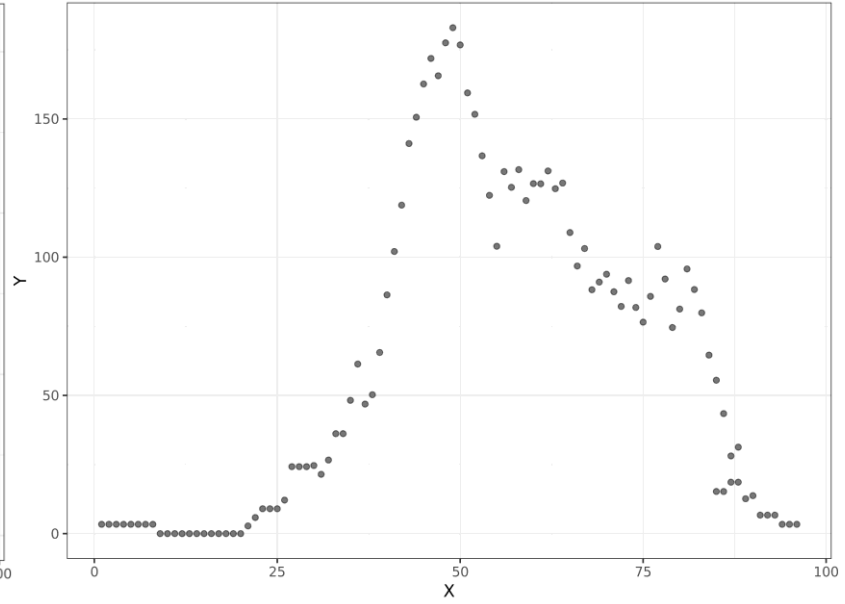
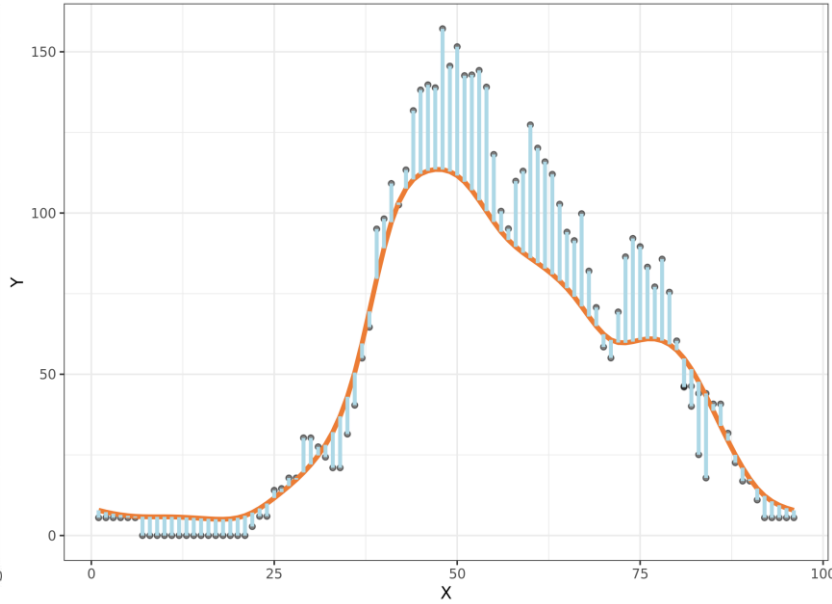
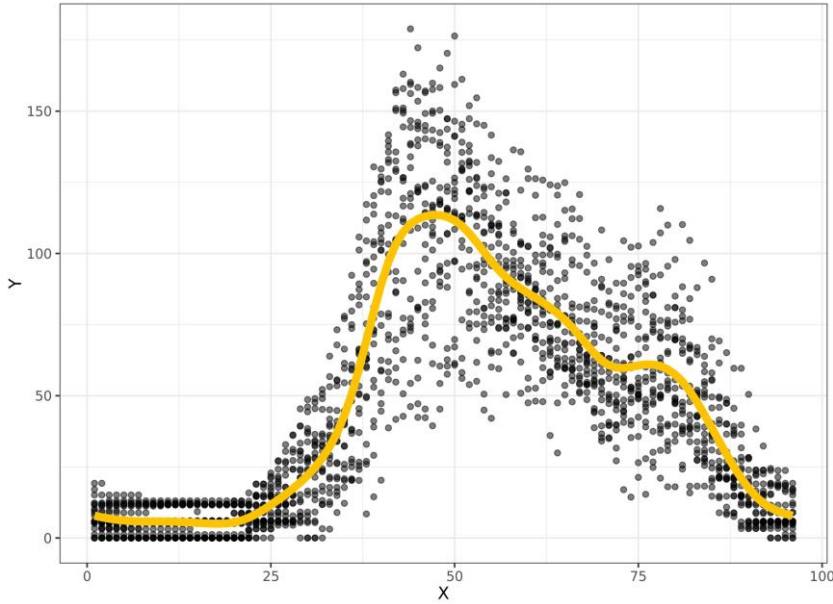


Split Conformal Prediction (SCP)

Training

Calibration

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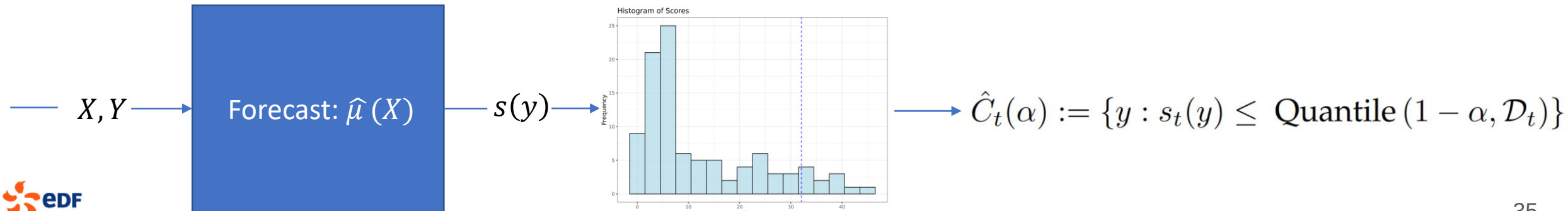
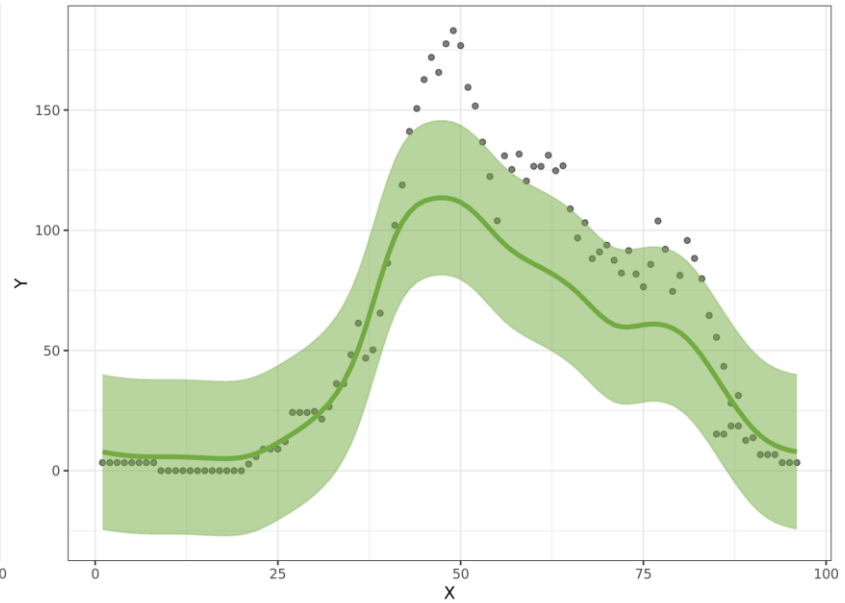
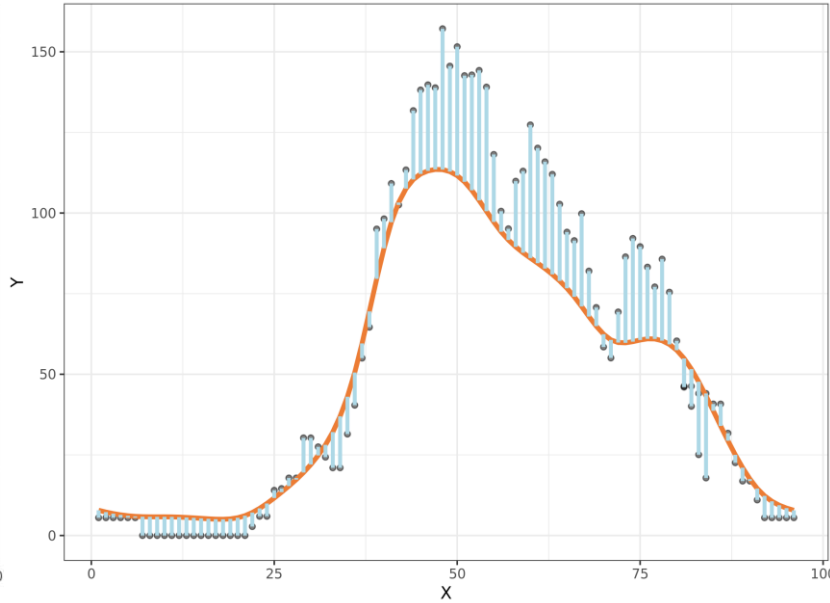
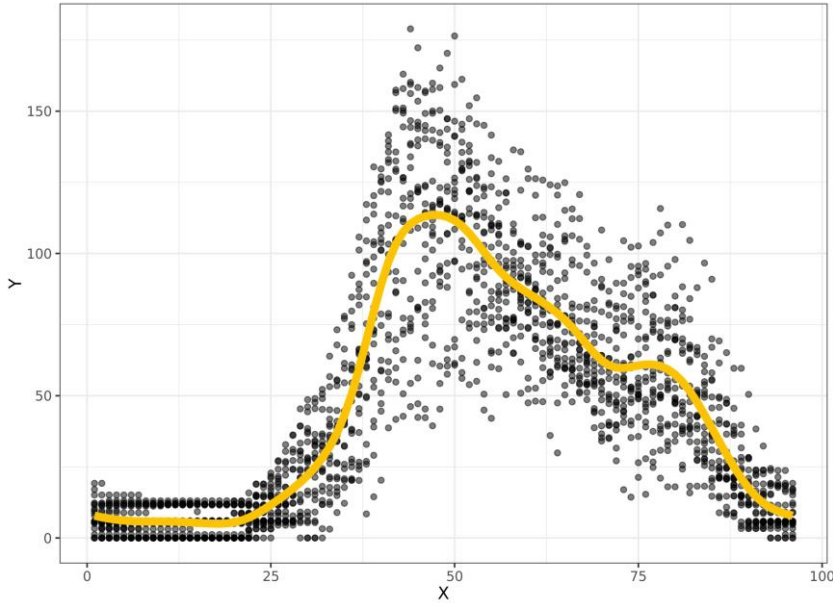


Split Conformal Prediction (SCP)

Training

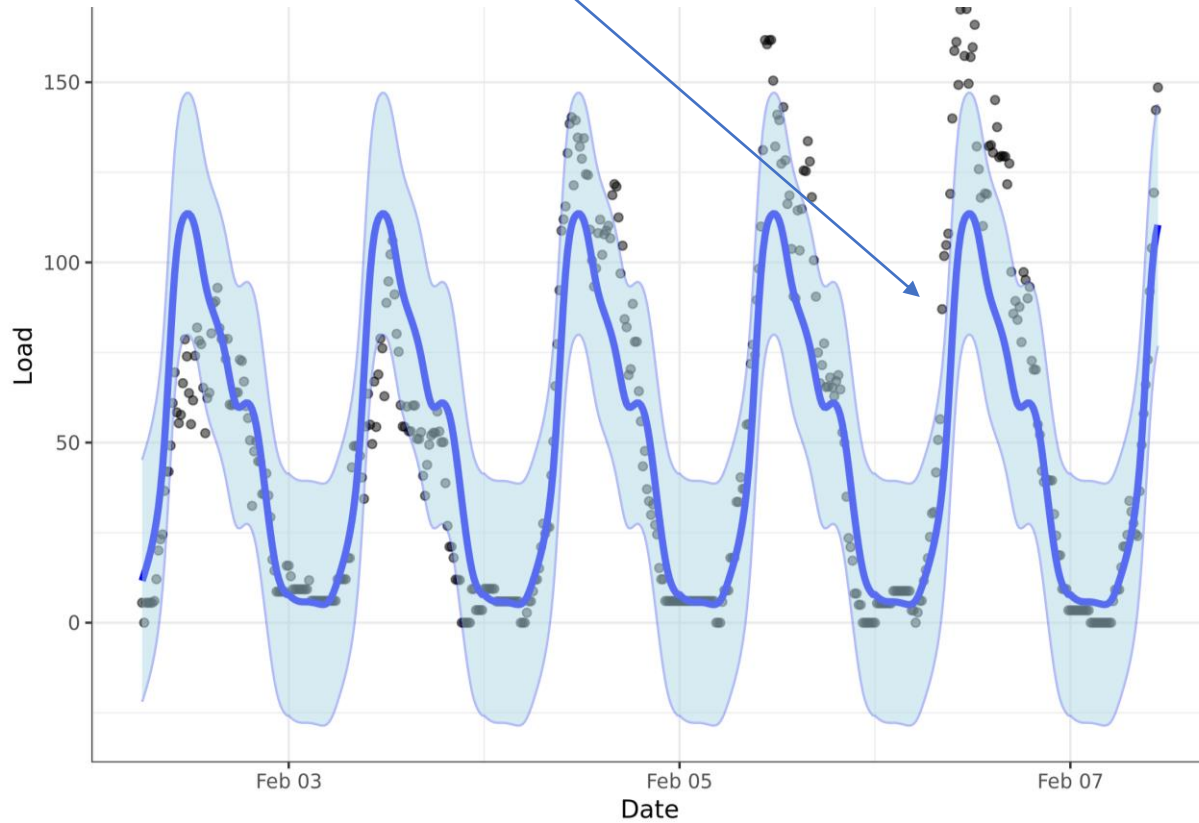
Calibration

Test

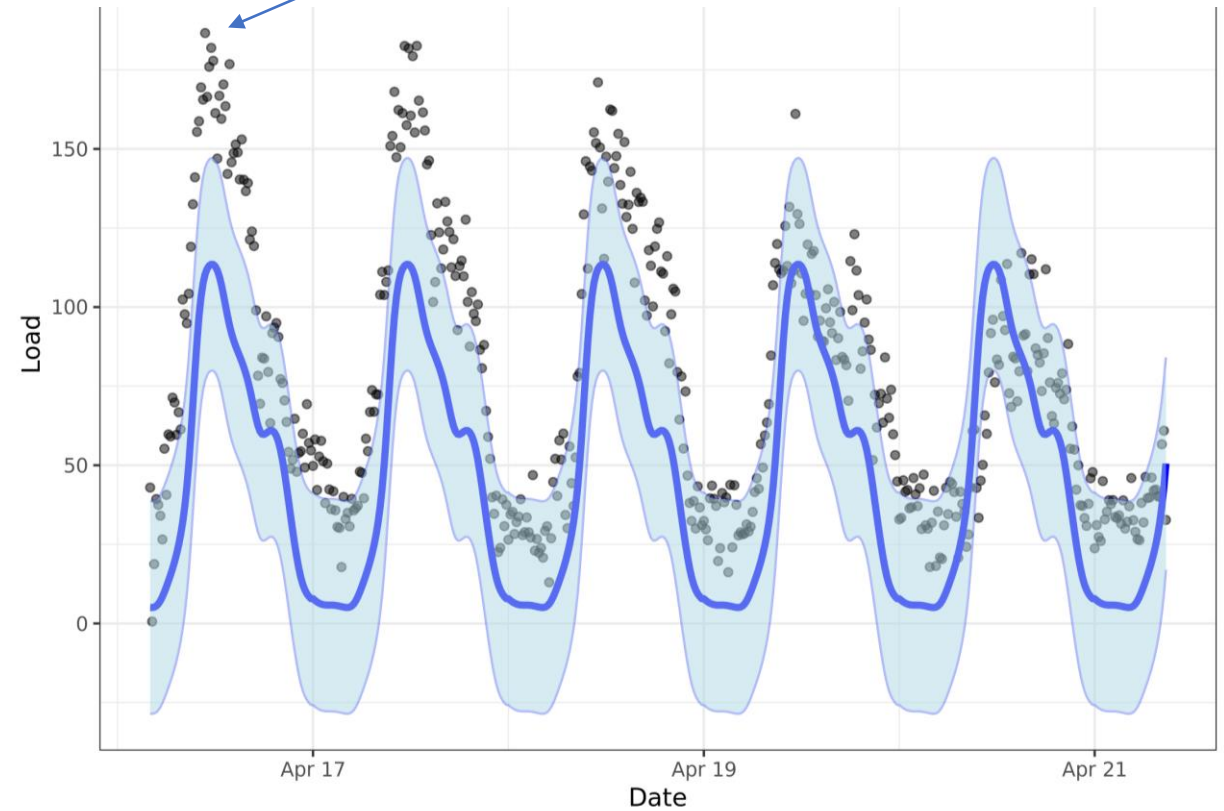


The Time Series setting

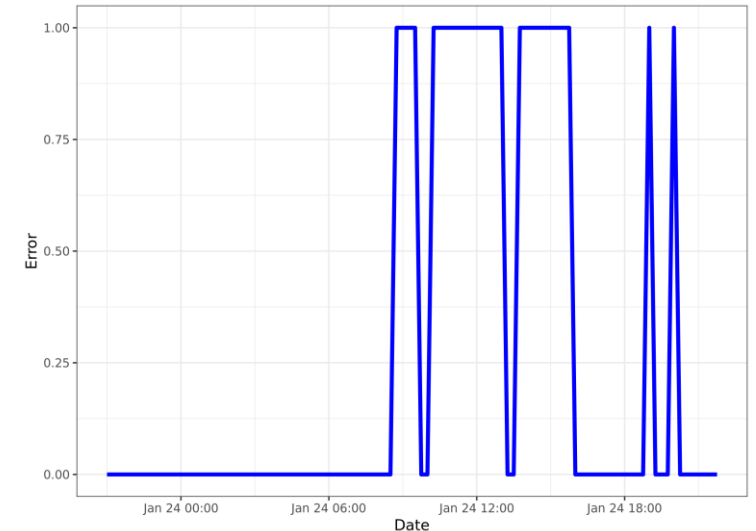
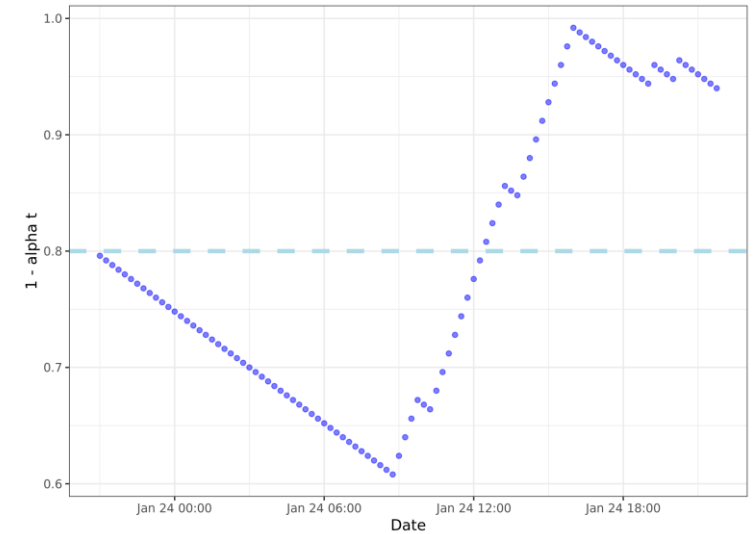
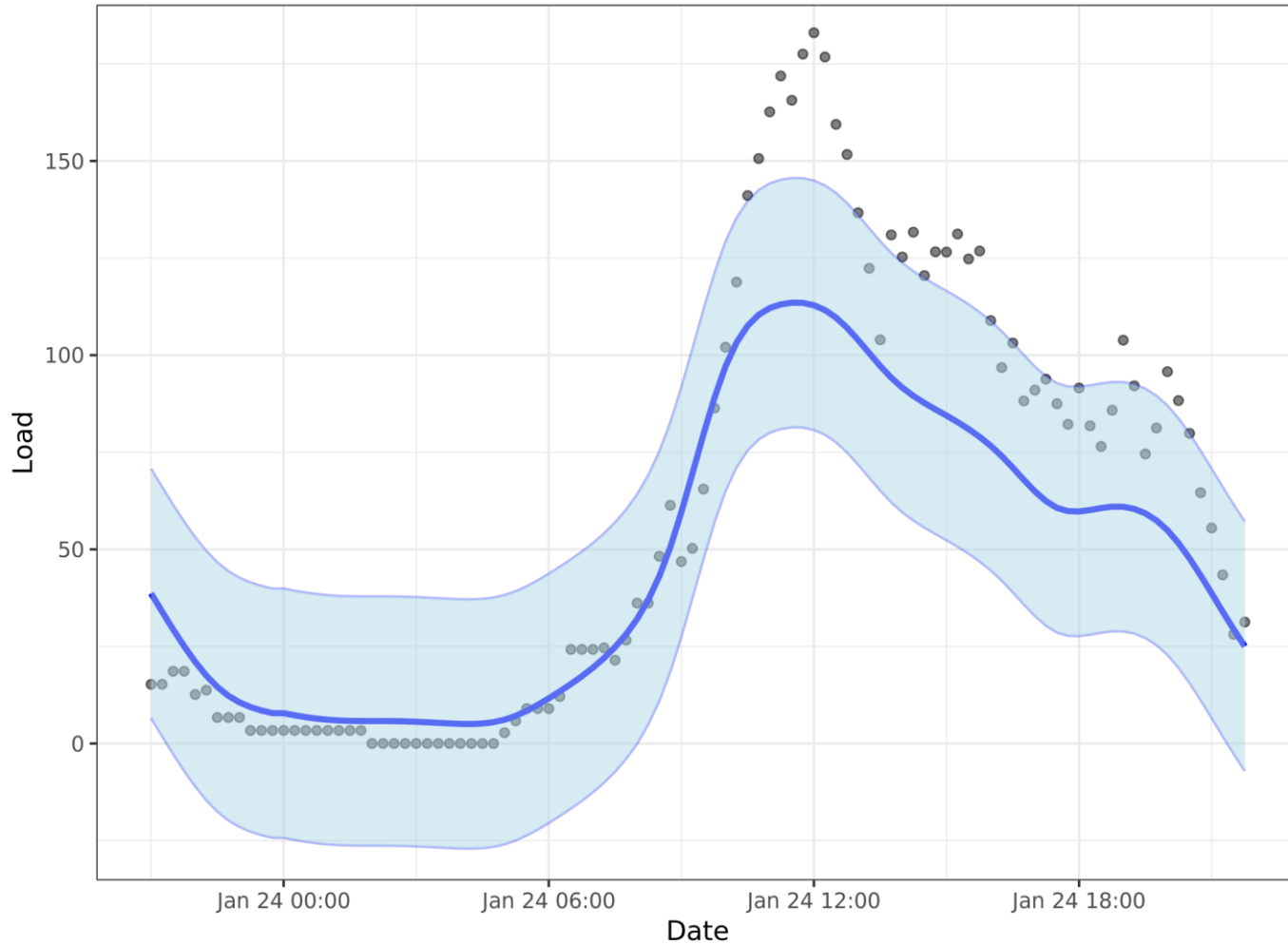
Dependency in high consumption events



Drift both in term of Mean and Deviation

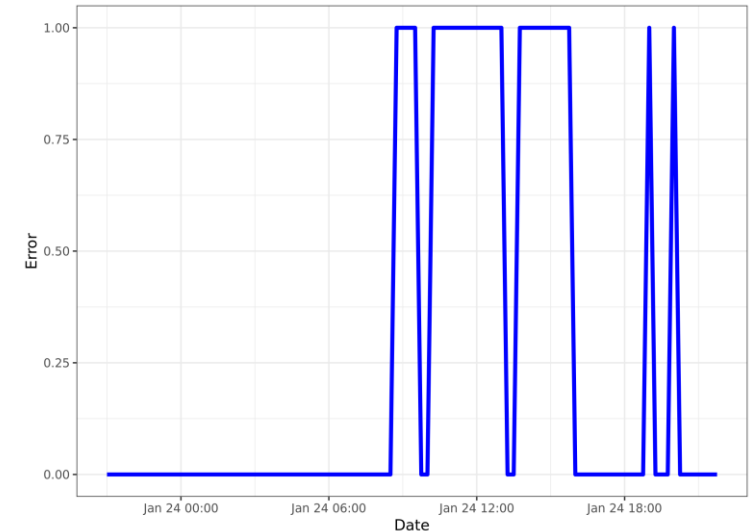
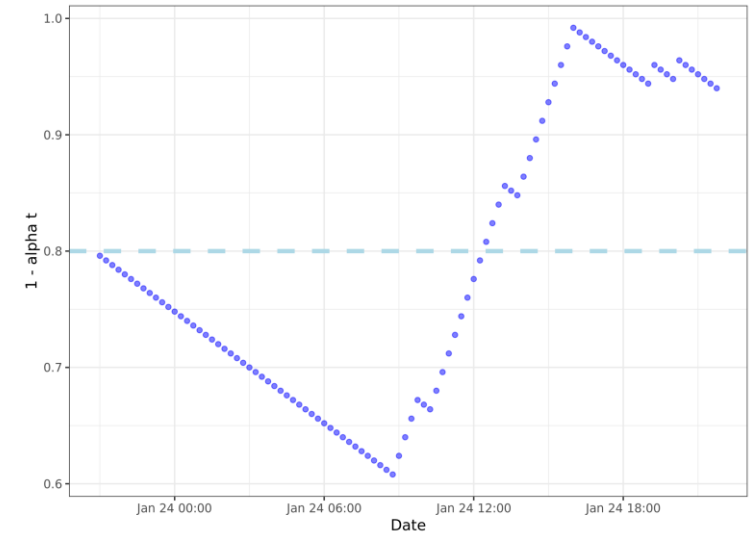
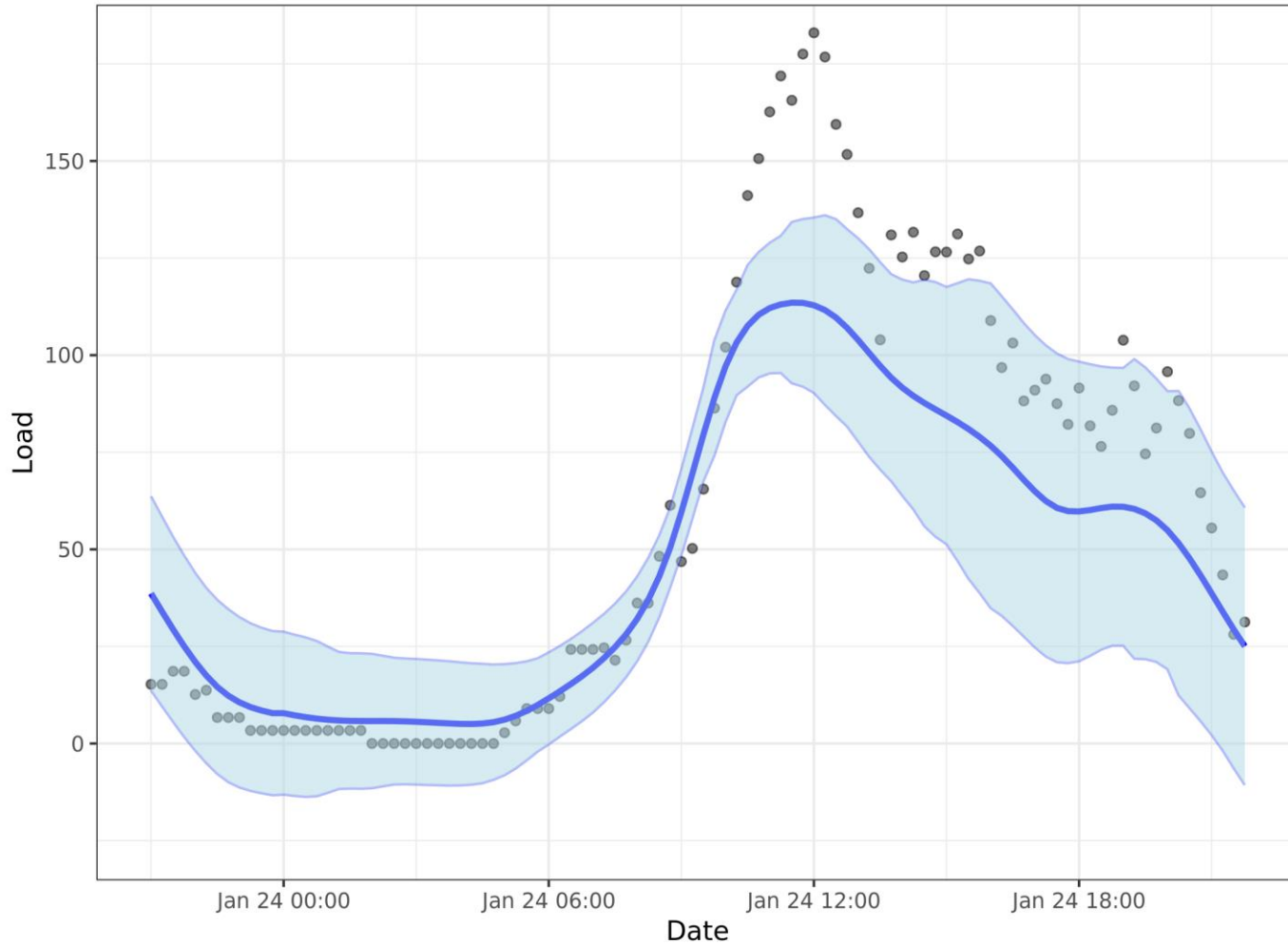


Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

Adaptive Conformal Inference: Gibbs & Candès (2021)



$$\alpha_{t+1} = \alpha_t + \gamma (\alpha - \text{err}_t) \quad \text{with } \text{err}_t = \mathbb{1}_{\{Y_t \notin \hat{C}_t^\alpha\}}$$

How to choose the gradient step size ?

- We want to maintain the validity while being more efficient
- To tackle this issue, we need to use a **dynamic step size**:

¹Zaffran et al. (2022)

²Gibbs & Candès (2022)

³Bhatnagar et al. (2023)

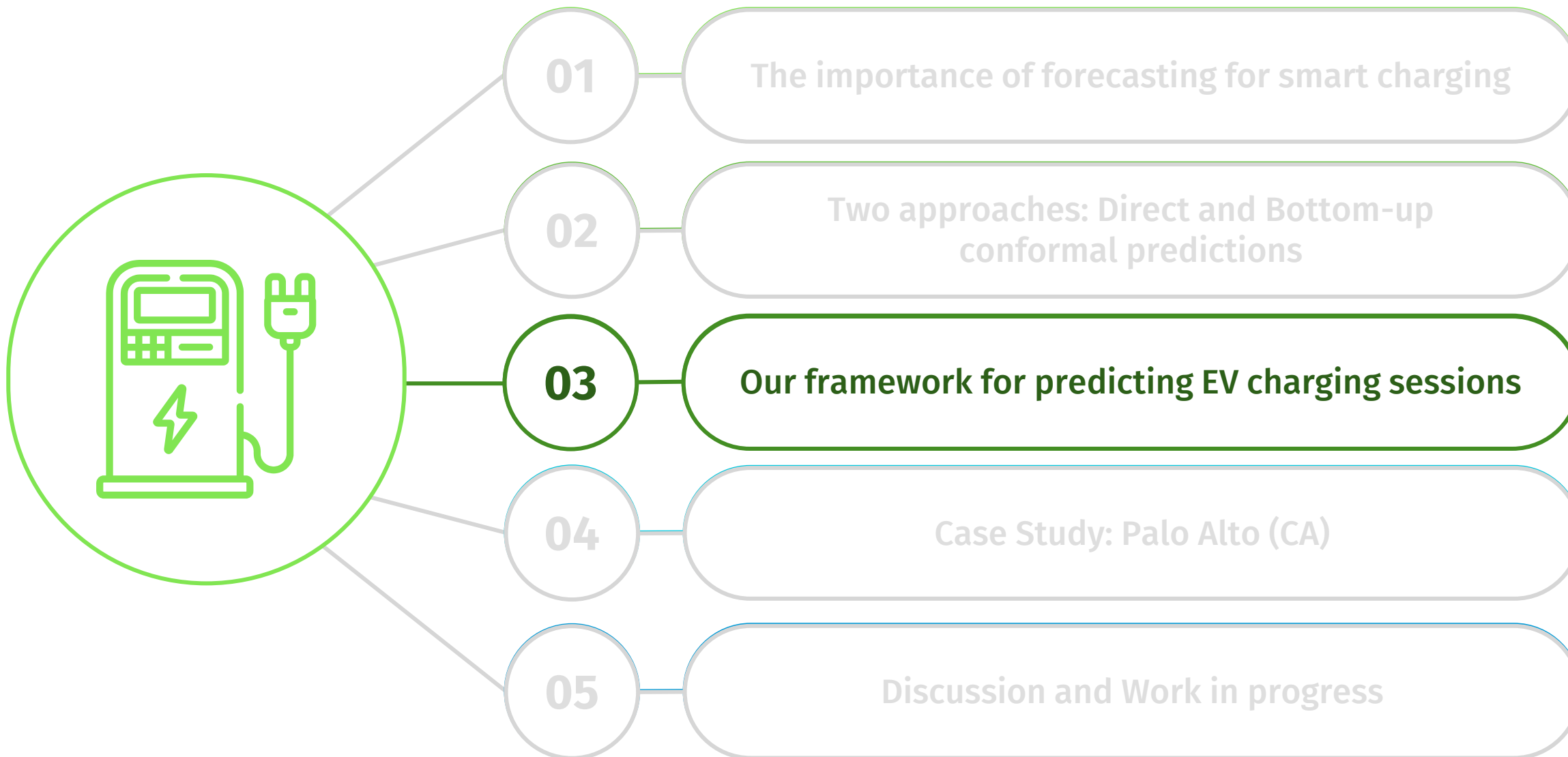
⁴Angelopoulos et al. (2024)

My presentation at JDS (2024)

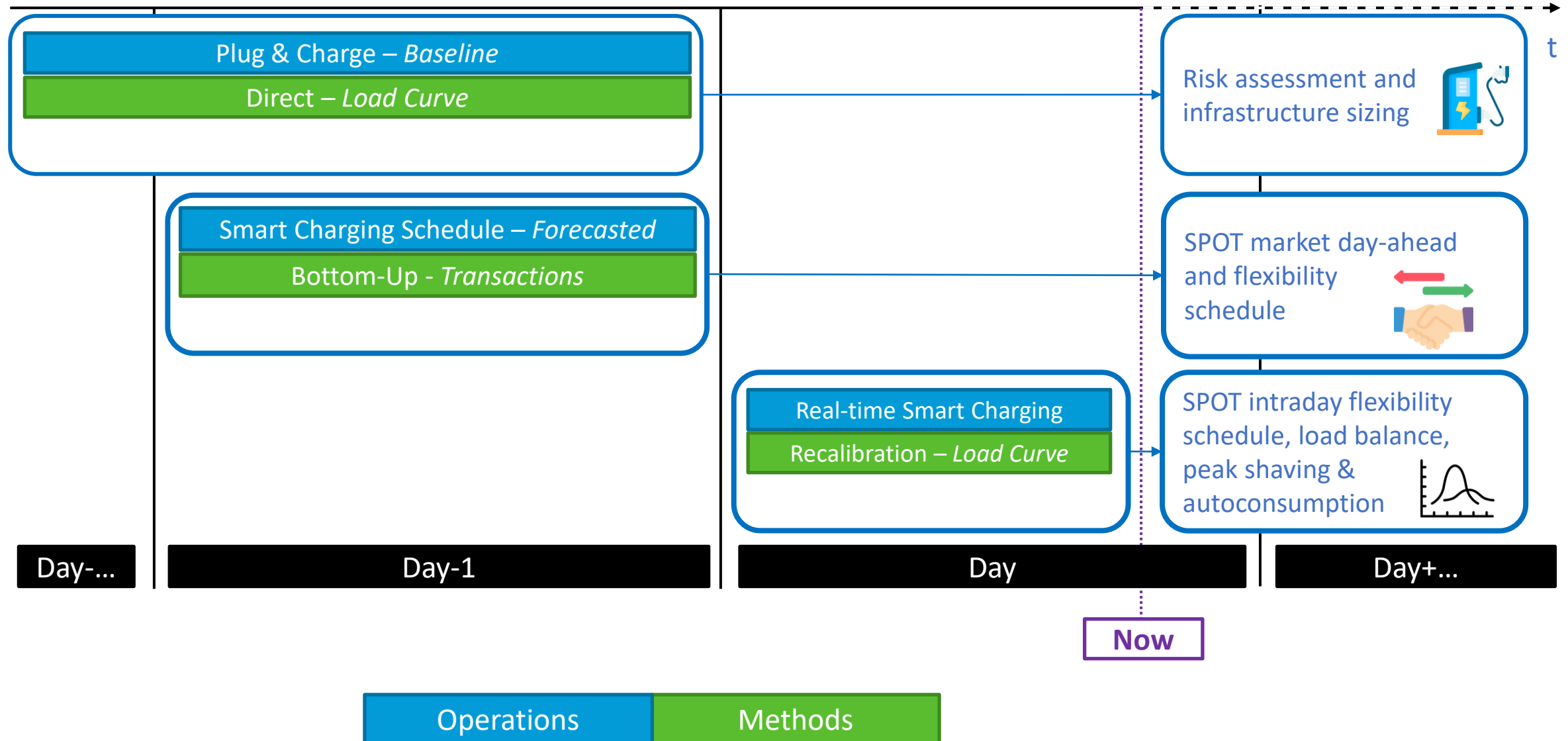
Meta algorithm to obtain γ_t

Explicit γ_t

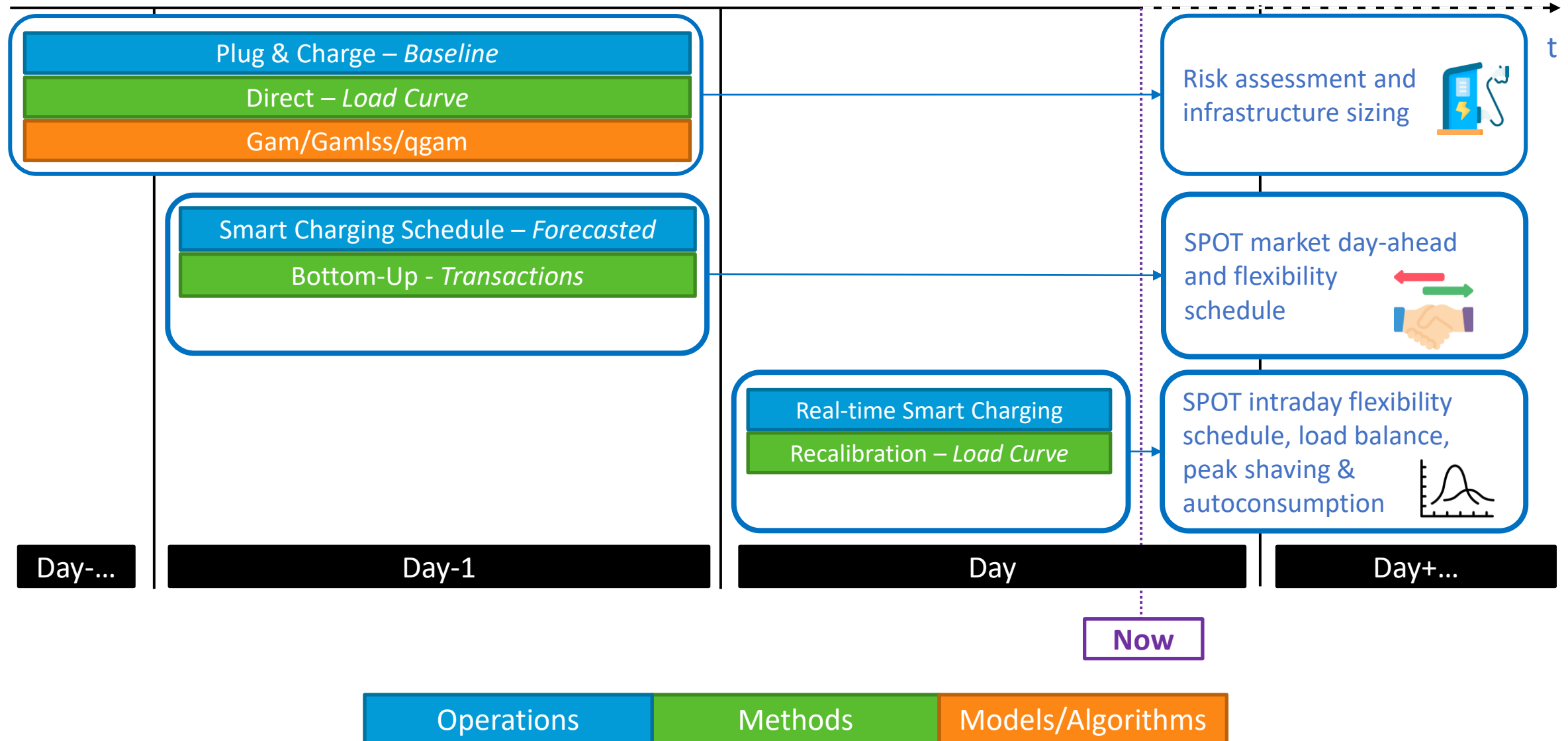
¹Adaptive conformal predictions for time series / ²Conformal inference for online prediction with arbitrary distribution shifts / ³Improved online conformal prediction via strongly adaptive online learning / ⁴Online conformal prediction with decaying step sizes



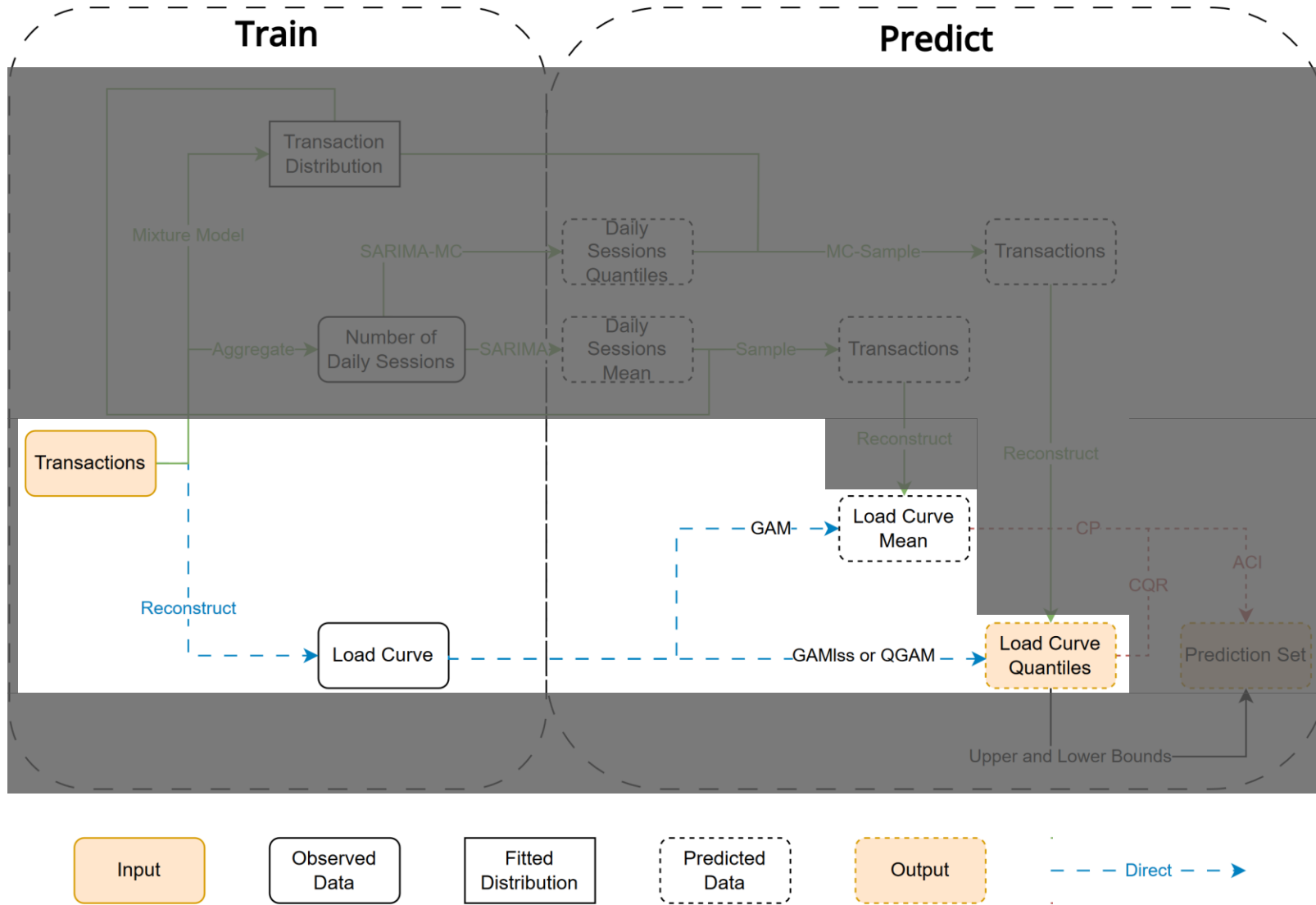
Operations, methods and models across the timeline



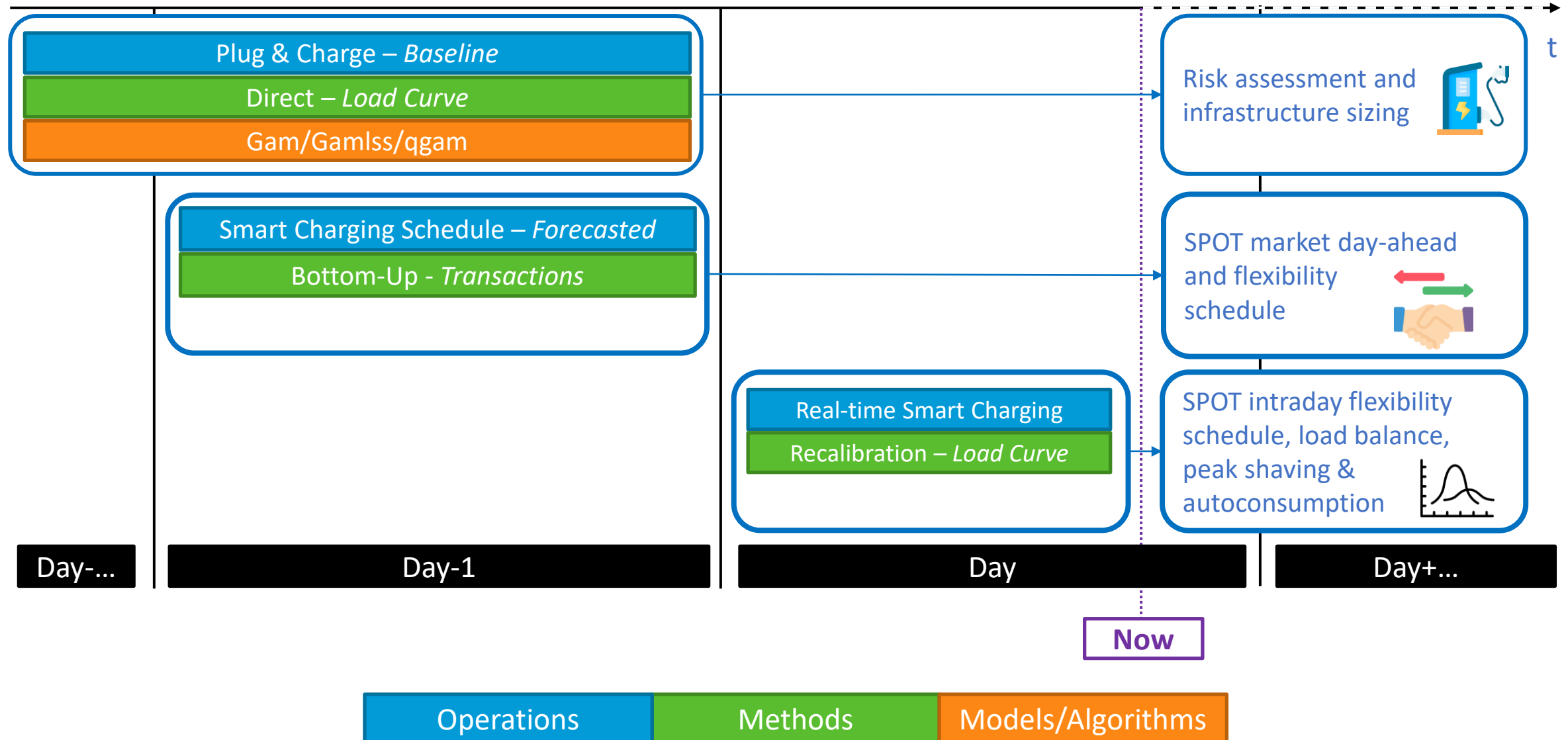
Operations, methods and models across the timeline



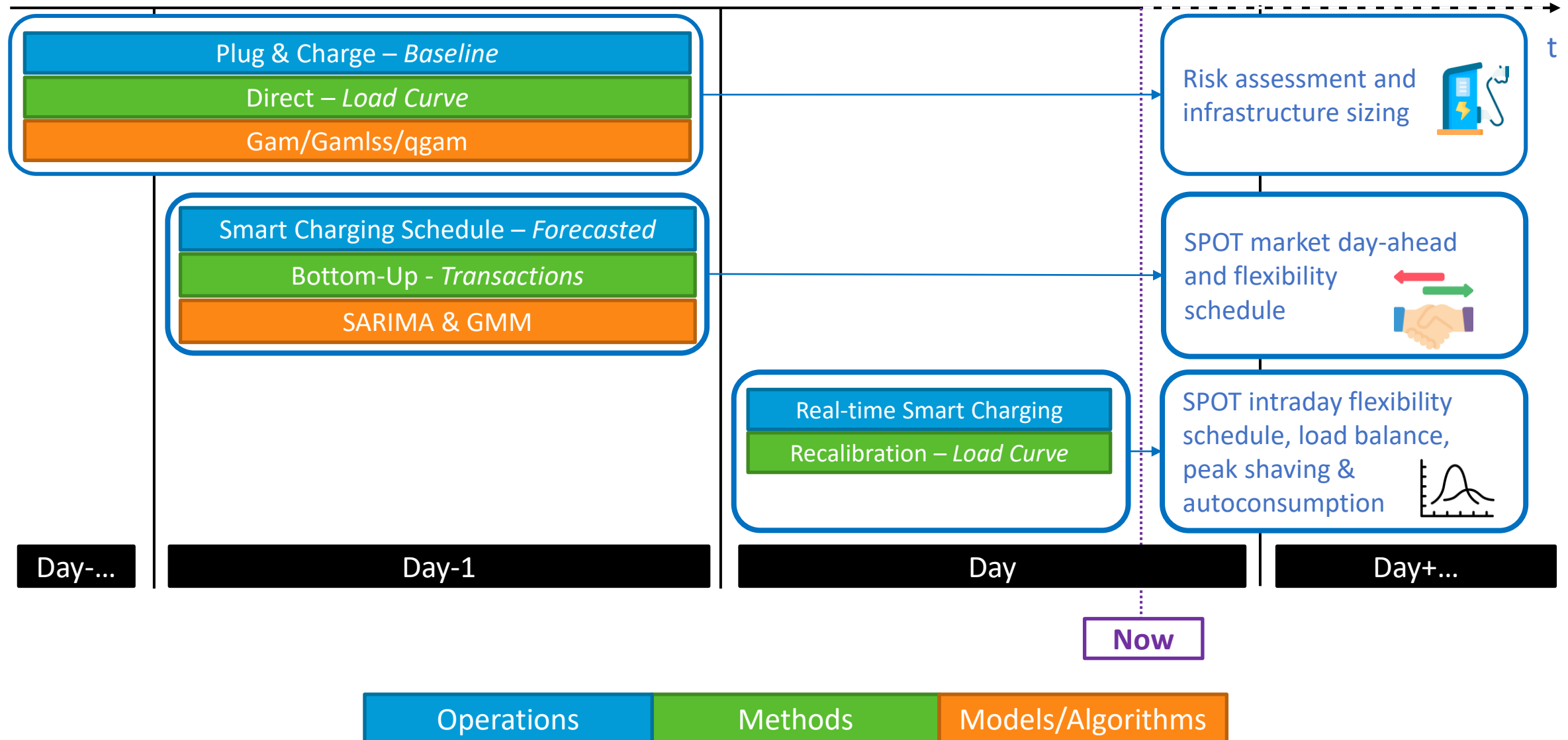
Prediction Framework



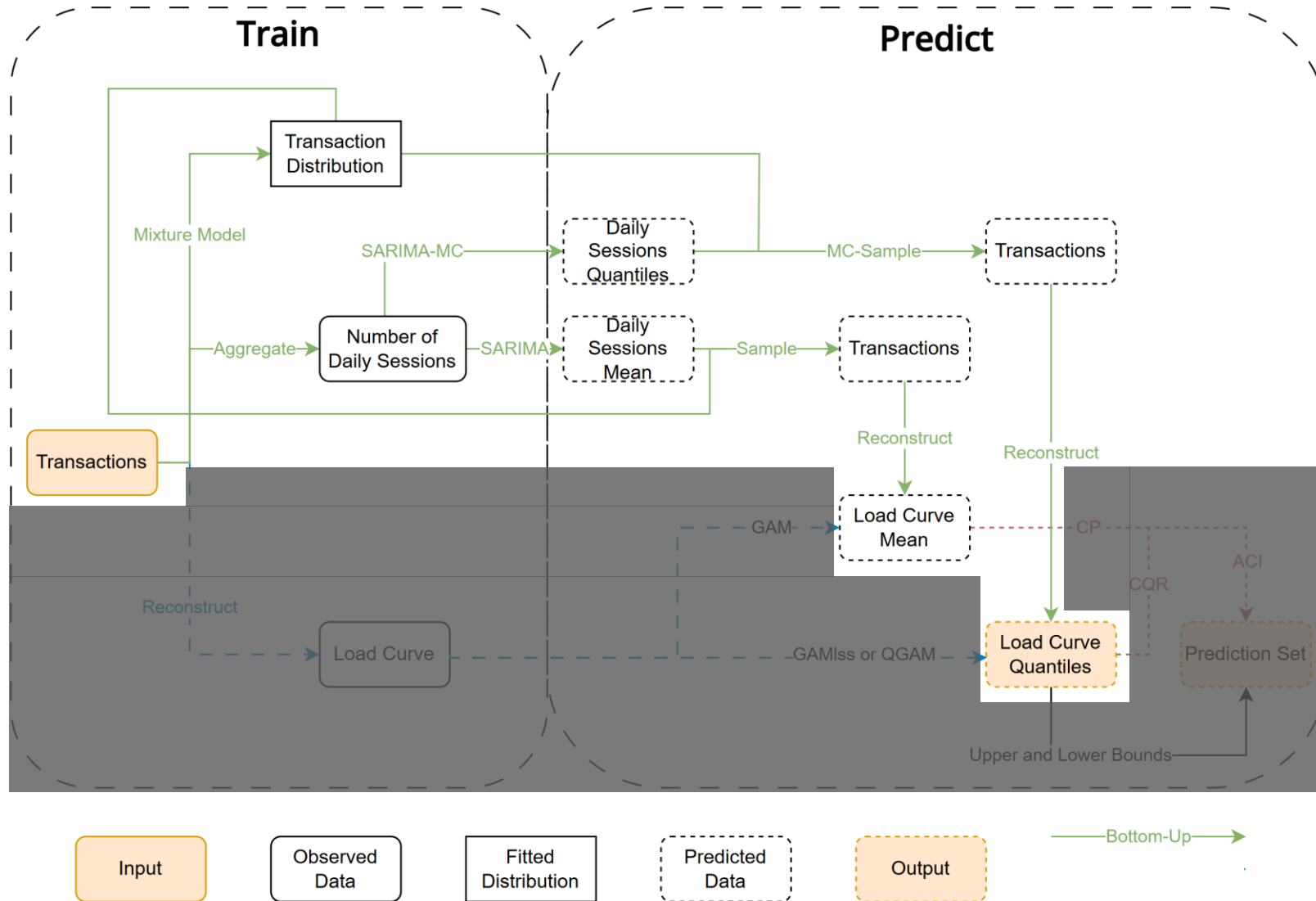
Operations, methods and models across the timeline



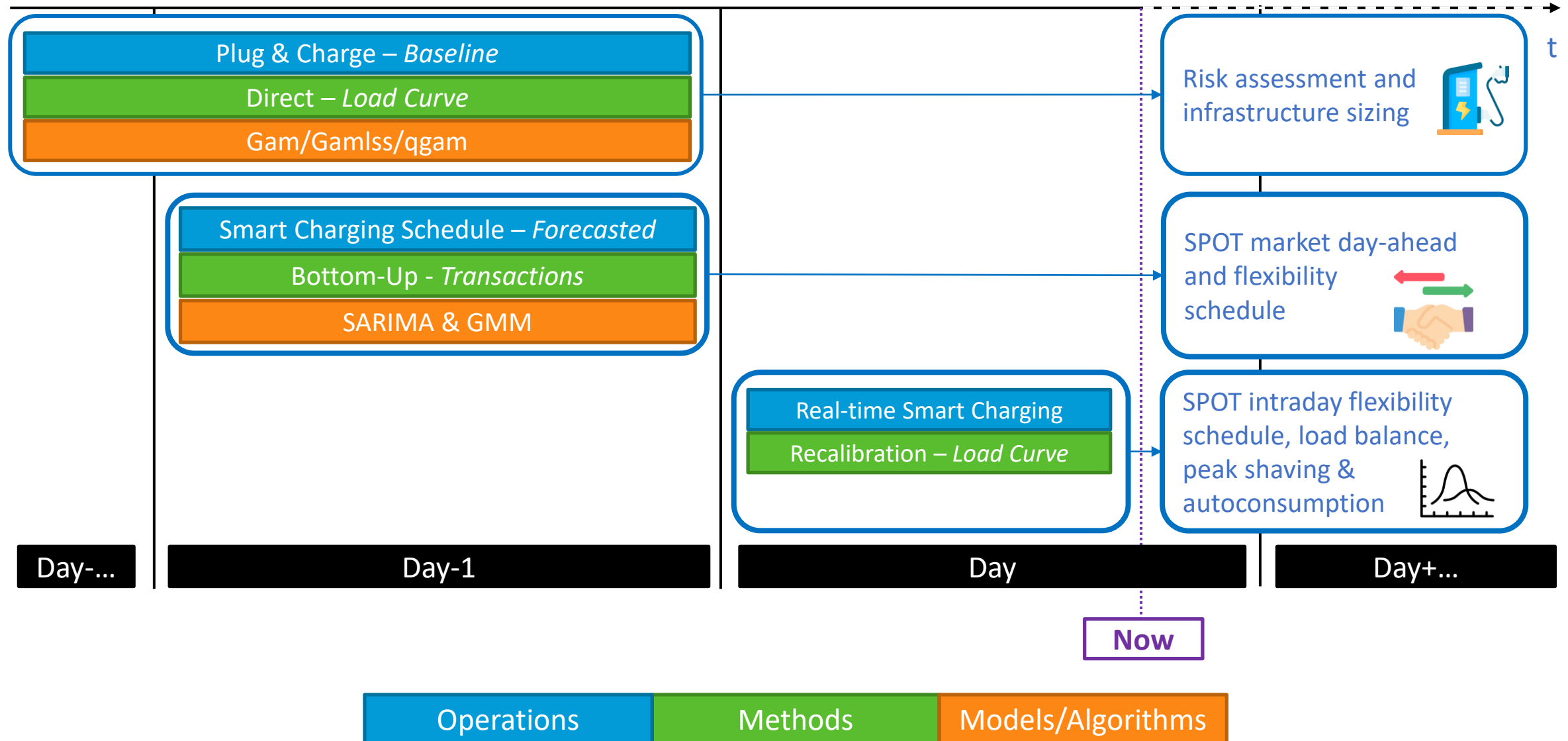
Operations, methods and models across the timeline



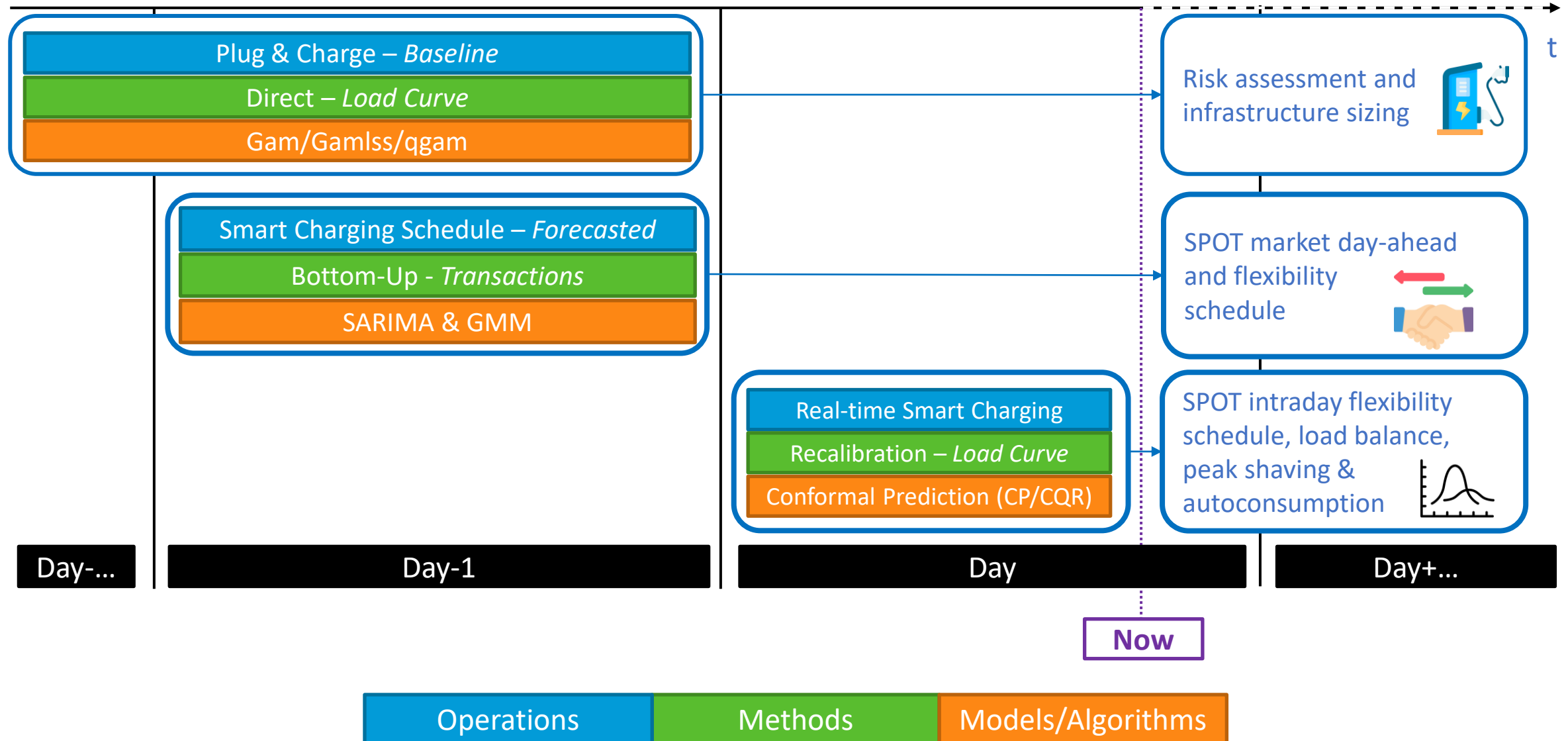
Prediction Framework



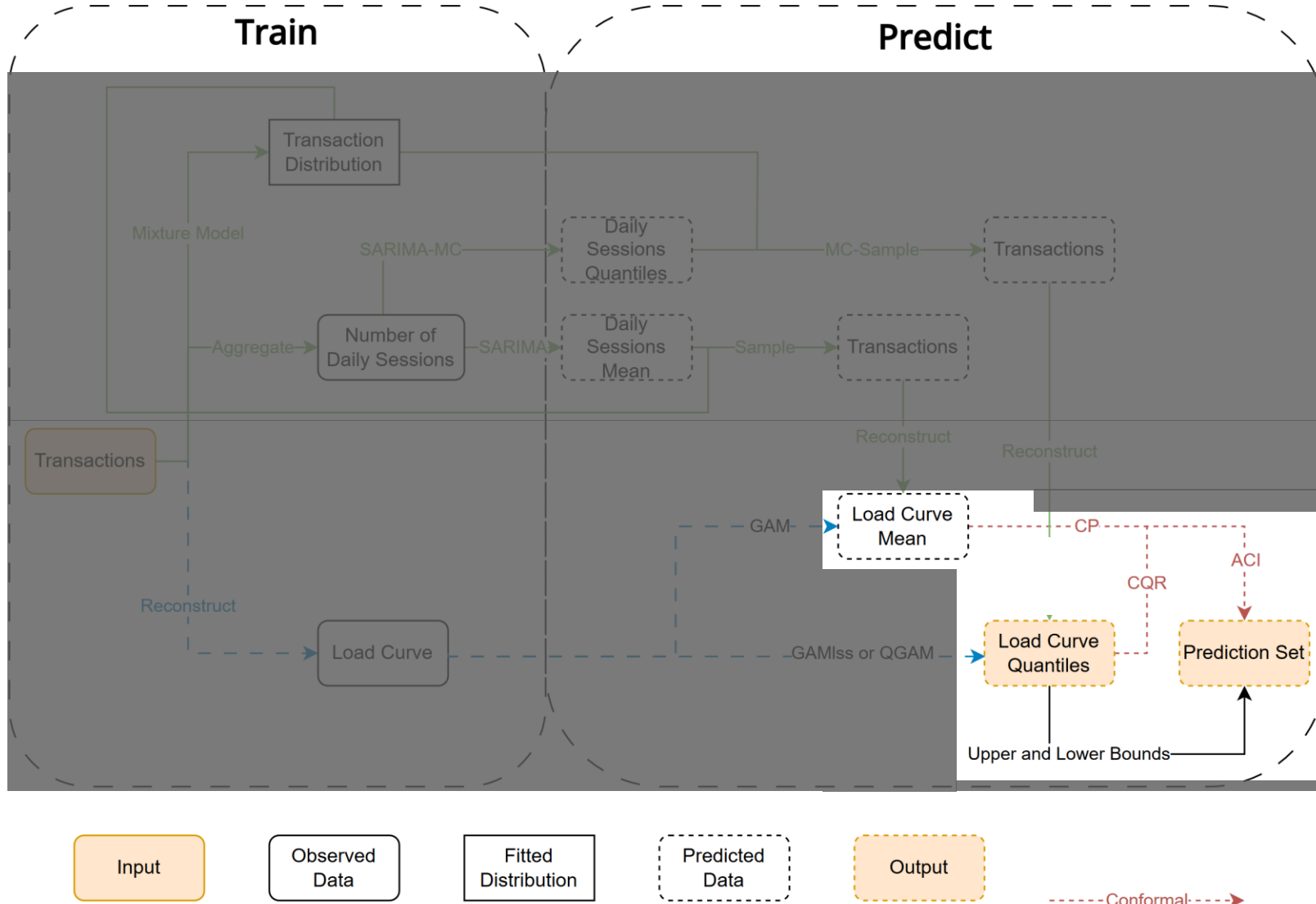
Operations, methods and models across the timeline

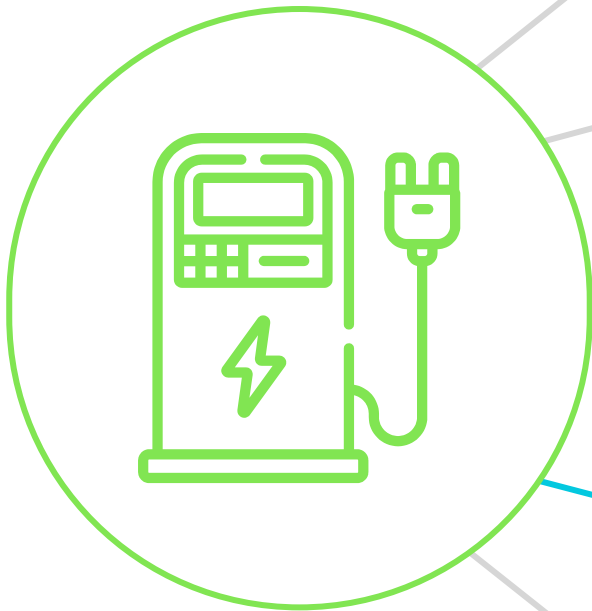


Operations, methods and models across the timeline



Prediction Framework





01

The importance of forecasting for smart charging

02

Two approaches: Direct and Bottom-up
conformal predictions

03

Our framework for predicting EV charging sessions

04

Case Study: Palo Alto (CA)

05

Discussion and Work in progress

Palo Alto Dataset

Public Charge

Boulder (USA)

Dundee
(UK)

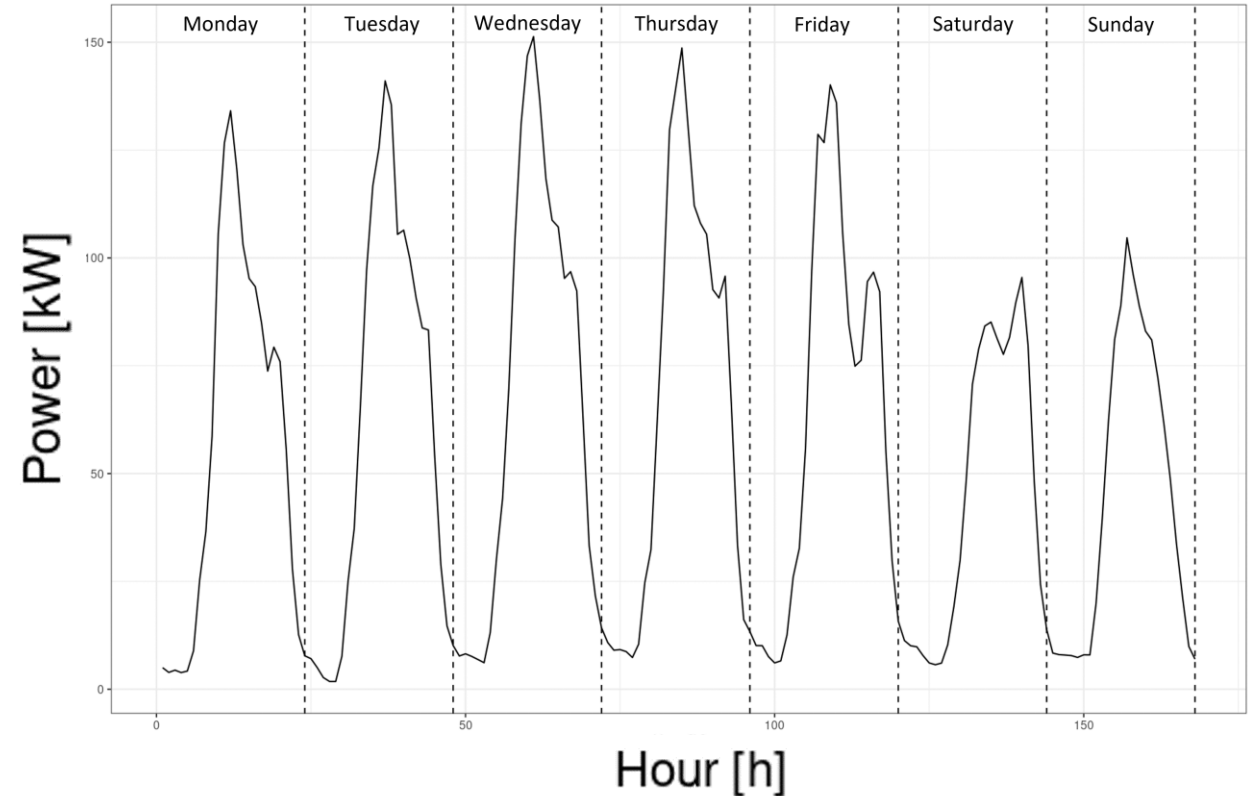
Palo Alto (USA)

Paris
(FRA)

Perth
(UK)



Open Data on public
charge available on GitHub:

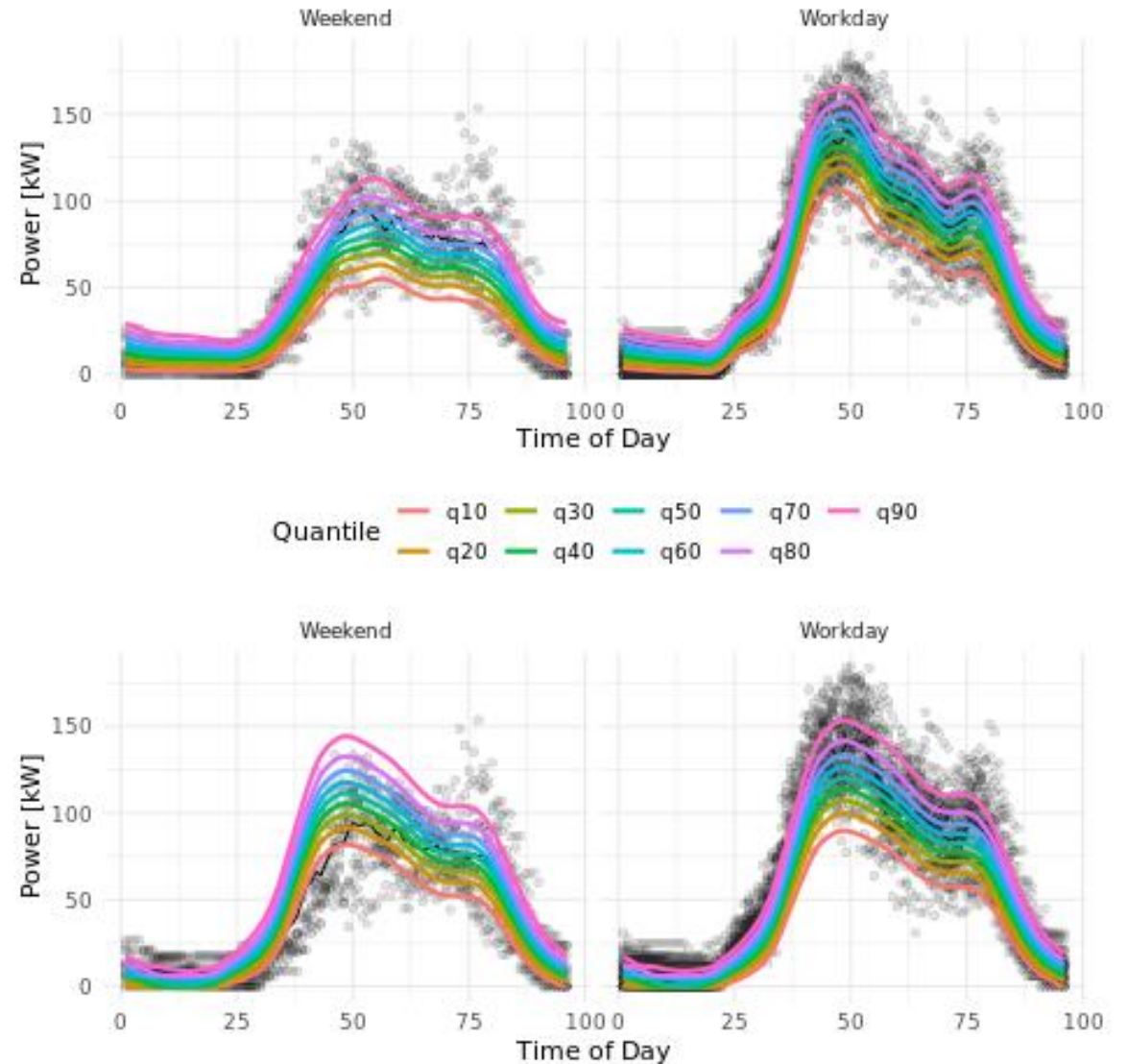


Load profiles for a typical week in Palo Alto (CA)

Uncertainty quantification from Direct and Bottom-Up

Results

- Direct Approaches give the best results in terms of day ahead forecasts for the plug & charge use case
- Bottom-Up gives good results on this task even though it is a more challenging problem
- Among the probabilistic methods, qgam is the best one over almost every quantile, both in accuracy and in sharpness



Averaged quantile forecasts (distinguished by type of day)

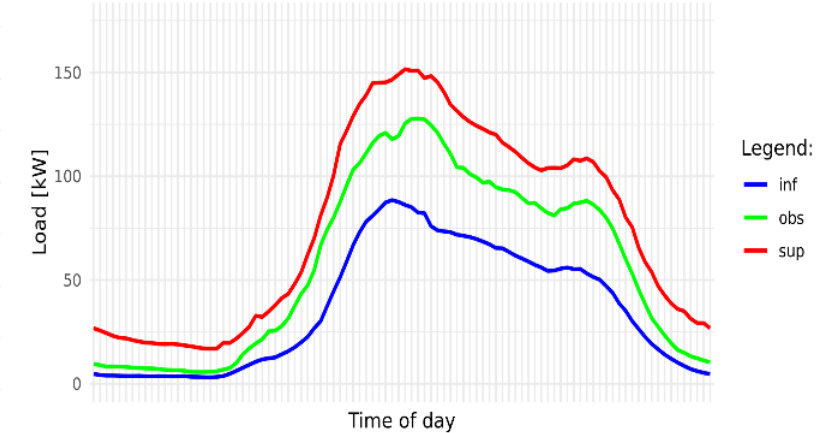
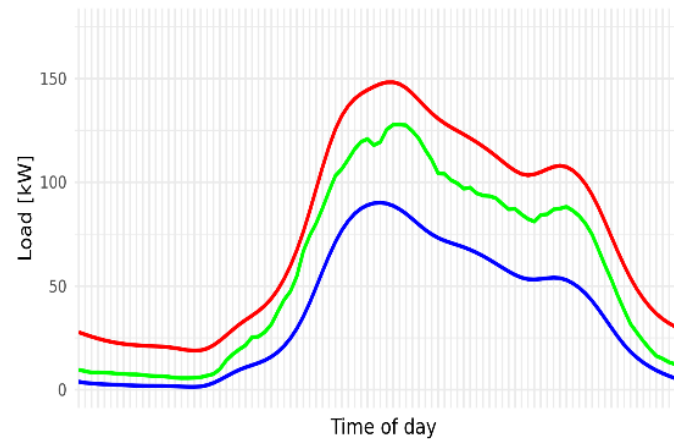
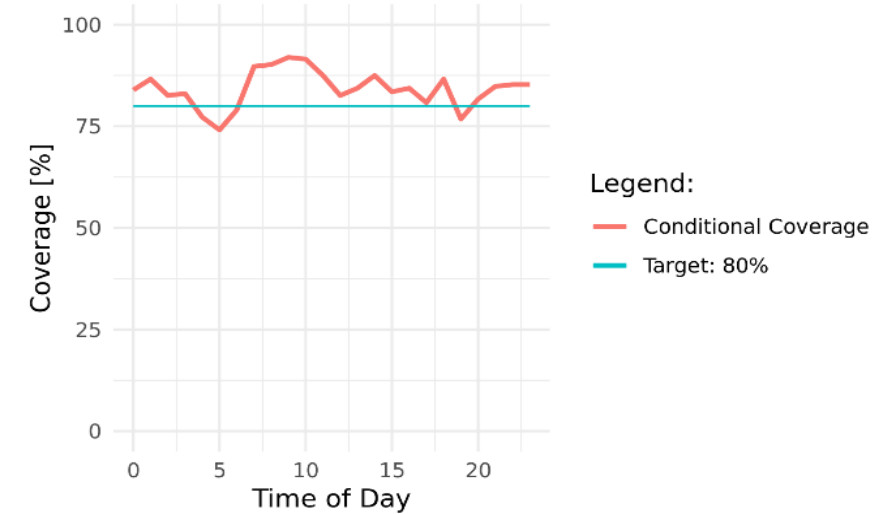
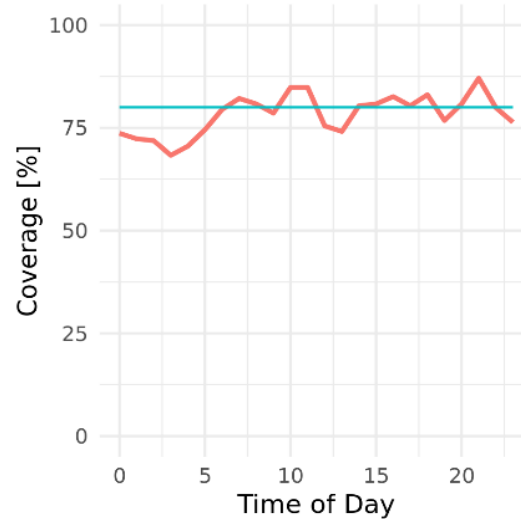
Improvement with Conformal Prediction

From static to adaptive forecasts

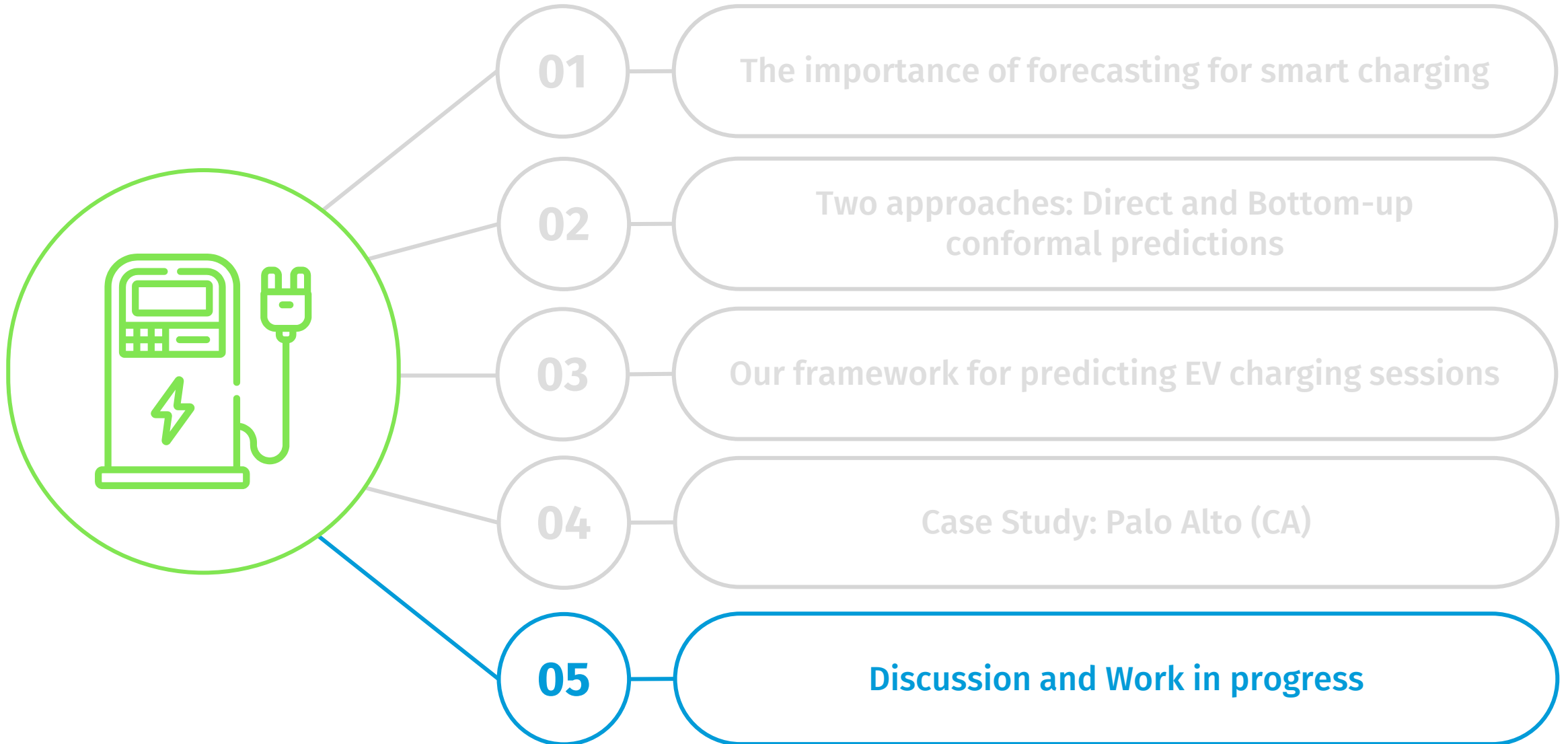
- Both Direct and Bottom-Up methods cannot adapt to behaviour changes
- This can lead to miscalibrated uncertainty in the real-time setting
- To adress this issue, we use Adaptive Conformal Inference methods

Results

- Higher Empirical Coverage while maintaining conditional Coverage
- Equivalent interval width on daily profile, even sharper during night



Conditional Coverage and daily profiles for qgam and CQR qgam

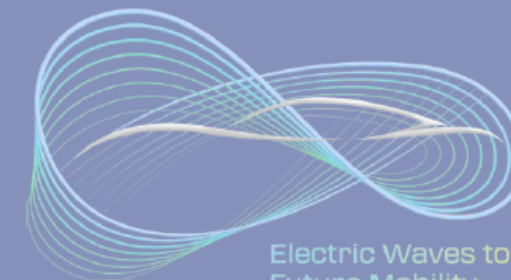


REX EVS

La Newsletter EVS37

Xueqi.miao@edf.fr
(EDF R&D China)

EVS37 NEWSLETTER



April 23 - 26, 2024 | COEX
The 37th International Electric Vehicle
Symposium & Exhibition

For more information, EDF R&D China Center Contact: xueqi.miao@edf.fr

ABOUT EVS37

The 37th Electric Vehicle Symposium & Exposition (EVS37) aims to address and shape the monumental movement of e-mobility under the theme of "Electric Waves to Future Mobility" with academic, government, and industry professionals involved in electric vehicle technologies. It was held from 23 to 26 April 2024 in Seoul, South Korea (Korea). EVS37 has received over 10,000 participants from 49 countries, with 328 papers from 28 countries presented.

EVS37 consisted of the exhibitions, plenary session, EV Tech summit and various scientific programs covering 12 major topics: Electric Vehicle Configuration, Energy Sources & Storage, Electric Vehicle Components, Electric Vehicle Control, Software Defined Vehicles, Electric Vehicle Manufacturing, Application Specific EVs, Infrastructure & Power Grid, Intelligent Transportation Systems, Policy, Standards & Regulation, Market Development, Environment & Social Contributions.

EVS37 Key Takeaways & Executive Summary

- EVS37 pictures the vision to future mobility: **more reliable charging infrastructure and more intelligent vehicles.**
- **Electric vehicle's (EV) potential participating in the electricity market and providing services to grid** are highlighted. The focus is on **load prediction and resource allocation optimization**
- The **unifying of the charging standards** and **interoperable charging** are recognized important to make the user experience reliable, easy and smooth.
- When it **comes to heavy mobility electrification**, the optimal **infrastructure provision (especially for long-haul transport) and business feasibility** remain as the key challenges. There is also growing interest in its **life cycle assessment (LCA)**.
- **V2G and V2H demonstration have been conducted by different stakeholder globally: automakers, charger OEMs, utility companies.** The market potential of the business (e.g., demand response, frequency regulation) is evaluated through various charging optimization algorithms of load.
- **Battery is another key focus at EVS37. All solid-state battery technology** is discussed, and first batch of **commercialization is expected in 2026** revealed by **Samsung SDI**.
- **Battery thermal runaway mitigation** is of great interests, studies are focusing on battery materials and structural design.
- The **new trend in the automotive industry is Software Defined Vehicle**. It continuously evolves vehicles to provide new experience customers even after purchasing the vehicle and hence new revenues opportunities for manufactures.



EDF R&D and international center presented at EVS37

EVS en chiffres



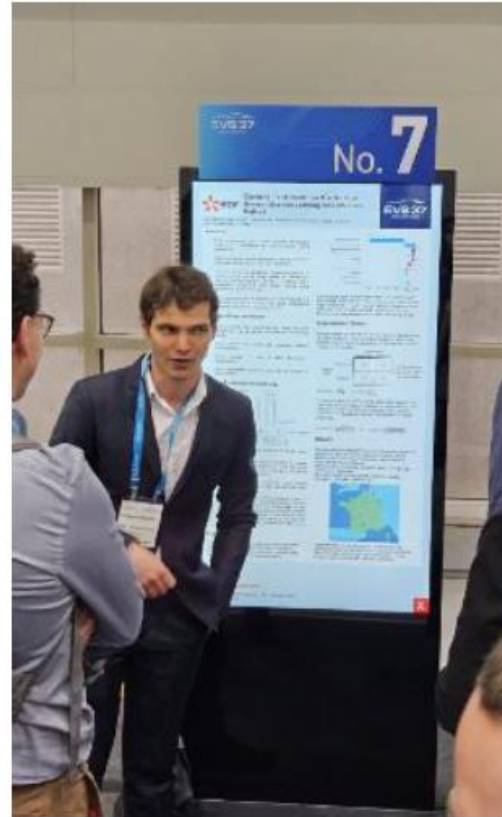
- 1 conférence
- 2 sessions plénières
- 2 sessions de discussions
- 3 sessions techniques VE
- 31 récompenses de posters
- 36 pays participants au symposium
- 37 récompenses de présentation orale
- 40 sessions techniques
- 53 pays participants au salon
- 146 posters techniques
- 172 présentations techniques
- 333 présentations
- 919 participants au symposium
- + 10000 participants au salon

Le groupe EDF bien représenté à EVS37



De gauche à droite : Alban JEANDIN (Izivia), Victor MACQUART (EDF R&D Osiris), Yvenn AMARA OUALI (EDF R&D Osiris), Guillaume PRINCIPATO (doctorant EDF R&D Osiris), Xueqi MIAO (EDF R&D China Center), Dominique BERTIN (EDF R&D Asie Pacifique), Bachir HAMROUCHE (EDF R&D Osiris),
 Thierry BRINCOURT (EDF R&D TREE - hors cadre)

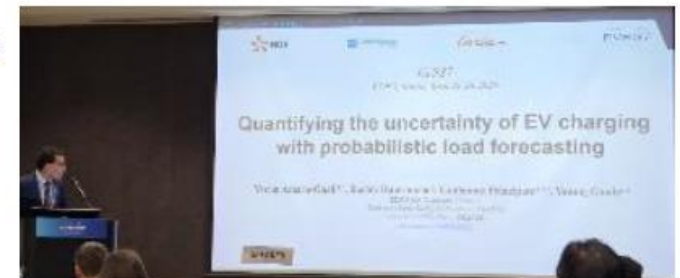
Slide tiré de la restitution de Thierry BRINCOURT (EDF R&D TREE)



Victor MACQUART présente le poster sur la maintenance prédictive des bornes Izivia par IA



Yvenn AMARA OUALI reçoit un prix d'éloquence pour sa brillante présentation sur la quantification des incertitudes de charge des VE.



Thématiques principales abordées



Electric Vehicle Configuration

- Personal e-Mobility
- Light/Heavy-duty EVs
- Electric Buses
- Purpose Built Vehicles (PBV)
- High Voltage E/E Architecture



Energy Sources & Storage

- Battery Technology
- Battery Management Systems
- Energy Storage Systems
- Fuel Cell Technology



Electric Vehicle Components

- Electric Propulsion Systems
- Electric Motors
- Power Electronics (Inverter/Converter)
- Auxiliary Components for EVs
- Chassis Systems for EVs



Electric Vehicle Control

- Energy Management
- Thermal Management
- Advanced Control for EVs
- Autonomous Driving



Software Defined Vehicles

- Platforms & Tools
- Human-Machine/Computer Interaction
- Vehicle Cloud & Cybersecurity
- AI for EVs
- Software Development Methodology



Electric Vehicle Manufacturing

- Materials for EVs
- Design & Manufacturing of EVs
- Recycling & Re-use

Thématiques principales abordées



Application Specific EVs

- Urban Air Mobility (UAM)
- Agriculture
- Construction
- Military



Infrastructure & Power Grid

- Charging Technology
- Charging Infrastructure
- Smart Grid Implementation
- Vehicle-to-Grid



Intelligent Transportation Systems

- EV Connectivity
- Shared Transportation
- Traffic Management
- Transportation Safety & Security



Policy, Standards & Regulation

- Government Policy
- Standardization of EVs
- EV Regulation
- Testing & Validation



Market Development

- Consumer Demand
- Trends & Forecasting of EVs
- Retrofitting EVs
- Business Transitions



Environment & Social Contributions

- Environmental Issues
- Climate Change
- Public Health & Equity
- Education

Etudes de marché et simulations long-terme

Mieux comprendre le profil des utilisateurs de véhicules électriques pour pouvoir orienter les politiques publiques d'incitation :

- Segmentation des clients en Europe (19000) sur la base de variables socio-démographiques et par pays
- Modélisation de l'adhésion client en Californie à l'horizon 2035 sur la base de différents scénarios de développement technologique, installation de bornes et politiques d'incitation pour évaluer la capacité de l'état à atteindre son objectif de vente 100% électriques à l'horizon 2035 : *objectif atteint sur la majorité des scénarios.*
- Etude de l'adhésion de la population Coréenne à une montée des prix des transports en commun en vue de leur décarbonation

Simulation long-terme des courbes de charge de véhicules électriques à différentes échelles et pour différents segments : focalisation importante sur les poids lourds et le transport de marchandises :

- Simulation de courbes en 2035 et 2050 de consommation du secteur de transport de marchandises sur la base de plusieurs modèles : simulation des opérations -> simulation du trafic -> hypothèses sur les véhicules -> génération des profils : objectif = *prendre des décisions d'investissement suivant les hypothèses prises.*
- Simulations sur la base de données de courbes de charges obtenues expérimentalement.

Smart Charging 1

Déploiement de solutions de smart-charging et d'autoconsommation et évaluation de leur viabilité économique et sociale

- Avancement du projet Européen SCALE dans lequel ENEDIS est impliqué :
 - Déploiement de plusieurs use-cases de smart charging (V2G) en Europe pour V2H, V2Public, V2Business, V2Depot.
 - Utilisation de standards ouverts et questionnement sur l'intérêt économique des solutions
 - Questionnement sur le passage à l'échelle de telles solutions
- Etude de l'acceptabilité des solutions de V1G et V2G dans les populations : Les principaux sujets d'inquiétudes sont la dégradation potentielle de la batterie et la disponibilité du véhicule en cas d'imprévu.
- Rentabilité d'un système de bornes pour poids lourds équipé de PV et de batteries stationnaires
- Proposition d'un framework analytique pour évaluer l'intérêt économique et environnemental du smart charging :
 - Utilisation de la load journalière et optimisation du placement des bornes avec cas d'usage au Colorado. Les résultats font état d'une économie de l'ordre de 3% du coût du réseau.

Le pb actuellement avec le V2G est qu'en prenant en compte les taxes et les frais de transport de l'électricité la solution est difficilement rentable (Danemark)

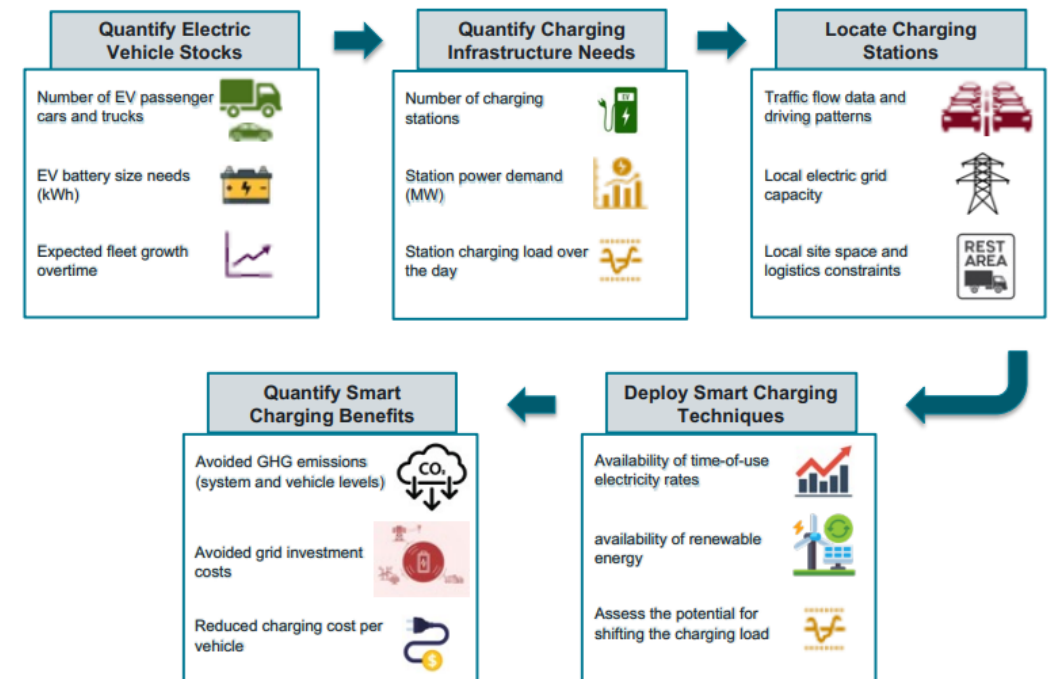


Figure 1. Framework to quantify the environmental and economic benefits of EV smart charging.

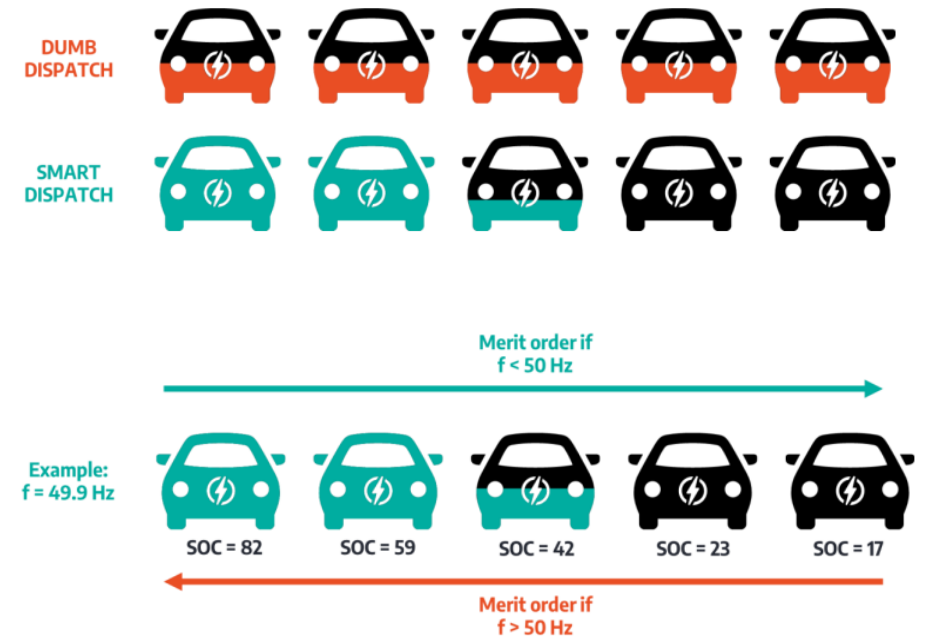
Smart Charging 2

Proposition de stratégies d'optimisation de la recharge :

- Réflexion autour de stratégies optimales de participation d'une flotte à la réserve primaire en maximisant le gain et en réduisant le nombre de cycles réalisés
- Real-Time Charging Scheduling Tool for Heavy-Duty EV: ils ont le SoC voulu et les heures d'arrivées et de départ et doivent définir un programme de recharge pour leur flotte sous contrainte du réseau et de vouloir avoir un système égalitaire. Leur optimisation n'est pas optimale mais tourne très vite (moins d'une seconde) pour pouvoir être utilisée en temps réelle.
- Prédiction du temps de charge par XGBoost à partir de de variables explicatives (choisies par SHAP values) avec un mix de différents critères comme loss.

Globalement les travaux de smart charging présentés supposent que les prévisions sont parfaites : connaissance à l'avance des arrivées, départs et énergies.

- Des chercheurs de LG proposent une application pour fournir des informations sur la dégradation de la batterie à partir de variables explicatives (météo, km, etc.) : *erreur d'estimation de SOH (state of health) est de l'ordre de 2%.*
- LADWP : étude de la dégradation des batteries en V2G en imposant des bornes de recharge sup et inf avec 3 cas d'usages : 1 charge + 1 décharge / 2 cycles / Fleet EVs in-service



Autres sujets

Software defined vehicle : voir le VE comme un centre logiciel avec une multitude de données de capteurs exploitées par machine learning pour :

- Améliorer la sécurité des usagers : vérifier la bonne tenue du volant, utilisation de la caméra pour détecter la somnolence...etc
- Améliorer les outils d'aide à la conduite : calibration des systèmes de freinage régénératif, détection d'obstacle sur des sites inconnus (par augmentation de données), estimation en temps-réel du poids du véhicule à partir de données de capteurs.

Pour plus d'informations :

- Restitution de Thierry BRINCOURT (plus orientée batteries)
- Catalogue des articles réalisé par EDF Asie Pacifique (dominique.bertin@edf.fr) qui contient tous les articles présentés.

	PL1 - Plenary Session 1 Electrification of mobility: Challenges and roadmap towards carbon neutrality Donghee HAN Hyundai Motor Company Republic of Korea Download
	PL1 - Plenary Session 1 Delivering the future of mobility I.P. PARK LG Sciencepark Republic of Korea Download
	IDS - Dialogue Session 1 Development of battery current ripple control method using PR Controller of V2L Bongyeon CHOI HL Mando Republic of Korea Download

TRAVAUX EN COURS

Projet Smart Charging

Présentation de DREEV

DREEV propose une plateforme « Cloud » permettant d'assurer la recharge des véhicules électriques tout en valorisant l'énergie présente dans les batteries en proposant des solutions de smart charging en V1G et V2G. (Travaux R36)

« La plateforme DREEV connecte le monde de la mobilité électrique (bornes, véhicules) avec le système électrique (système électrique local et marchés de l'énergie nationaux et européens). Grâce à des algorithmes d'optimisation avancés, nous optimisons la recharge (V1G) voire la décharge (V2G) des véhicules électriques pour satisfaire les besoins de charge à un coût inégalable et une empreinte carbone minimale. »

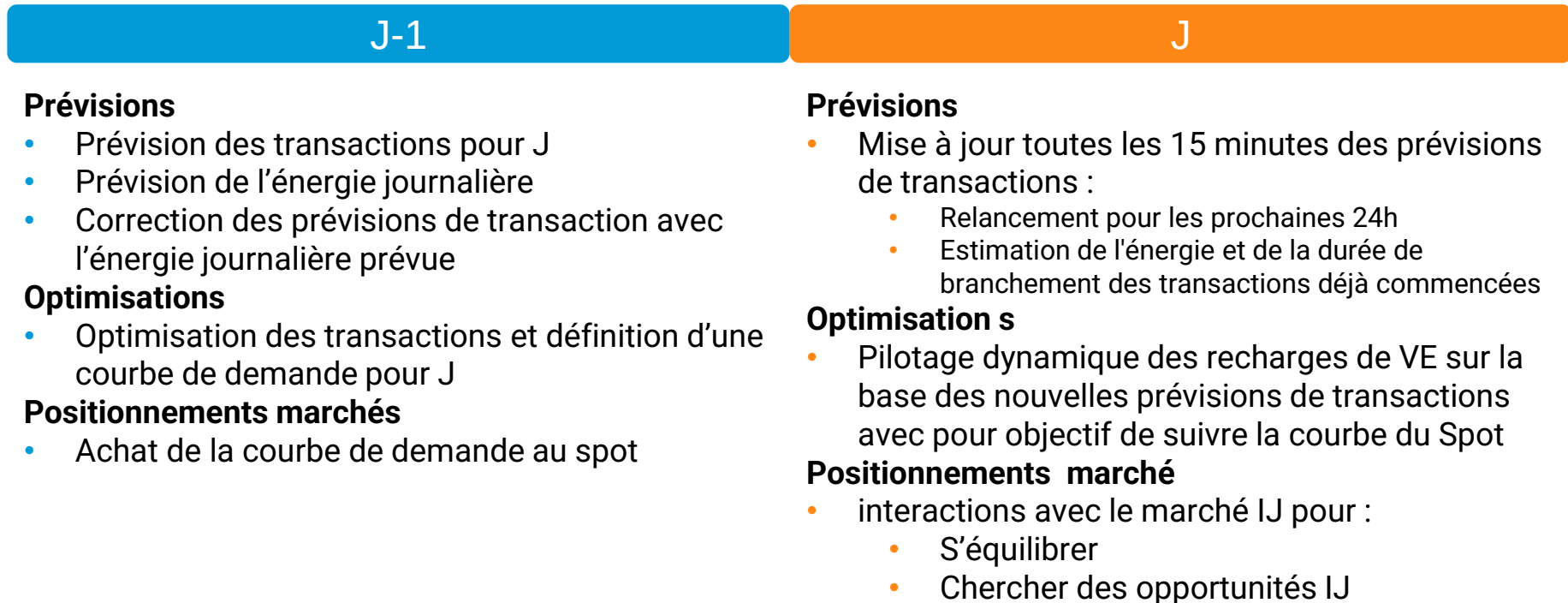
➡ **Plénière 12/2024**



Problématique métier

Pour optimiser la recharge d'une flotte de vans pour une entreprise de livraison, DREEV a besoin de modules de prévision pour pouvoir se placer sur les marchés spot et IJ

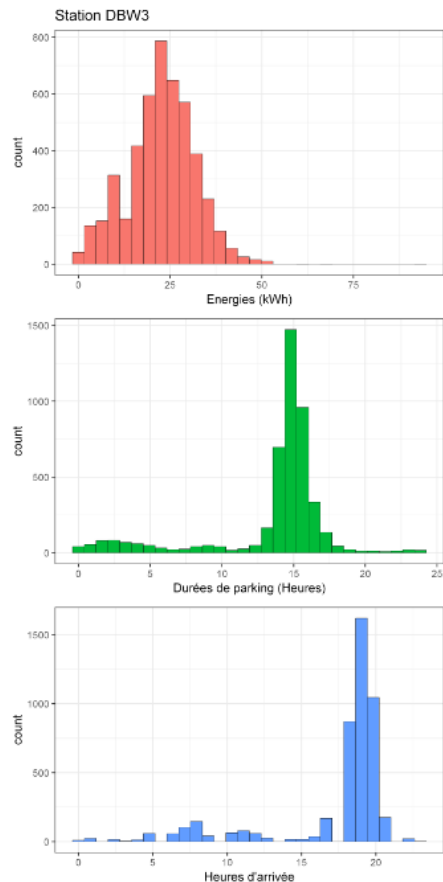
Processus Métier



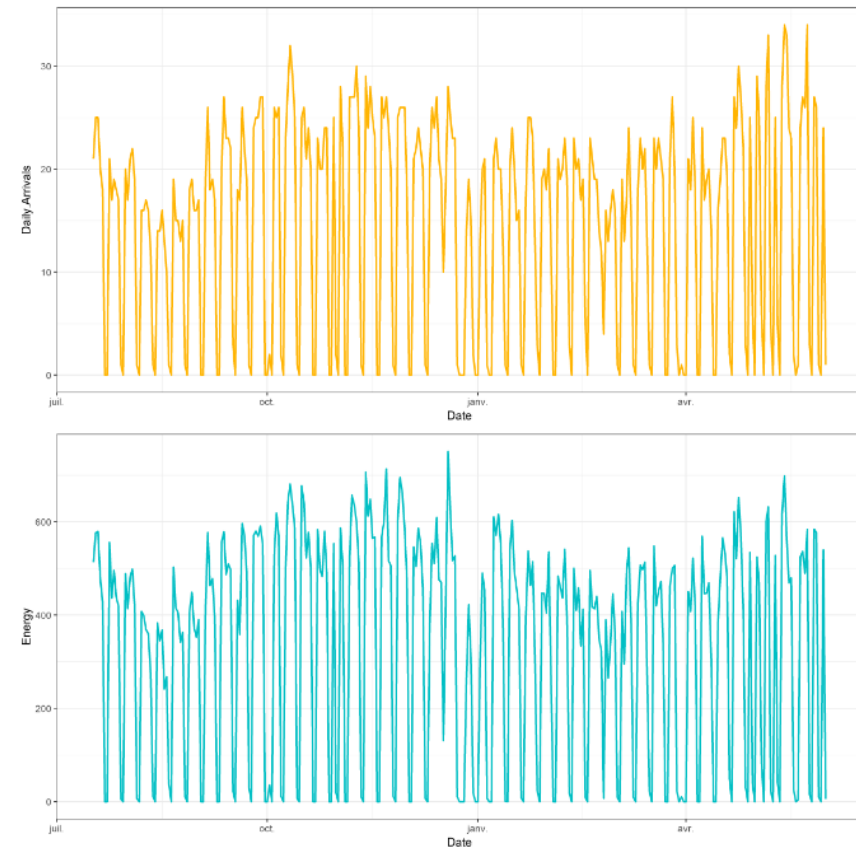
Données disponibles

- On s'intéresse à la prévision de sites Allemands : 33 stations
- Période 01/07/2023 – 31/05/2024 (historique réduit et séries non stationnaires : plusieurs ruptures à cause des changements de tailles de flottes et d'infrastructures)

Distribution des transactions



Profils journaliers



Module de prévision des transactions

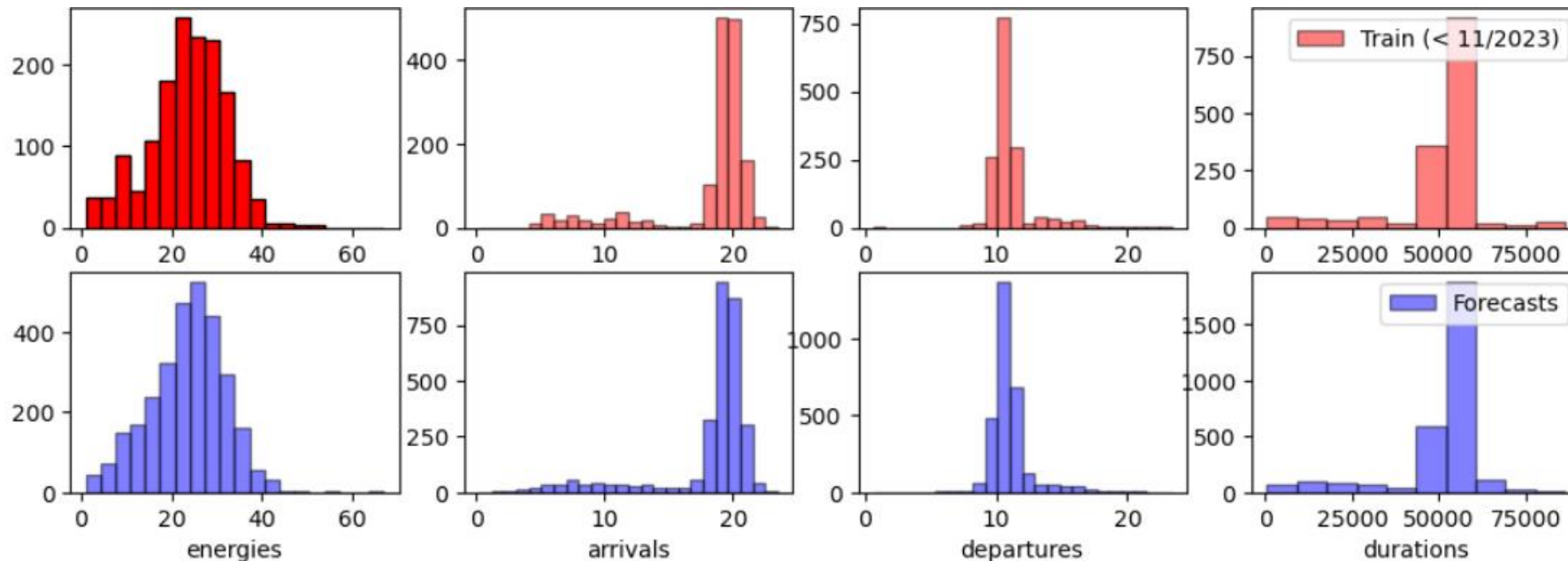
1^{er} module livré : module Python de prévision des transactions J+1 et J+2 par un modèle bottom-up

- Prévion SARIMA du nombre de branchements journaliers
- Estimation de la distribution conjointe (A, D, E) par GMM par type de jours

Estimation des modèles

- Modèle SARIMA réestimé au fil de l'eau
- Modèle GMM estimé sur une période fixe mais ré-estimable directement par DREEV

Prévisions obtenues



Module de prévision de l'énergie journalière

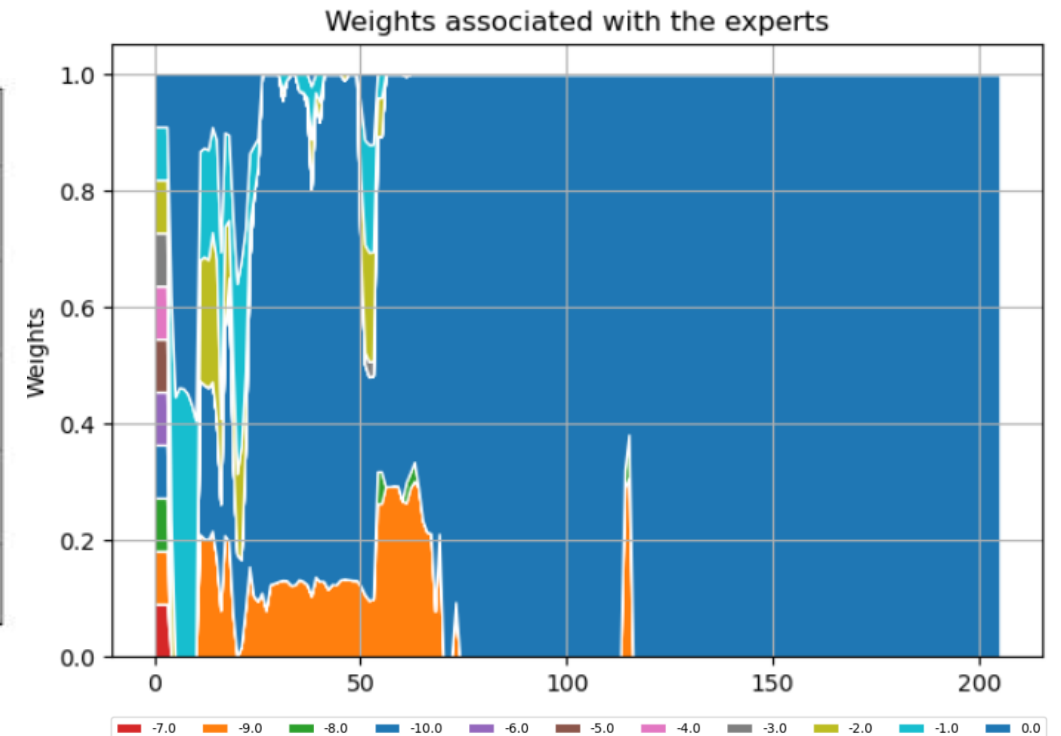
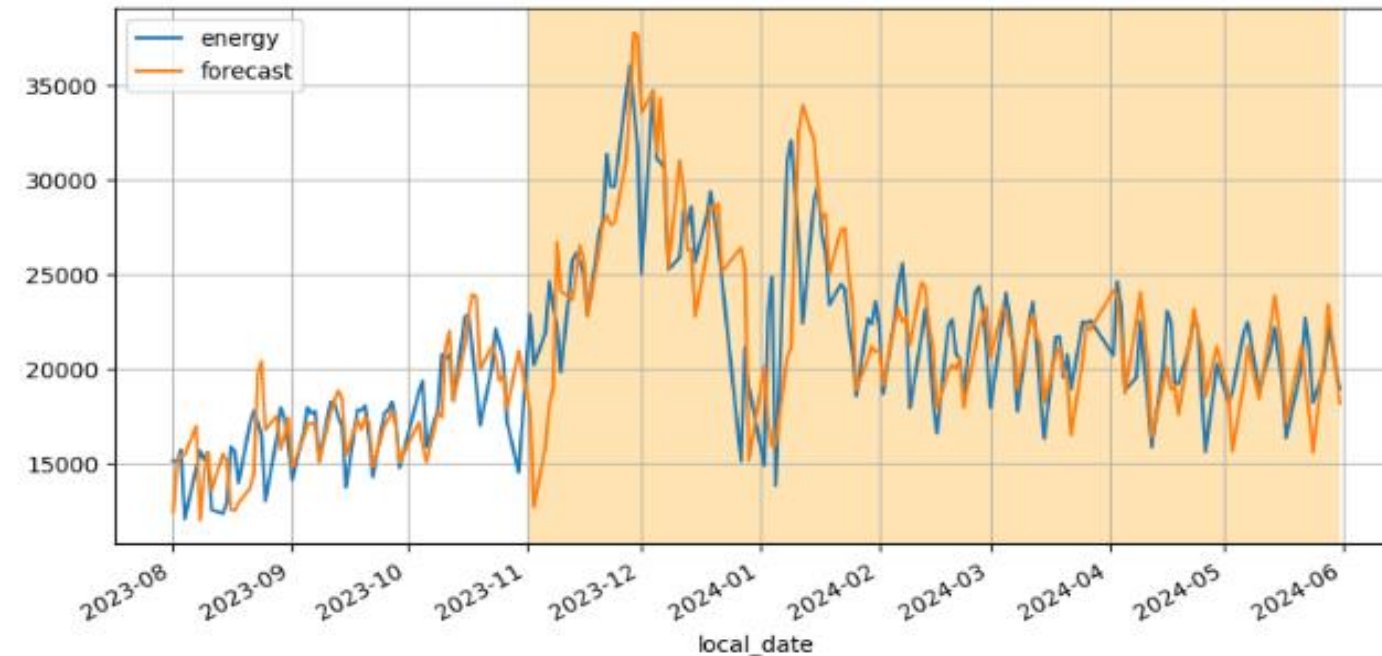
2^e module livré : module Python de prévision de l'énergie journalière

- Prévion par un mélange de filtres de Kalman scalaires sur les covariables

Avantages de la solution

- Pas d'apprentissage, les prévisions utilisent les filtres de Kalman utilisent tout l'historique pour se corriger online
- Les filtres de Kalman permettent une bonne adaptation aux non-stationnarités des données

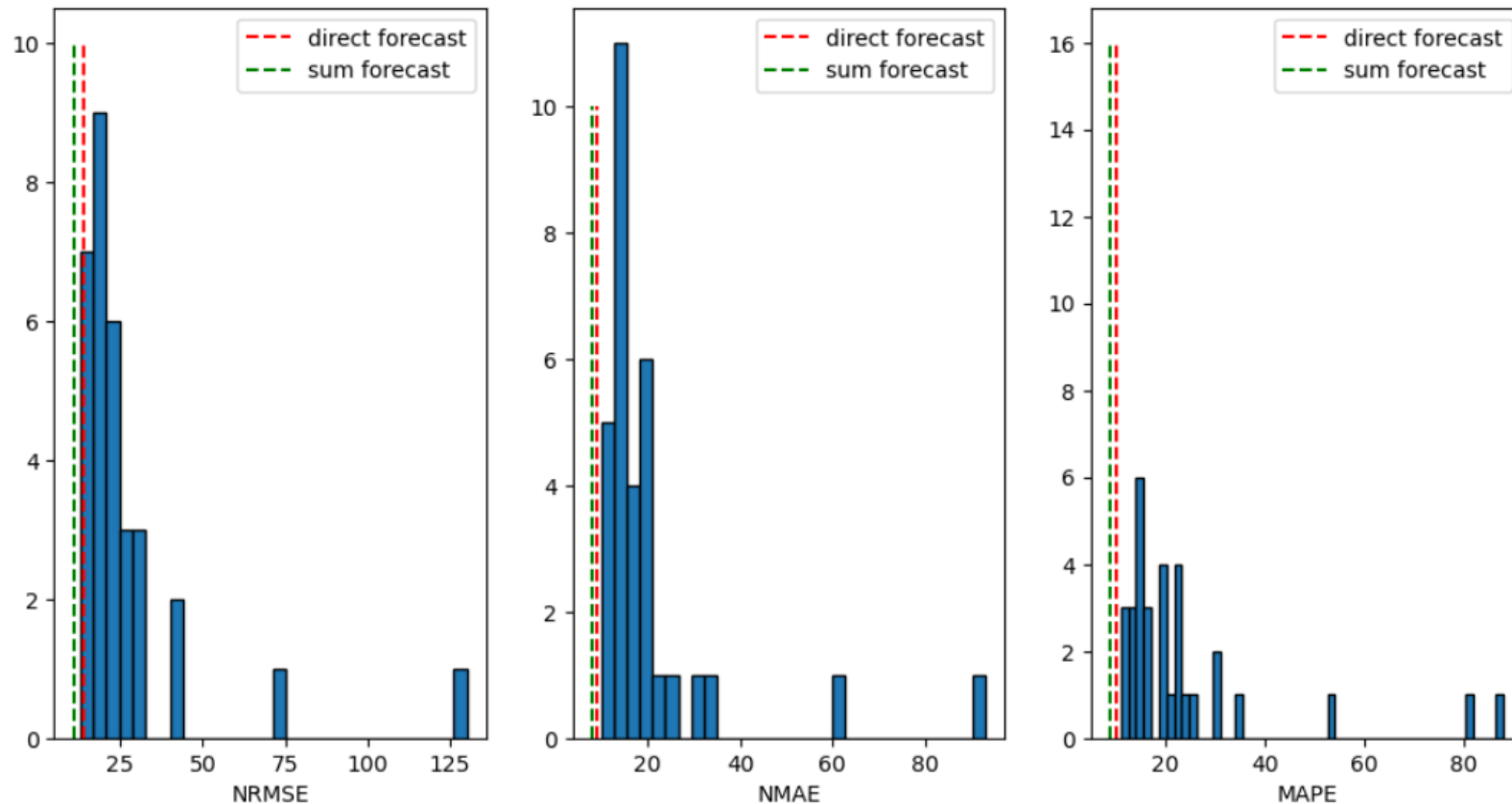
Prévisions obtenues



Prévision nationale vs prévision des stations

Questionnement : Faut-il prévoir les stations ou prévoir l'énergie nationale directement ?

Analyse comparative



!\\ Le foisonnement permet d'améliorer les performances des prévisions.

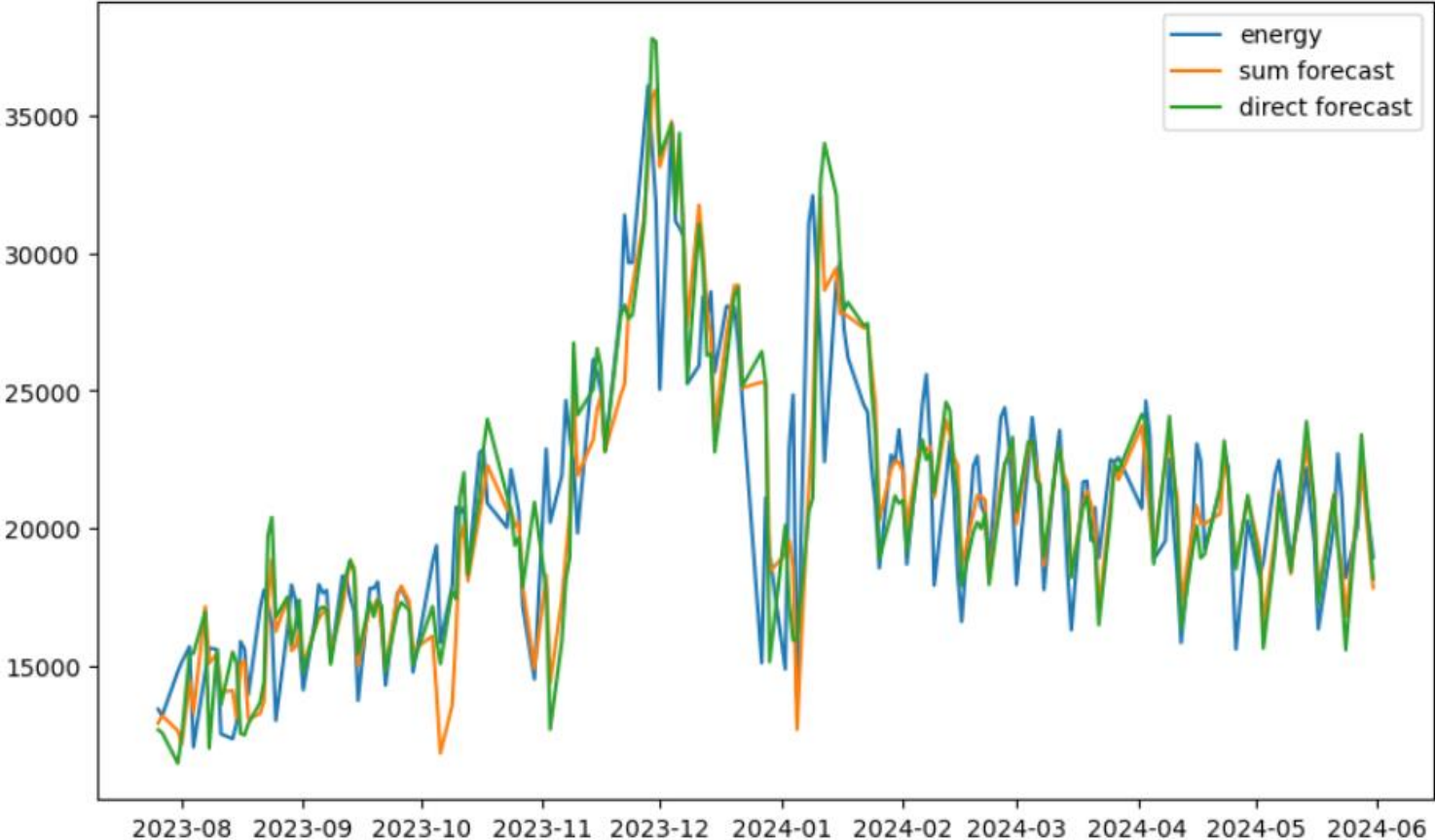
Prévision nationale vs prévision des stations

Questionnement : Faut-il faire la somme des prévisions des stations ou prévoir l'énergie nationale directement ?

Analyse comparative

Modèle	NRMSE	NMAE	MAPE
Somme prévisions	11%	8%	9%
Prévisions directes	14%	9%	10%

Prévision directe vs somme des prévisions des stations

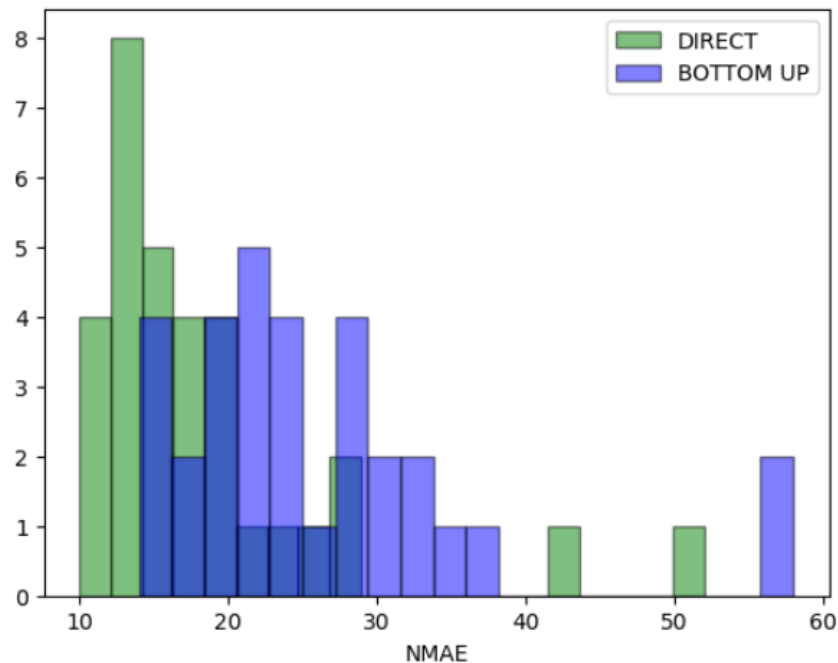


Agrégation des prévisions bottom-up vs prévisions directes

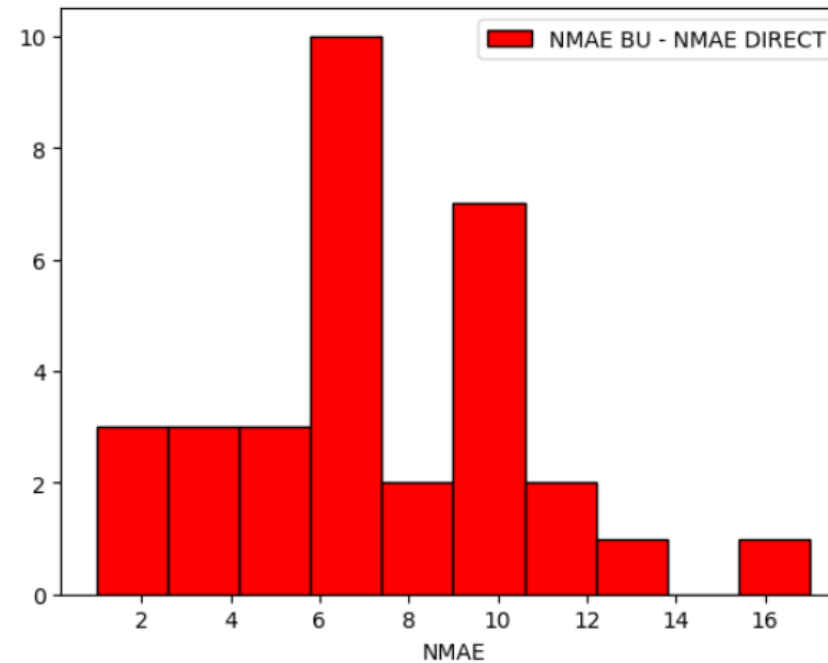
Questionnement : Faut-il prévoir l'énergie ou agréger la prévision des transactions ?

Analyse comparative

Distribution des NMAE par station pour les modèles directes et bottom-up



Distribution des écarts de NMAE par station entre les modèles directes et bottom-up (BU)



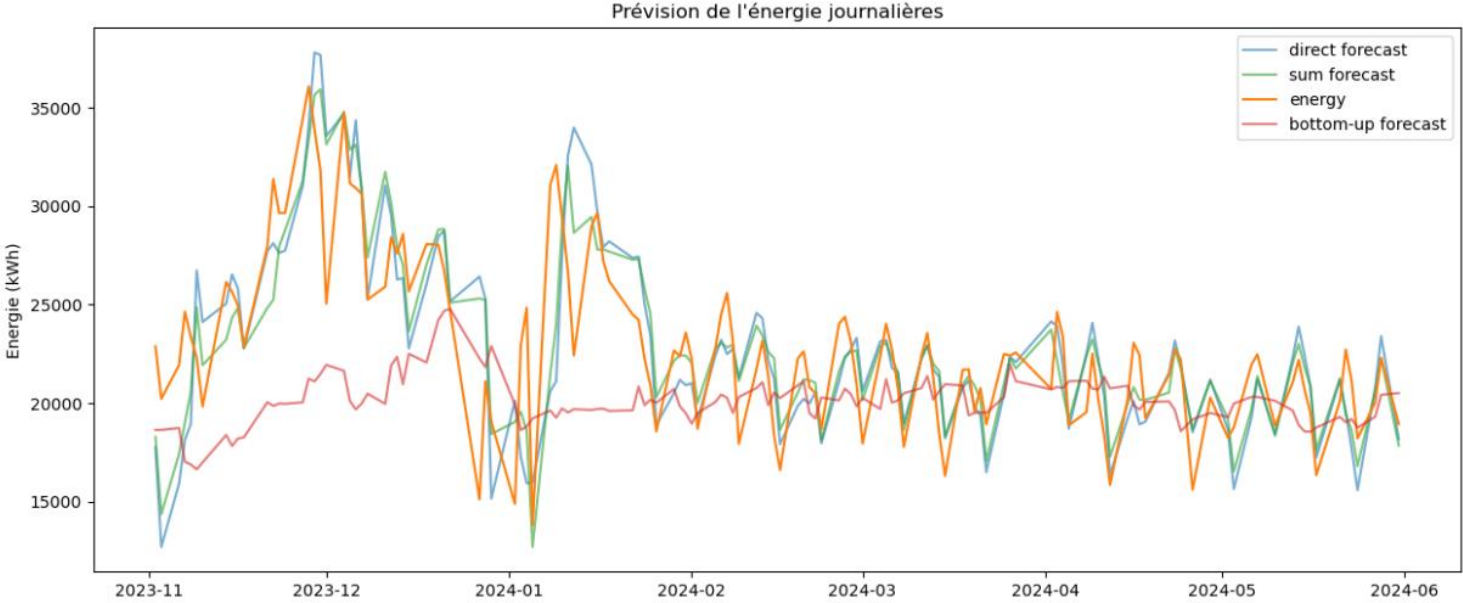
Agrégation des prévisions bottom-up vs prévisions directes

Questionnement : Faut-il prévoir l'énergie ou agréger la prévision des transactions ?

Analyse comparative

Modèle	NRMSE	NMAE	MAPE
Somme prévisions	11%	8%	9%
Prévisions directes	14%	9%	10%
Bottom-up	22%	17%	16%

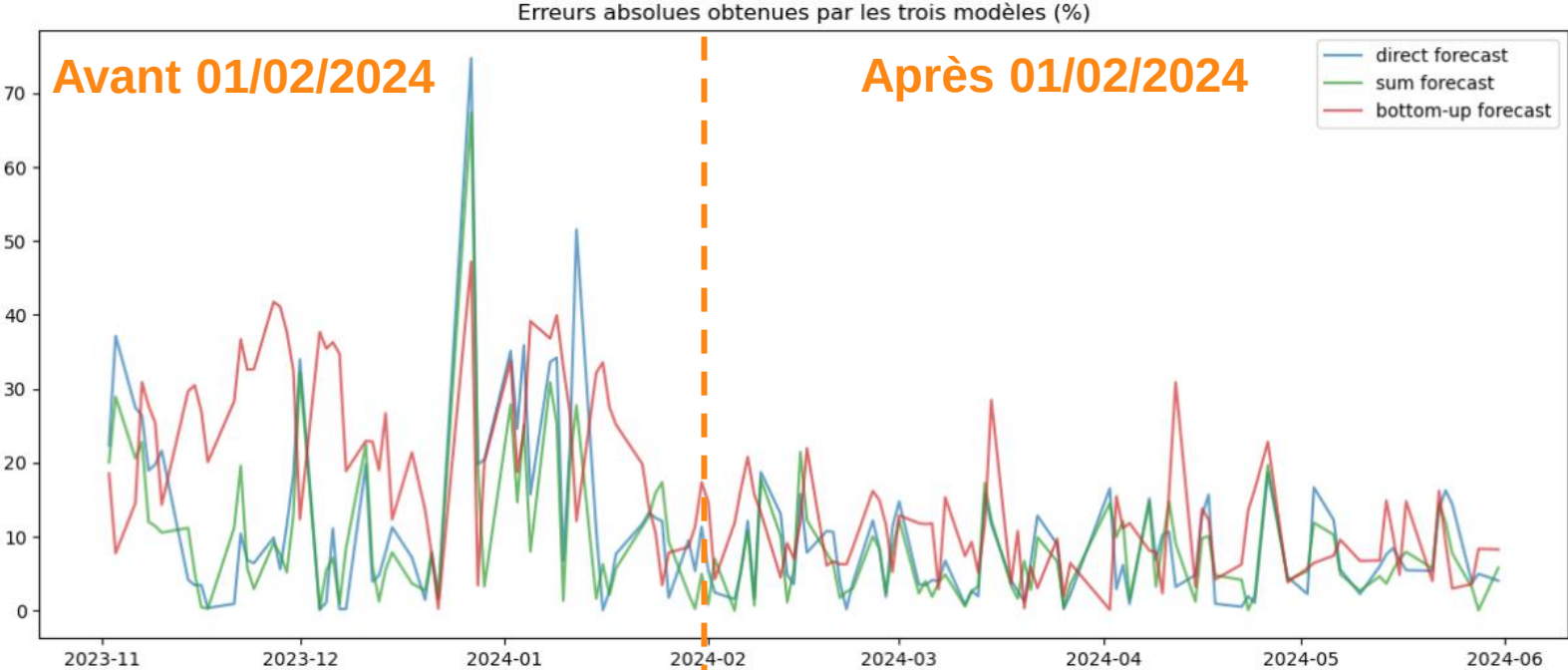
Prévision directe vs somme des prévision bottom-up



Agrégation des prévisions bottom-up vs prévisions directes

Questionnement : Faut-il prévoir l'énergie ou agréger la prévision des transactions ?

Prévision directe vs somme des prévision bottom-up



Modèle	NRMSE	NMAE	MAPE
Somme prévisions	14	11	12%
Prévisions directes	17	13	14%
Bottom-up	28	24	24%

Modèle	NRMSE	NMAE	MAPE
Somme prévisions	7	6	7%
Prévisions directes	8	6	7%
Bottom-up	11	9	10%

Questions en cours de traitement

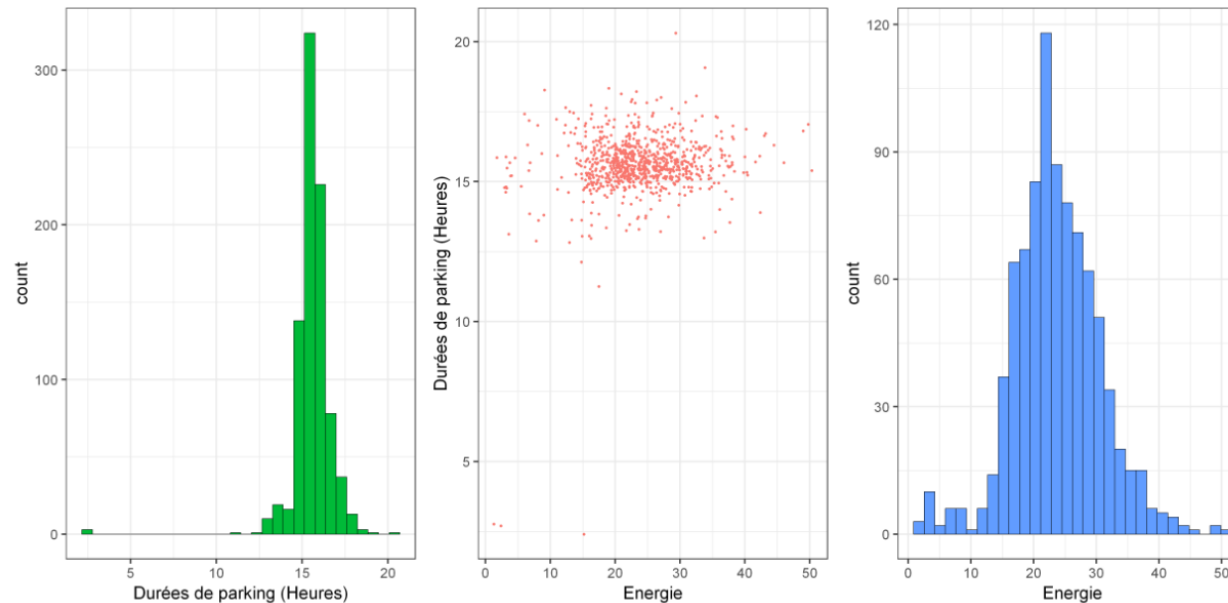
Questionnement : Comment faire des prévisions de transactions infra-journalières ?

Développement de deux modules IJ supplémentaires :

- Un module permettant de relancer des prévisions à l'instant T pour les prochaines 24h en utilisant le nombre de transactions déjà commencées dans la journée et en tirant dans la loi jointe : $(A, D, E \mid A > T)$
- Un module permettant de prévoir la durée de branchement et l'énergie nécessaire à une transaction en cours en prenant un quantile favorable dans la distribution jointe $(D, E \mid A = T)$

Qui devront être améliorés dans les années à venir.

Distributions marginale et jointe des energies et durées de parking pour les véhicules arrivés à 18h



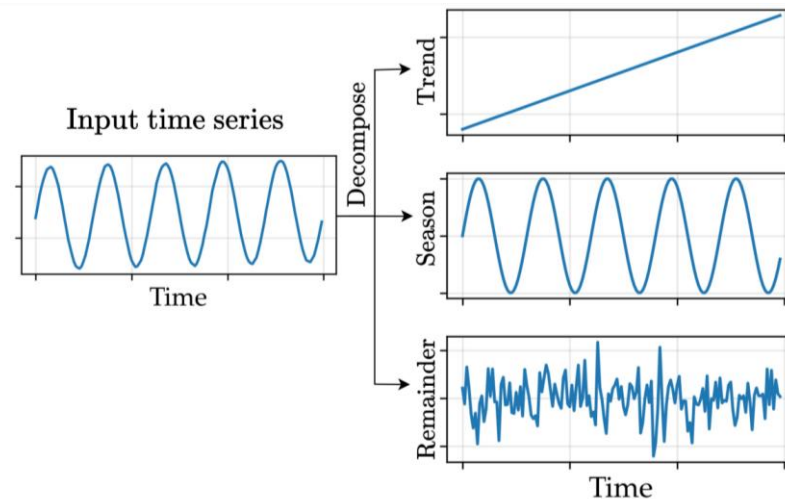
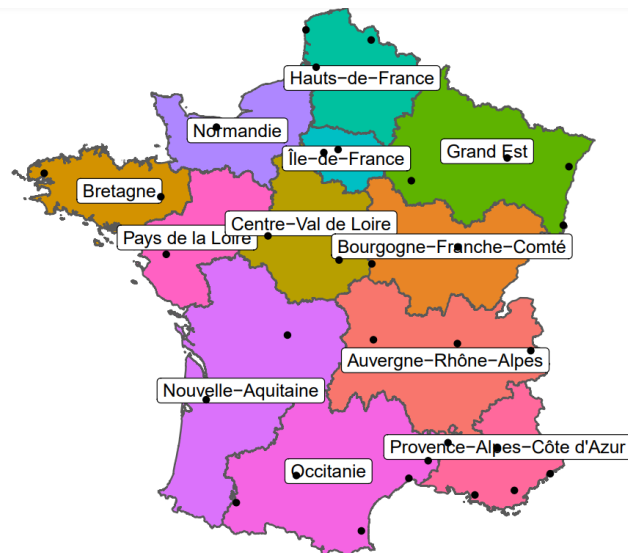
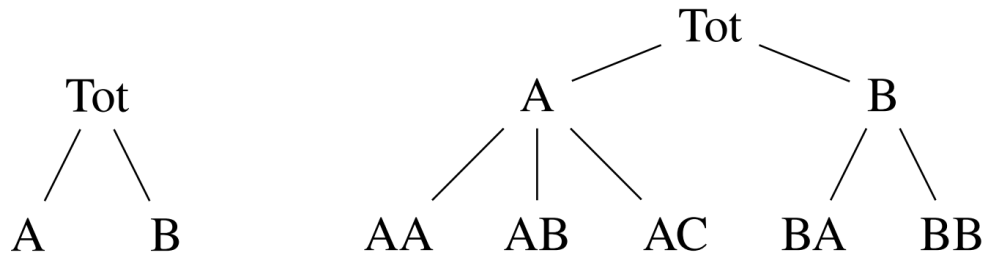
Questions ouvertes

- Comment réconcilier les prévisions au niveau des stations avec la prévision nationale ?
- Comment réconcilier les prévisions au niveau des transactions avec la prévision journalière en énergie ? Comment recalculer les prévisions de transactions à partir de la prévision en énergie plus précise ?
- Comment associer les transactions prévues aux bornes ?
- Comment passer ces modèles à l'échelle au moment où DREEV gèrera +3000 stations ?

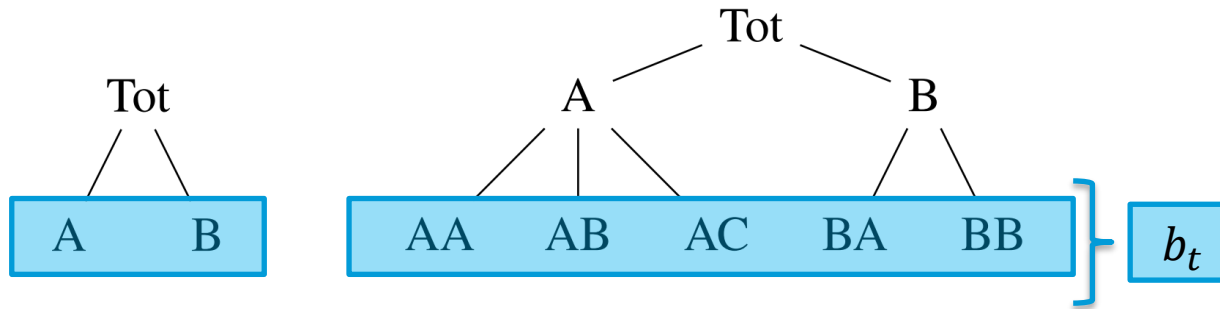
TRAVAUX EN COURS

Projet Energy Data Lab

Perspectives: How to leverage the hierarchical structure?

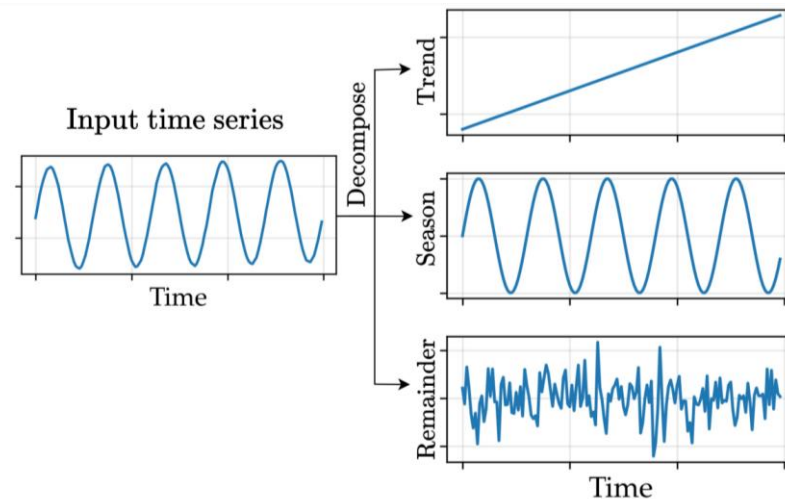
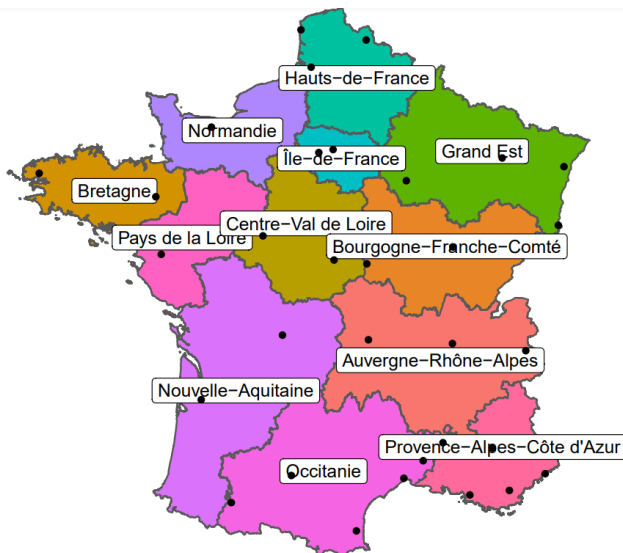


Perspectives: How to leverage the hierarchical structure?



$$S = \begin{pmatrix} 1 & 1 \\ \text{Id}_2 \end{pmatrix} \quad S = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ & & \text{Id}_5 \end{pmatrix}$$

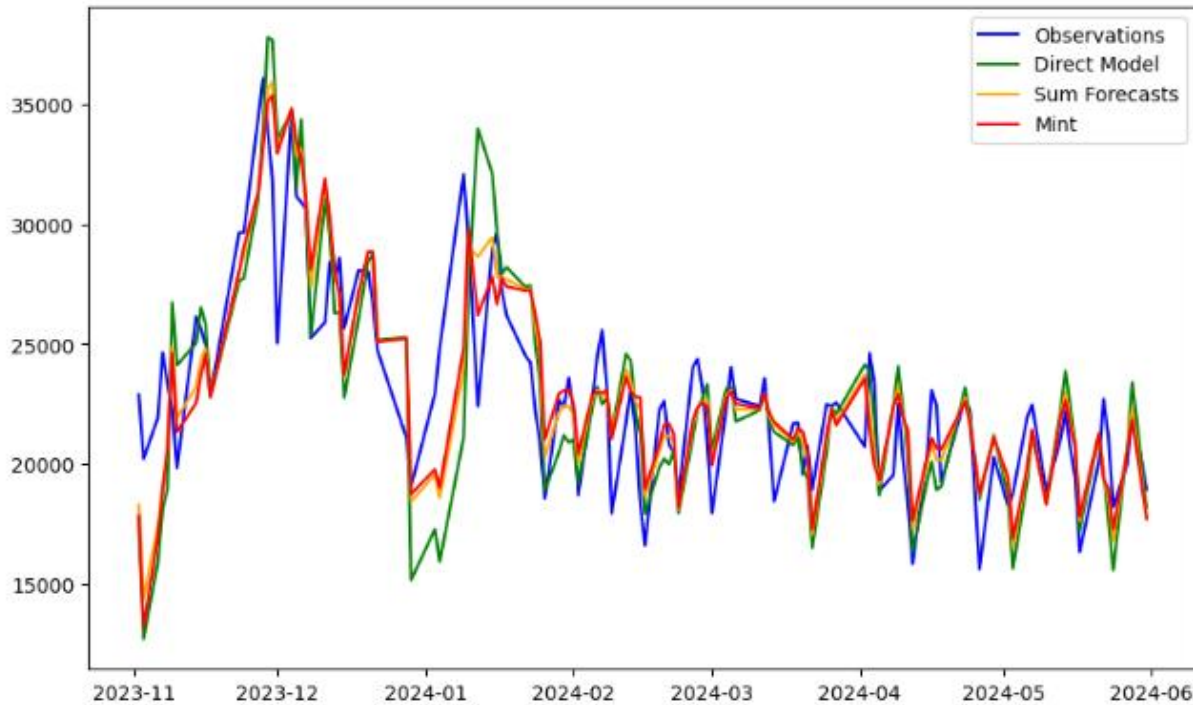
$$\begin{pmatrix} \text{Tot} \\ \text{A} \\ \text{B} \\ \text{AA} \\ \text{AB} \\ \text{AC} \\ \text{BA} \\ \text{BB} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ & & \text{Id}_5 & & \end{pmatrix}}_S \underbrace{\begin{pmatrix} \text{AA} \\ \text{AB} \\ \text{AC} \\ \text{BA} \\ \text{BB} \end{pmatrix}}_{b_t}$$



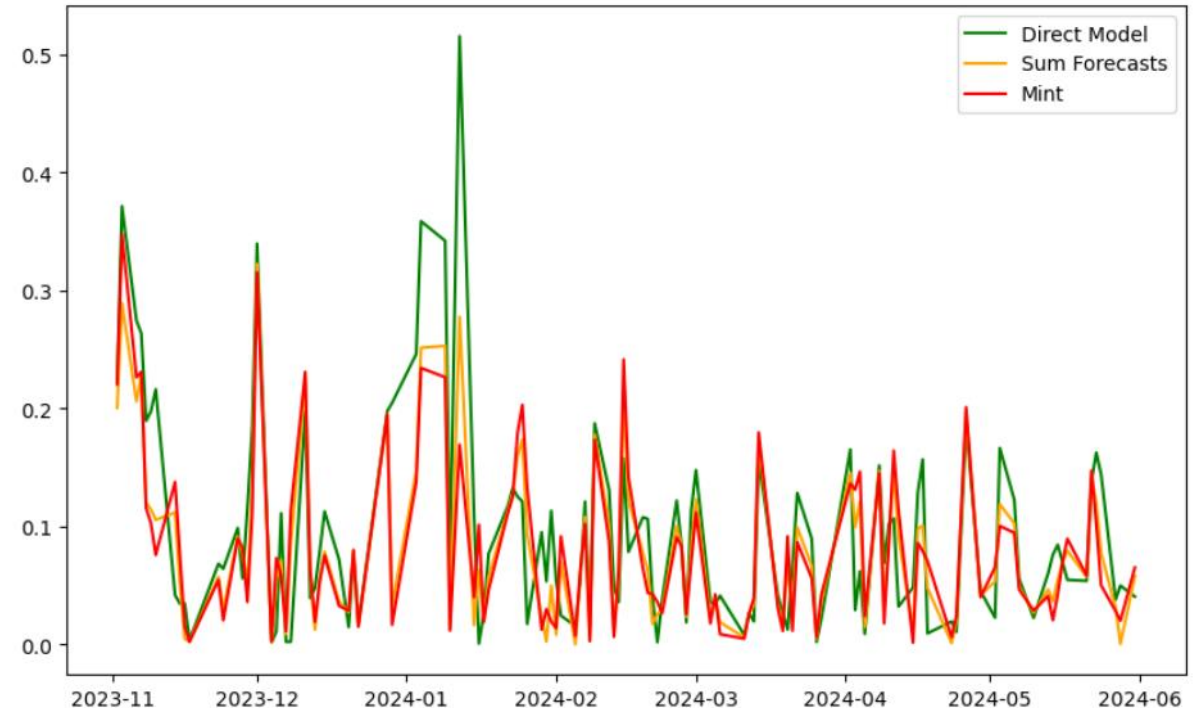
Réconciliation hiérarchique stations – national

Questionnement : Comment réconcilier les prévisions au niveau des stations avec la prévision nationale ?

Prévision directe vs somme des prévisions des stations vs réconciliation hiérarchique

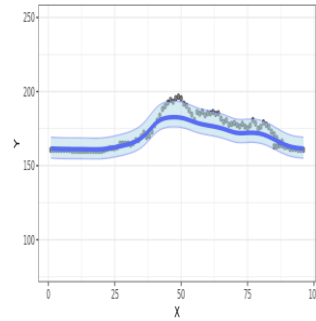
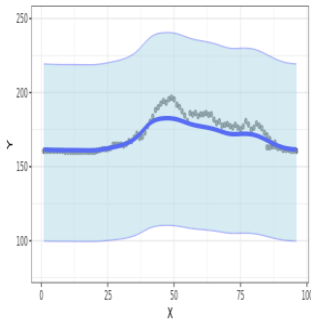
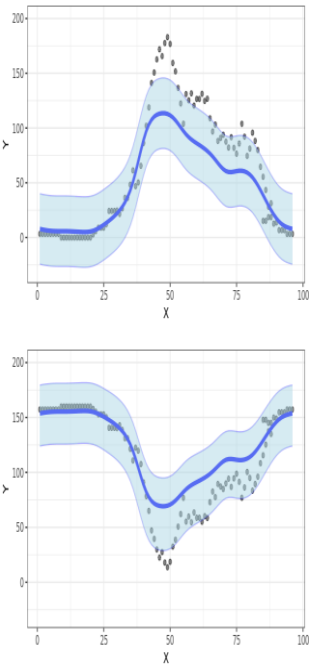


Erreurs absolues des prévisions des stations vs réconciliation hiérarchique



Perspectives: How to leverage the hierarchical structure?

- A well studied problem in point forecasting (⁵Athanasopoulos et al. (2024))
- Recent interest in probabilistic prediction (⁶Panagiotelis et al.(2023),
⁷Wickramasuriya (2024))



Conformal Prediction links point forecasting and prediction set.

What's it for ? CP already provides valid prediction set. Yet, there is still room for improving their efficiency

⁵Forecast reconciliation: A review/ ⁶Probabilistic Forecast Reconciliation: Properties, Evaluation and Score Optimisation/

⁷Probabilistic Forecast Reconciliation under the Gaussian Framework

Perspectives

